Labour Scarcity, Capital Deepening, and Productivity: Insights from the Last Nordic Plague

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Abstract

I study the causal impact of labour scarcity on productivity, a key driver of long-run growth. Exploiting a natural experiment, a 1710s plague outbreak, I show that labour-scarce plagued regions shift into capital-intensive goods and expand their exports. Using port-level trade data and a model of technology adoption, I trace the chain from labour scarcity through capital deepening to productivity growth. Parish-level production data confirm capital deepening and TFP growth. Technology adoption accounts for 51-75% of the change in comparative advantage, which reflects temporary labour scarcity for almost a century. Labour scarcity raises productivity where higher wages reach producers.

Keywords: Labour Scarcity, Capital Deepening, Technology Adoption, Productivity, Growth
JEL Classification Codes: F16, O47

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1 Introduction

What determines economies' long-run growth rate? One of the main drivers of growth are sustained increases in total factor productivity. The relationship between productivity and labour scarcity has taken on an increased importance as many economies face both low productivity growth and shrinking workforces. One view holds that labour scarcity hampers growth as firms' production is constrained and competitiveness is harmed by higher wages (OECD, 2024). Another hypothesis holds that labour scarcity spurs the adoption of capital-intensive techniques and thus productivity growth (Habakkuk, 1962). While the first view supposes fixed productivity, labour scarcity leads to productivity growth under the second view.

Evidence on the causal relationship between labour scarcity and productivity remains scarce. Exogenous variation in labour scarcity is rarely observed: Allen, 2009b, e.g., attributes the Industrial Revolution to high wages in levels. Where such variation is available, the literature focuses on causal evidence for intermediate outcomes in the causal chain, such as the adoption of labour-saving technology (Caprettini, Trew, and Voth, 2026). Yet economic growth requires that such technology choice translates into capital deepening and productivity growth (Acemoglu et al., 2026). There is little causal evidence connecting labour scarcity all the way to productivity growth through a general equilibrium mechanism. Since workforces are shrinking and the productivity effects of capital-deepening AI investments are hotly debated (Acemoglu, 2025), this tension is relevant both historically and today.

My paper closes this gap by tracing the chain from labour scarcity through technology adoption and capital deepening to productivity growth, combining granular causal evidence with a general equilibrium model. I study a natural experiment, the last Nordic plague (1708-1712), which induced localised labour scarcity whose effects I document in novel port-level trade data. Plagued regions shift their exports into capital-intensive goods and conquer larger destination market shares. I also find capital deepening and productivity growth in parish-level production data. In a trade model of endogenous technology choice, I rationalise my findings: higher wages induce the adoption of capital-intensive technology, which links labour scarcity to capital deepening and productivity growth. Motivated by capital deepening and productivity growth outlasting population recovery by decades, my model also predicts that temporary factor shocks can lead to persistent comparative advantage shifts.

The historical setting of this natural experiment is uniquely suited to study this question. To identify the causal effect of labour scarcity on productivity growth, an ideal experiment would randomly reduce populations in a subset of otherwise identical regions, while keeping demand, market access, and institutions in place. The different paths of affected and unaffected regions would then identify how factor proportions change production, technology choice, and productivity. The last Nordic plague comes close to this ideal experiment: it killed 36% of affected regions' populations, thus leading to labour scarcity. The plague's spread closely tracked army routes (rather than trade connections) during the Great Northern War (1700-1721) between Sweden and Russia, and a naval blockade reduced transmission through trade, which allows me to identify the causal effect of labour scarcity. While earlier plagues, such as the 1340s Black Death, affected all of Europe, this outbreak left export markets and most neighbouring regions unaffected. These nearby regions form the comparison group. Plague incidence varies so granularly that I control for institutions, policies, and other confounders by comparing ports in the same area. The setting decades before the Industrial Revolution allows me to abstract from innovation and to focus on technology choice from a fixed menu.

Exceptionally detailed trade data allow me to track plagued regions' trade for decades while incorporating the gravity structure of trade and separating supply-side from demand-side factors. I assemble novel bilateral commodity-level trade data from the Sound Toll records,

covering virtually all trade between the Baltic and North Seas (Marczinek, Maurer, and Rauch, 2025). Three features make these records the ideal source. First, origins and destinations were recorded at the level of ports rather than countries. I therefore observe trade at the same granular spatial resolution as the plague treatment. Second, cargoes were disaggregated into goods: I map 143,855 Old Danish cargo descriptions to 227 goods, which I assign into labour- and capital-intensive sectors. Third, the records span almost two centuries, including four decades before treatment, so that plagued and unplagued ports are observed side by side in the same destination markets before and long after the plague. I complement these trade records with Swedish parish-level production data that allow me to document the composition of production and factor usage and to estimate productivity. In sum, these data permit an unprecedentedly detailed analysis of trade and production following a labour supply shock.

The credibility of this natural experiment rests on the plague having struck regions for reasons unrelated to their economic trajectories. I argue that plague exposure resulted coincidentally from proximity to armies that spread the plague (Frandsen, 2009) while a Danish naval blockade unrelated to the plague reduced transmission through trade. I study differences over time between plagued and unplagued regions in the same NUTS 1 or 2 area, where area-year fixed effects absorb common war disruptions, famines, market access changes, institutions, and policies. I subject this design's identifying assumptions to a range of tests: to-be-plagued regions show no pre-trends in population, export volume, market shares, export composition, or comparative advantage. Additionally, I model plague exposure as a function of observable determinants and find that controlling for, reweighting on (Abadie, 2005), and restricting to similar observable determinants (Crump et al., 2009) leaves my estimates statistically unchanged. Proximity to armies and sieges does not itself predict the shift into capital-intensity and army movements in later wars do not differentially affect plagued regions. I also construct a continuous treatment variable and find support that a region's trade responds to its own mortality shock but not to neighbouring outbreaks. Together, these results support a causal reading of labour scarcity's effects on trade and productivity.

I establish three novel empirical findings. First, plagued regions' share of capital-intensive exports rises by 11pp, compared to a pre-plague mean of 16%. As capital became relatively more abundant, this is reminiscent of the Rybczynski theorem, which states that a relative rise in a factor's endowment leads to an expansion in sectors using that factor intensively. Second, plagued regions expand their destination market shares by 1pp, compared to a pre-plague mean of 1.7%. This is surprising as labour scarcity should increase wages and reduce competitiveness. With fixed productivity, higher wages raise costs across sectors: through differing factor intensities, factor proportions shift the composition of exports but cannot raise them. The export expansion, which holds across sectors and factor intensities, therefore suggests productivity growth. Third, these trade results display persistence beyond population recovery. By 1750, four decades after the plague, populations had returned to their pre-plague trends. However, trade, the capital-to-labour ratio, and comparative advantage patterns reflect temporary labour scarcity for almost a century. Thus, adjustments to labour scarcity produce persistent changes in comparative advantages.

To complement the trade findings, I draw on historical production data that reveal both capital deepening and productivity growth. Agricultural data from Sweden show capital deepening in labour-scarce parishes. Estimated production functions show that, after the plague, the output elasticity of labour falls and that of capital rises, most strongly in capital-intensive agriculture, while land per worker remains unchanged. Reminiscent of the persistent trade changes, the capital-to-labour ratio was still higher after a century. Agricultural production shifted from labour-intensive arable farming into capital-intensive animal husbandry. In data on ship ownership and imports of capital goods, I show that plagued regions accumulate larger capital stocks. Finally, I estimate TFP growth, which is significantly larger in plagued parishes

and in capital-intensive sectors. Taken together, the trade and production data show that labour scarcity leads to persistent capital deepening and productivity growth.

I use heterogeneity from pre-existing factor market frictions to narrow down the mechanism and propose capital deepening and technology adoption as necessary ingredients to explain my empirical findings. First, I consider serfdom as a labour market friction. While mortality and labour scarcity were most severe in cities, serfs could not move there, which kept urban wages artificially high and strengthened incentives to adopt capital-intensive technology. This is reflected in plagued regions with serfdom shifting more strongly into capital-intensive exports in (urban) manufacturing. The mechanism therefore runs through the transmission of labour scarcity into factor prices. Second, I construct a proxy for the depth of ports' pre-plague capital markets from the average period of production of their exports (Antràs and Kulesza, 2026, Böhm-Bawerk, 1889). Financially more developed ports shift more strongly into capital intensity and see a larger export expansion. Thus, the mechanism operates more strongly where pre-plague capital frictions were less severe, and hence runs through capital deepening. Finally, since capital intensity and competitiveness remain elevated long after population recovery, the mechanism needs to produce persistent changes from a temporary factor shock.

Different standard theories fail on at least one of these counts. A Ricardian model of fixed productivities mechanically links higher wages to lower exports, which is contradicted by the documented export expansion. A Heckscher-Ohlin model can explain the shift into capital-intensive exports, yet cannot produce the export boom, and also implies shifting relative imports, which is at odds with my data. I find no evidence for a demand-side channel of non-homotheticity (Voigtländer and Voth, 2012, Fieler, 2011) and directed technical change (Acemoglu, 2002) is implausible in the early 1700s, when the menu of technology choices was fixed (Atack, Margo, and Rhode, 2019, Gallardo and Sauer, 2018, Coleman, 1956). I also argue against venting out (Almunia et al., 2021), market power, human or social capital deepening, and plague-related trade cost changes. Instead, to rationalise my empirical findings and incorporate the evidence from serfdom, financial development, and persistence, it takes a model in which factor prices induce the adoption of existing capital-intensive technology, which persistently raises productivity and shifts comparative advantage.

To explain my findings, I propose a general equilibrium trade model of endogenous technology choice. Labour scarcity triggers technology adoption from a fixed menu, placing this model in the tradition of Habakkuk, 1962 and Hicks, 1932. By linking productivity-enhancing technology choice to wages and thus factor proportions, this is a hybrid model. The pattern of trade is determined by relative productivity, which is the Ricardian element, and factor proportions affect trade both directly (higher wages harm competitiveness) and indirectly (through technology choice). The model therefore contains two channels, which I separate below: the direct factor price channel and the endogenous technology channel.

Productivity consists of exogenous draws and endogenous technology choice (Bustos, 2011). Producers can adopt lower marginal cost capital-intensive technology, which requires a sunk investment that generates a cutoff productivity draw. This cutoff drops as wages rise, inducing technology adoption. The technology set is an ordered menu of such techniques: producers climb it rung by rung as wages rise, each step governed by the same cutoff structure. The sunk adoption investment generates a band between the entry and exit cutoffs and turns a temporary factor price shock into a persistent change in productivity and comparative advantage.

Mapping this model to the data implies that adoption responds more strongly in capital-intensive sectors, where more technologies existed that were worth a sunk investment, such as furnaces, dry docks, and livestock housing. While I do not observe technology adoption directly, I observe its two footprints, capital deepening and productivity growth, under one exogenous shock. The model connects the two through adoption and I argue that nothing else does.

Notably, the model contains a feedback loop: higher wages induce adoption, raise productivity, and raise labour demand and wages again. I derive a condition on this loop gain under which the equilibrium is unique and show that it holds at the estimated magnitudes.

I propose a mechanism that proceeds in three steps, each of which I validate with historical data. First, labour scarcity leads to higher wages, absolutely and relative to capital returns. Second, rising relative wages lower the adoption cutoff, so more varieties are produced under capital-intensive technology, which increases the sectoral capital-to-labour ratio. The model attributes the export shift into capital-intensive goods to three channels: sectors' different labour intensities, their different adoption responses, and relative price effects. Third, technology adoption results in productivity growth, which explains the export expansion and the parish-level TFP estimates.

Taking the model to the data, I recover a triple difference of sectoral competitiveness from gravity (Costinot, Donaldson, and Komunjer, 2012). I show that capital-intensive agriculture and manufacturing see significantly faster competitiveness growth than their labour-intensive counterparts. The model factors this recovered competitiveness into a productivity component, which is the endogenous adoption response, and a factor proportions wedge, which depends on the trade elasticity, the labour-share gap between sectors, and wage growth. I bound the wedge with pre-industrial labour shares, wage growth after previous plagues, and seminal estimates of the trade elasticity: 51-75% of the differential reflects technology adoption under my proposed mechanism. This bound concerns the relative shift between sectors: the level expansion of exports across sectors is productivity growth alone. Estimated production functions for Swedish parishes corroborate this decomposition independently. After the plague, output elasticities shift from labour to capital, most strongly in capital-intensive agriculture, and the implied widening of the cross sector labour share gap falls inside the model-consistent range.

Serfdom provides a second moment and further validation to my mechanism. My argument is that exposure to factor price changes drives adoption. Indeed, in agriculture, serfdom removes more than half of the differential, though the interaction is imprecisely estimated, whereas in urban manufacturing, untouched by serfdom, it is unchanged. This is precisely what my mechanism predicts: serfdom switches off the factor price transmission and the adoption response while the technology primitive remains the same. The mechanism thus reconciles all empirical findings: the shift into capital-intensive exports reflects lower labour shares and stronger adoption in capital-intensive sectors; the export expansion and falling export prices reflect productivity growth; persistence arises from sunk adoption capital.

This paper makes four contributions. First, I provide causal evidence that labour scarcity leads to productivity growth. A long literature focuses on the first step in this chain, typically in a closed economy, showing that labour scarcity leads to technology adoption (Habakkuk, 1962; Acemoglu, 2010 on the factor bias of technological progress; Allen, 2009b on high wages causing the Industrial Revolution, criticised by Humphries and Schneider, 2020 and Stephenson, 2018). Recent causal evidence identifies this step, typically in partial equilibrium: Caprettini, Trew, and Voth, 2026 use variation in warships' coastal access, Franck, 2024 exploits temperature variation, and Andersson, Karadja, and Prawitz, 2022 use an instrument for emigration. Börschlein, Bossler, and Popp, 2024 and Groiss and Sondermann, 2023 study firms' responses to labour scarcity in the modern context. My paper is the first test of the Habakkuk thesis in an open economy general equilibrium framework, addressing the criticism of Temin, 1966 (see also James and Skinner, 1985). I also show that the feedback loop between wages and adoption is consistent with a unique equilibrium.

A smaller literature studies the relationship between labour scarcity and (productivity) growth. While some consider labour scarcity detrimental to growth (OECD, 2024, Caldara, Iacoviello, and Yu, 2025), Acemoglu et al., 2026 document that falling birth rates are followed by

higher TFP. My contribution is to carry the chain from labour scarcity to technology adoption through to its endpoint. I offer a unified framework explaining how labour scarcity leads to technology choice, capital deepening, and productivity growth. To do this, I draw on an exogenous mortality shock and granular data, in which I observe the intermediate step, capital deepening, and the final outcome, productivity growth. Like Acemoglu et al., 2026, who turn to wartime deaths, I use a mortality shock. However, this plague's granularity allows me to compare ports in the same area, differencing out institutions, policies, and other confounders. The trade and production data track local economic adjustments for decades and separate supply-side from demand-side factors. Therefore, my paper contributes an empirically grounded and unified framework to answer whether and why labour scarcity leads to productivity growth.

Second, I offer a factor price-driven model of technology adoption. I formalise the causal chain from labour scarcity to productivity growth, quantify the mechanism, and prove uniqueness. Making productivity respond to factor prices generates a feedback loop that structurally mirrors scale economies. Kucheryavyy, Lyn, and Rodríguez-Clare, 2023 characterise when trade models with external economies of scale are well-behaved. Here, the measure of varieties is fixed and the externality is linked to factor prices rather than industry size, leaving the adoption loop as the only scale force. I state uniqueness as a condition on the slope of labour demand and confirm existence and uniqueness. The model is distinct from the universal gravity framework of Allen, Arkolakis, and Takahashi, 2020 and Allen, Arkolakis, and Li, 2024, which requires a single aggregate good. I also provide a method to separate productivity from factor proportions in gravity-recovered competitiveness. The latter factors into two components and the factor proportions wedge can be bounded with labour shares, the trade elasticity, and wage growth. This applies directly to other settings where factor supplies shift, such as migration, demographic change, or capital flows.

Third, my paper contributes to the literature on path dependence in comparative advantage: Krugman, 1987 studies learning-by-doing, Hausmann and Rodrik, 2003 stresses cost discovery, Loecker, 2013 export learning, and Allen and Donaldson, 2022 build a model for persistent effects of temporary shocks. Several papers attest to the empirical relevance of path-dependent comparative advantage (Lane, 2025, Juhász, 2018). Hanson, Lind, and Muendler, 2018 stress the empirical relevance of dynamic comparative advantage. I show that a shock to factor proportions is followed by a lasting shift in comparative advantages, where studying a natural experiment exempts my findings from the identification criticism of Juhász, Lane, and Rodrik, 2024. On the theoretical side, I propose a general equilibrium framework in which technology adoption produces persistent comparative advantage changes from a temporary factor shock.

Fourth, my paper relates to the literature on plagues. Several papers study population recovery and wages (Jedwab, Johnson, and Koyama, 2024, Álvarez-Nogal, Prados de la Escosura, and Santiago-Caballero, 2020), agricultural production (Gelman, 1982), long-run development and changing demand (Voigtländer and Voth, 2012), and the political economy of plague outbreaks (Bosshart and Dittmar, 2025, Dittmar and Meisenzahl, 2019). I provide detailed trade findings and a micro-foundation for economic recovery (Davis and Weinstein, 2002).

This paper proceeds as follows. Section 2 introduces the context and data, Section 3 the identification strategy, and Section 4 the empirical results. Section 5 narrows down the mechanism, Section 6 presents the model (existence and uniqueness in Appendix G), Section 7 quantifies it, and Section 8 concludes.

2 Context and Data

This paper studies the last plague outbreak in Northern Europe, which coincided with large-scale warfare. The war that broke out in 1700 between Sweden, Russia, and their allies affected most countries on the Baltic Sea. Early campaigns led the Swedish army into the Baltics and later deep into the Polish-Lithuanian Commonwealth. After the decisive Swedish defeat in Poltava in 1709, the Danish navy blocked ships between Sweden and its Northern German possessions, and the Russian navy raided the Baltic coast. As a result, ship passages at the Sound collapsed. As the war turned in Russia's favour, Swedish troops receded to Northern Germany, and Russian troops marched far into Sweden. This so-called Great Northern War ended in 1721 and marked the end of Sweden as a great power, as it lost the Baltic to Russia and parts of Pomerania to Prussia. The war was accompanied by a plague outbreak which followed army routes (Appendix Figure A.2b), reaching East Prussia in 1708, most of the Baltic Sea by 1711 and Hamburg by 1712. Notably war, not trade, spread the plague (see section 3).

2.1 The Soundtoll Data

A novel granular trade panel is constructed based on the Soundtoll data (see also Marczinek, Maurer, and Rauch, 2025). I report key descriptives in Appendix Table A.1. Virtually all trade between the Baltic and North Seas is recorded.

Classification of Goods I map 143,855 Old Danish cargo descriptions to 227 goods and to five sectors: labour-intensive agriculture, capital-intensive agriculture, labour-intensive manufacturing, capital-intensive manufacturing, and remaining unclassified goods. As a robustness exercise, I use Claude's Opus 4.6 model to classify cargoes. This simple exercise produces the same goods classification as my manual efforts for 84% of traded value, and for 96% of traded value after some inconsequential differences[1] are accounted for (Appendix A.1). Re-estimating the shift into capital-intensive exports with Claude's classification in place of mine reproduces the baseline result (Appendix Figure A.4e).

Below, I argue that the differentiation between labour- and capital-intensive goods is crucial, and that land intensities are not a driving factor.[2] I classify goods by recognising that capital-intensive goods require a high baseline amount of capital, contained in buildings, furnaces, machinery, tools, carts, and animals. Labour-intensive goods, on the other hand, are dominated by labour-intensive tasks and limited in the degree to which labour can be saved.

Labour-intensive agriculture is arable farming. While using some capital such as horses, carts, and ploughs, arable farming is dominated by the labour-intensive tasks it entails, such as ploughing, sowing, weeding, harvesting and threshing (Heblich, Redding, and Zylberberg, 2025). Pastoral farming, mining, and processed foods are capital-intensive agriculture due to the infrastructure and tools required to mine and process foods. For pastoral farming, livestock itself is capital (Broadberry et al., 2014, Allen, 2005). Livestock require capital-investments for grazing (fencing, drainage, and roads) and for milking and slaughtering (buildings and tools) and otherwise graze freely in fields, requiring little labour input. In manufacturing, I classify textiles as labour-intensive (Broadberry and Gupta, 2009, Pomeranz, 2006). Pre-industrial production methods were labour-intensive, with tasks such as spinning, weaving, and dyeing. Iron and metal works and ship building, on the other hand, are naturally more capital-intensive in that they use furnaces, buildings, docks, and tools.

Appendix Table A.2 ranks the top 30 goods, accounting for 90% of traded value during

[1] These are iron works/iron, planks/wood, goods/mercery, cloth/sail, meat/beef.

[2] I find in Appendix Figure A.4c that there is no relative shift into land-intensive commodities (arable crops, livestock and animal products, plantation goods, raw forestry) after the plague.

this period, and shows their assignments. I run a robustness exercise in which I reassign four debatable goods: wine and tow to labour-intensive agriculture and planks and ash to labour-intensive manufacturing. This does not change my results.

Classification of cities, duties, currencies, and units The Soundtoll data record origins and destinations at the port level. I map 90,737 original port names to 3,085 unique port identifiers and assign them into one of 56 areas. Areas are mostly equivalent to NUTS 2 regions; for the United Kingdom, France, Germany, and Spain, they are NUTS 1 regions to ensure similarly sized areas and to avoid areas with a single port (Appendix B). However, my estimates are robust to grouping ports into NUTS 1 or NUTS 2 regions. Only ships passing the Sound are recorded, so only trade between the Baltic Sea and the North Sea is recorded. I confirm my findings in trade data for Hamburg (Appendix Table A.10) for which I see trade with all partners. I also present simulation results in Appendix Figure A.11 comparing the hypothetical and the observed sample, with no evidence for bias.

Details on tolls and duties are given in Appendix A.2. Currency concerns are minor as 99.76% of transactions are in Danish coin. I also use shipments' weights to construct unit prices (Appendix Table A.3). A truth-telling mechanism was in place to mitigate incorrect reporting: the Crown could purchase goods at the stated value to discourage underreporting.

2.2 Plague & Mortality Data

Appendix Table A.4 presents the plague data I collected.[3] A region is coded as plagued when there is a documented outbreak, almost always in its port though sometimes for the entire region. Mortality rates, which are available for half of plagued cities with the median mortality rate at 36%, may not be reliable (Dittmar and Meisenzahl, 2019, Roosen and Curtis, 2018), pointing to the plague dummy as being more robust; I show results for both. Where unavailable, I predict mortality rates with the timing of the plague, geographical controls, and the proximity to army routes dated before the local outbreak (Appendix Table A.5). There is no evidence for demographic heterogeneity in mortality (Raster, 2023). Morever, there is no evidence that the plague was spread through specific commodities.[4]

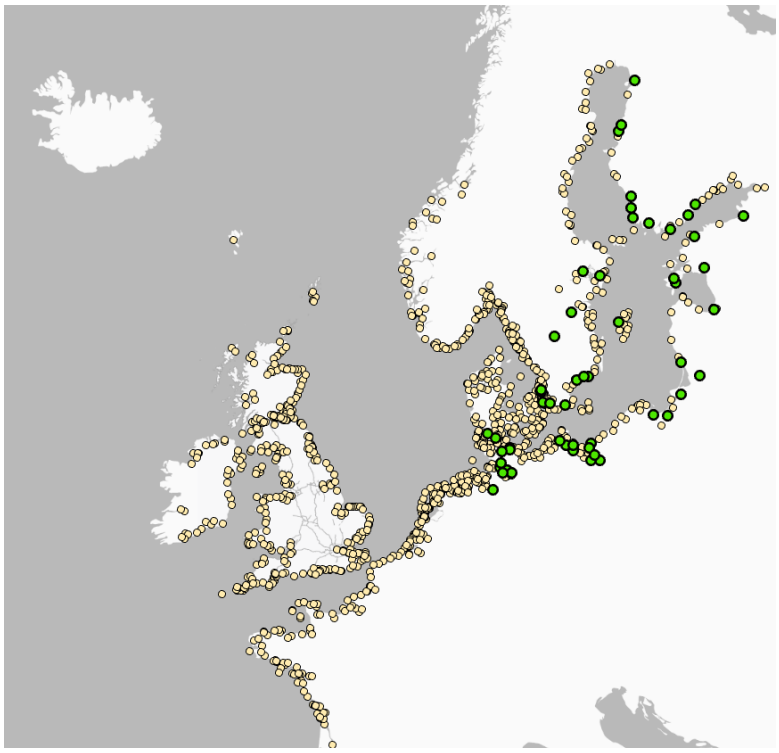
Figure 1 and Appendix Table A.4 show the 38 plagued regions. As only seven plagued regions were additionally directly besieged (of which the sieges of Copenhagen, Narva, and Riga occurred before the plague), I expect the destruction of physical capital to play a minor role. Appendix Table A.6 shows that plagued cities were not significantly larger before the plague and not significantly smaller by 1750. To test regional populations, I find that plagued Swedish parishes had recovered their populations by 1750 (Appendix Table A.6).

The main analysis views treatment as an outbreak in a region. As a robustness check, I incorporate outbreaks across all of Europe. I construct an indirect plague treatment variable that assigns higher weights to closer by and larger plagued regions (Appendix C.2). In Appendix Table A.9, I show that the indirect plague treatment variable is mostly insignificant and of inconsistent sign. The export expansion established for the direct plague treatment goes through. Therefore, I continue to focus on the direct effects of the plague.

[3]Raster, 2023 presents rural mortality data for Northern Estonia; this effort cannot be extended to the entire region. For towns covered in both the trade data and in Raster, 2023, the plague status overlaps precisely.

[4]Frandsen, 2009 mentions that in some places, textiles of those who perished were burnt as they were believed to be contagious. However, the finding in Appendix Figure A.10 that relative demand did not shift into textiles after the plague speaks against the salience of this anecdote.

Figure 1: Plagued regions



Notes: **Green circles:** plague. **Beige circles:** no plague.

2.3 The Little Ice Age, Serfdom, and Famine

The period after the plague coincides with warming (Waldinger, 2022), for which I control and account below with area-(sector)-time fixed effects. In temperature data by Xoplaki, 2005 and Luterbacher et al., 2004, I match regions to the four closest observations, weighting by the inverse distance to these grid points and defining spring and summer as the growing season.

Serfdom hindered rural-to-urban labour mobility by binding serfs to their lords' lands. Skill and language barriers further prevented serfs from moving to cities. I conceptualise serfdom as a mobility restriction and collect data on its presence from Raster, 2023 and Peters, 2022 (see Appendix E.1). Raster, 2023 argues that labour coercion increased in Northern Estonia as a result of post-plague labour scarcity; Denmark reintroduced serfdom in 1733 (Gary et al., 2022). Throughout, the empirical specifications absorb potential time variation in serfdom.

Before and during the Great Northern War, some areas around the Baltic Sea witnessed famines. Given only vague geographical delineations of where famines hit, I propose three approaches. Throughout, I control for growing season temperatures to capture poor harvests. Additionally, I control for 10- and 20-year lagged temperatures (Appendix Figure A.8c) to more flexibly control for temperature-driven bad harvests. Finally, I include area-year fixed effects throughout, which control for area-wide famine incidence.

3 Identification

This paper asks how economies respond when labour becomes scarce. The ideal experiment would remove a large share of workers from some regions and not from otherwise identical neighbours, leaving demand, market access, institutions, and available technologies untouched. Comparing treated and untreated regions selling in the same markets would then trace out how

factor prices move production, technology choice, and productivity. The Great Northern War plague outbreak comes close to this experiment. It killed a large fraction of the population in affected regions, struck ports that had traded side by side for decades, and left neighbouring competitors untouched. The Sound Toll registers consistently record the trade of both groups before and after the plague.

The strength of this setting lies in how the plague travelled. In this outbreak, propagation followed army movements rather than commerce (Frandsen, 2009).[5] The earliest outbreak occurred in a Swedish military hospital in Southern Poland (Frandsen, 2009). The disease then moved with troops to the Baltic coast. The frontier of infection, too, traces army operations rather than merchant networks. Trading links reached deep into Western Europe, yet the West stayed unplagued. Instead, the westernmost outbreak coincides with the westernmost advances of the Danish army. Further, a Danish naval blockade unrelated to the plague reduced the trade channel of transmission (Appendix Figure A.3). Its footprint is measurable: significantly less ships sailed from ports closer to the Danish base at Christiansø during the blockade years (Appendix Table A.14). Armies rarely targeted cities, and there is no evidence of the plague being used as a biological weapon. I argue that exposure was a byproduct of campaign routes, and which regions caught the plague reflects army itineraries and epidemiological chance rather than regional fundamentals. Exposure to labour scarcity was therefore plausibly assigned by a process orthogonal to the sectoral composition, productivity, and trading positions that form my outcomes, meeting the requirements of a natural experiment.

My design exploits this variation conservatively. I compare differences over time between plagued and unplagued regions within the same area, so that anything common to an area, including war disruptions, famines, market access, and policy, is differenced out. I confirm that the data behave as the experimental reading predicts: to-be-plagued regions display no pre-trends in population, export volumes, market shares, export composition, or competitiveness (and thus comparative advantage). However, to-be-plagued regions have more exports in levels and are closer to army routes (Appendix Table A.7), which motivates this paper's empirical design. The level difference is not a threat to the below identifying assumptions and absorbed by fixed effects in every specification. Additionally, a propensity score design, which explicitly accounts for the level difference and army proximity[6] (Appendix Table A.8), leaves my estimates statistically unchanged. I also confirm that proximity to armies and sieges alone predicts no shift in export composition and that later wars did not disproportionately return fighting to plagued regions. The remainder of this section states the identifying assumptions formally and maps each to this evidence.

Specification and estimand

The below regressions are special cases of the following common event study specification, where i denotes origins, j are destinations, t are years, k are sectors, f are factor intensities, and a are areas. Let K_{it} be the time difference to the plague. For an outcome y_c in cell c (origin-sector (ik) for compositional outcomes, origin-destination (ij) for bilateral outcomes) I estimate:

$$y_{ct} = \sum_{l=-20\setminus -1}^{30} \beta_l \, \mathbb{1}\{K_{it} = l\} + \phi_{ct} + \gamma x_{it} + \varepsilon_{ct}, \tag{1}$$

[5]Propagation channels differ across outbreaks (Alfani and Murphy, 2017). Trade has been proposed as the main channel for the Black Death (Benedictow, 2020), while armies spread the plague during the Thirty Years War (Eckert, 1996). This outbreak belongs to the second type.

[6]This is measured as the distance to the nearest army movement dated before the local outbreak, so that the covariate is strictly pre-determined.

where ϕ_{ct} is a vector of fixed effects, x_{it} are time-varying origin controls and standard errors ε_{ct} are clustered on origins. Interacting the β_l with a capital-intensity indicator yields the triple difference used for Fact #1 and for productivity growth.

In most regressions, I include area-time fixed effects. Thus, identification comes from differences between plagued and unplagued regions within the same area (see Appendix B). This pre-empts several confounders, as anything common to an area is differenced out, such as area-wide market access changes, war disruptions, famines, warming, and policies. It extends to country-wide policies, for which even variation across a country's areas is absorbed. The remaining concern is thus about region-idiosyncratic confounders correlated with plague status.

I capture plague-induced labour scarcity with a plague dummy. Alternatively, I also show results for the mortality rates, which is a noisier measure (Roosen and Curtis, 2018) and imputed for half the sample (Appendix Table A.5). Further, I incorporate plague outbreaks across Europe to construct an indirect continuous plague treatment variable . As I find that this continuous treatment has no predictive power beyond the own plague dummy, I choose the latter as my main treatment variable. Treatment is absorbing and staggered (1708-1712). A decomposition (Goodman-Bacon, 2021) attributes 99.1% of the identifying variation to comparisons of eventually- against never-treated ports, so bias from treatment-effect heterogeneity is minimal. I also confirm robustness to the estimators of Borusyak, Jaravel, and Spiess, 2024 and Chaisemartin and D'Haultfoeuille, 2022.

Identifying assumptions

Let $y_{ct}(0)$ and $y_{ct}(1)$ be potential outcomes and $D_i = \max_t D_{it}$ indicate ever-plagued regions. Identification of β_l rests on three assumptions[7]:

(A1) Conditional parallel trends. Given controls and fixed effects, the untreated potential outcomes of ever- and never-plagued regions would have evolved in parallel,

$$\mathbb{E}[y_{ct}(0) - y_{c,t-1}(0) \mid D_i = 1, \phi_{ct}, x_{it}] = \mathbb{E}[y_{ct}(0) - y_{c,t-1}(0) \mid D_i = 0, \phi_{ct}, x_{it}] \,.$$

This is validated by the absence of pre-trends: to-be-plagued regions show no differential growth in population (Appendix Table A.6), export volumes (Appendix Figure A.7a), market shares (Figure 3), capital intensity (Figure 2), or competitiveness (Appendix Figure A.14). Additionally, a propensity score analysis shows that controlling for, reweighting on (Abadie, 2005), and restricting to similar observable determinants of plague exposure (Crump et al., 2009) delivers estimates that are statistically indistinguishable from my baseline estimates (see Appendix C). Thus, observable selection does not drive my results. As Abadie, 2005 notes, selection on observables is a special case of this conditional parallel trends restriction, so the propensity score evidence speaks directly to the credibility of (A1).

(A2) No anticipation. $y_{ct} = y_{ct}(0)$ for all t prior to onset: before the plague, regions do not adjust. I find support for this below in the absence of differential pre-trends.

(A3) No interference (SUTVA). A region's plague does not alter other regions' potential outcomes. A continuous treatment incorporating exposure to plagues in neighboring regions (Table A.9) has no consistent effect and leaves the baseline point estimates virtually unchanged for the own-plague coefficient. Trade responds only to a region's own mortality shock and not to its neighbours', supporting (A3). I also present evidence against (unobserved) market reallocation in the form of level effects, simulations on trade selection (Appendix G.5), and a robustness check for Hamburg, for which I see trade with all partners (Appendix Table A.10). Finally, I test whether ports whose hinterlands are tied to them by low-cost upriver transport, where switching export harbours is most costly, show a differential post-plague export response.

[7] All details and further results are in Appendix C.

Appendix Table A.11 rejects this, contradicting a rerouting channel from hinterlands to different export harbours after the plague that might violate assumption (A3). Throughout, I find evidence that interference is no concern.

Assumptions (A1)-(A3) identify β_l as the causal effect of plague status on y_{ct}. To interpret β_l as the effect of plague-induced labour scarcity, I impose one further restriction:

(E) Exclusion. Conditional on x_{it} and ϕ_{ct}, plague status affects trade only through labour scarcity. War- and policy-related exposure correlated with plague status has no direct effect on trade. In support of this assumption, I run a placebo test: proximity to armies and sieges does not predict a shift into a higher capital intensity of exports (Appendix Table A.12). The robustness of my estimates to the propensity score exercise, which incorporates army movements, also supports this assumption. Given the inclusion of area-time fixed effects, I argue that policies are unlikely to violate the restriction. Further, I find no evidence that army movements in later wars pose a threat to identification as exposure to battles and sieges in the War of the Polish Succession and the Russo-Swedish War of 1741-1743 is balanced across plague status (Appendix Table A.13). Finally, the naval channel behaves the same way: proximity to the Danish blockade lowered exports during the blockade years, with no differential effect by plague status or sectors' factor intensity (Appendix Table A.14).

4 Empirical Results

In this section, I present empirical results on the effects of labour scarcity. The trade findings are complemented by production findings, which confirm and strengthen my results. I begin by forming a set of hypotheses. Labour scarcity[8] increases wages, making capital relatively cheaper.[9] I classify goods as labour- or capital-intensive. For agriculture, arable farming is labour- and pastoral farming is capital-intensive (Heblich, Redding, and Zylberberg, 2025, Broadberry et al., 2014, Allen, 2005). For manufacturing, I distinguish labour-intensive production of textiles from capital-intensive metal works and ship building (Broadberry and Gupta, 2009). Details in section 2.1.

Three adjustments could appear. First, an adjustment *within* sectors. If factor prices are not equalised across regions, a higher capital-to-labour ratio will be employed: capital deepening. In arable farming, this might take the form of draught animals or metal ploughs; in manufacturing, machinery and tools. This adjustment is difficult to observe in trade data, so I document it in production data below.

The second adjustment is a compositional effect *across* sectors and as such observable in trade data. Labour scarcity should induce regions to relatively increase their production of capital-intensive goods, which is predicted by the Rybczynski theorem as part of factor content theory. At constant relative goods prices, it states that the rise in the endowment of capital, compared to labour, leads to a more than proportional expansion of output in capital-intensive sectors.[10] I propose a hybrid model below which also predicts an adjustment *across sectors*.

Third, labour scarcity may lead to productivity growth. If it does not, exports decline in every sector as wages rise since the first two adjustments only lead to changes in the capital intensity and composition of exports. If labour scarcity does lead to productivity growth,

[8]The terms scarcity and shortage are used interchangeably (Caldara, Iacoviello, and Yu, 2025). I speak of scarcity throughout, as labour became scarcer compared to capital.

[9]In Appendix Table A.29, I follow Keller, Shiue, and Wang, 2021 in inferring interest rate from monthly grain price variation, and find that interest rates do not rise after the plague.

[10]This change in regional relative factor endowments does not lead to changes in regional factor prices as long as the region is sufficiently small. Factor price equalisation is rejected by the data, though Trefler, 1993 reconciles this with factor content theory by introducing factor-augmenting productivity differences.

one should observe both a compositional change and an expansion of exports. Productivity growth might take time to materialise as best practices spread slowly (Juhász, Squicciarini, and Voigtländer, 2024). Further, trade was severely depressed during the war: after 1709, the Danish navy blocked ships and the Russian navy raided many Baltic coastal areas and intercepted vessels. This reduced the number of passages at the Sound (Appendix Figure A.3). I collect detailed data on Russian naval raids and blocked ports and exploit the fact that the Danish navy was operating its blockade out of the island of Christiansø. In Appendix Table A.14, I show that halving a port's distance to Christiansø is associated with about 27% fewer ships sailing from that port during the blockade, controlling for other direct blockades, Russian raids, their interactions with plague status and factor intensity, latitude and longitude interacted with the war period, and a host of fixed effects. Crucially, this adverse effect on trade does not vary by a port's plague status or sectors' factor intensity. The increased risk of shipping affected both plagued and unplagued ports and limits our ability to compare their trade in the years immediately after the plague. Therefore, most effects become visible only after the peace treaty of 1721. This is confirmed by studying the export volumes and composition of the main plagued port on the North Sea, where shipping was not disrupted: Hamburg's trade (see Appendix Table A.10) shows effects on impact.

Fact #1: Capital-intensive sectors expand relative to labour-intensive sectors.

To test whether the plague's effects on trade differ by sectors' factor intensities, I estimate an event study using OLS on log exports, weighted by exports in levels:[11]

$$T_{ikt} = \sum_{l=-20\setminus -1}^{30} \beta_l^f \cdot \mathbb{1}(K_{it} = l) + \alpha_{ik} + \alpha_{kt} + \alpha_{at} + \gamma_f x_{it} + \epsilon_{ikt}, \tag{2}$$

where i are origins, k sectors, f is factor intensity (capital- or labour-intensive), a are areas, and t are years. T_{ikt} are (log) exports. K_{it} is the time difference to the plague, α_{ik} are origin-sector, α_{kt} sector-time, and α_{at} area-time fixed effects. x_{it} are temperatures. Standard errors are clustered at the origin level. Due to the area-time fixed effects, identification comes from comparing plagued and unplagued regions within the same area in the same year, so that area-wide shocks to market access, war disruption, famines, warming, and policy are absorbed.

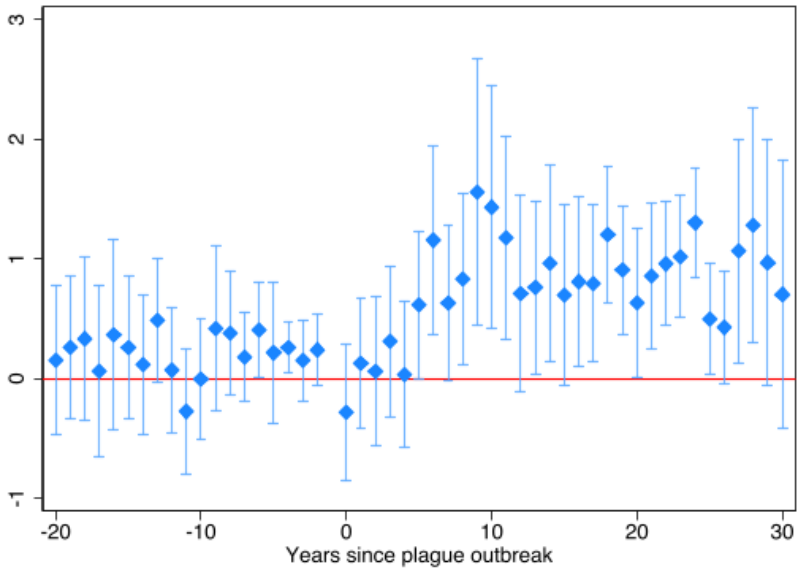
Figure 2 shows the difference between β_l^C and β_l^L: after the plague, capital-intensive exports expand relative to labour-intensive exports. The estimates imply a 11pp increase in the share of capital-intensive exports, compared to a pre-plague mean of 16% among plagued ports.[12]

Appendix Figure A.4a shows that an alternative assignment of goods produces almost identical results. Appendix Table A.15 shows PPML results in a DiDiD set up, both overall and within each agriculture and manufacturing. Appendix Table A.12 shows that army proximity and sieges are not associated with a shift into capital-intensive exports, which also serves as a placebo test. Appendix Figure A.4b confirms the finding using the imputation estimator in Borusyak, Jaravel, and Spiess, 2024. Further, Appendix Figure A.4c shows no shift into land-intensive exports after the plague. While populations had recovered by 1750 (Appendix Table A.6), this finding outlasts the recovery by four decades (Appendix Figure A.4d).

[11] This recovers OLS estimates similar to PPML estimates. Head and Mayer, 2013 show that PPML places more weight on high trade flows, so weighting OLS on log trade by levels brings OLS close to PPML results (Mayer, Vicard, and Zignago, 2019).

[12] As discussed above, the shock becomes particularly visible after a few years, when war-induced shipping disruptions unrelated to the plague had been resolved (Appendix Figure A.3 and Appendix Table A.14).

Figure 2: Capital-intensive exports expand relative to labour-intensive exports



Notes: Estimation of equation (2). Standard errors clustered at the origin level.

Capital-Intensity Shift and Capital Deepening: Evidence from Production Data

Beyond trade data, Swedish production data show the plague shifting production technology directly. For Swedish parishes observed in 1690 and in 1750 (Andersson Palm, 2025), I estimate production functions in which every factor's output elasticity may change after the plague,

$$\ln Y_{pt}^{s} = \sum_{f \in \{L,K,T\}} \left(\beta_{f}^{s} + \beta_{f,post}^{s}\, \mathbb{1}_{t} + \beta_{f,pl}^{s}\, P_{p} + \delta_{f}^{s}\, \mathbb{1}_{t} P_{p}\right) \ln X_{fpt} + \lambda^{s} \mathbb{1}_{t} + \mu^{s} P_{p} + \varphi^{s} \mathbb{1}_{t} P_{p} + \varepsilon_{pt}^{s}, \quad (3)$$

where s indexes harvests (labour-intensive agriculture) and animal output (capital-intensive agriculture), the factors f are labour (population), capital (horses), and land (sowed acreage), $\mathbb{1}_{t}$ indicates the 1750 census and P_{p} a plagued parish. By the 1700s, horses had become the main draught animal in much of Europe (for England, see Langdon, 1986). Further, horses were needed to use agricultural tools (Andersson, Karadja, and Prawitz, 2022). Thus, more horses per capita reflect capital deepening in agriculture (Olsson and Svensson, 2008).

The two observations lie sixty years apart, so the triple coefficients δ_{f}^{s} measure the long-run change in the output elasticity of each factor in plagued relative to unplagued parishes. I find that in capital-intensive agriculture the labour elasticity falls by 0.63 and the capital elasticity rises by 0.45. In labour-intensive agriculture, the labour elasticity falls by only 0.27 (Appendix Table A.17). Land per worker does not move while capital per worker rises by 13%.

For robustness, I also report results for all 2,344 Swedish parishes in 1690, 1750, and 1810. Equation 3 can only be estimated for 869 parishes and up to 1750 given data limitations. Using only data on horses and population, Appendix Table A.18) shows that plagued parishes use significantly more horses, controlling for population, both in 1750 and in 1810.

More granularly, I find in farm-level production data from Scania (Olsson and Svensson, 2017) that farms shift out of labour-intensive agriculture (rye and barley) and into capital-intensive agriculture (calves) (Appendix Figure A.4f).[13] To test the link with labour scarcity, I construct for each farm:

[13]The same pattern was observed in English agriculture after the Black Death (Clark, 2016).

$$plaguedist_{it} = \Sigma_j Distance_{ij} * Plague_{jt}. \quad (4)$$

Farms located closer to plagued cities have lower values of $plaguedist_{it}$, which I expect to be associated with higher mortality and a stronger shift into capital-intensive agriculture. Proximity to plagued cities is indeed associated with higher growth in calves production and lower growth in rye and barley production (Appendix Table A.16).[14] This shift out of rye and into foals is driven by farms with good soil close to plagued cities (Appendix Table A.19). This is capital deepening in arable agriculture: horsepower is being substituted for lost labour, and particularly so with rich soil, where the acquisition of foals and horses is easiest to recoup. Soil quality does not interact with other animals, supporting that this is about draught animals specifically. In plagued Scania, there is evidence of capital deepening through metal ploughs, early threshing machines and modern farm buildings (Magnusson, 2007).

Capital accumulation and deepening can be inferred beyond agriculture. As no data on capital stocks exist, I use ship ownership as a proxy. Ships are an expensive investment and highly mobile. An increasing number of ships is indicative of growing capital stocks, as the export expansion is not mechanically linked to an accumulation of ships since merchants from other regions could sail to competitive locations. I find that for two decades after the plague, significantly less ships are registered (Appendix Figure A.5). 50 years later, there were significantly more ships registered in previously plagued regions, which lasts until the 1850s. Beyond ships, I find that plagued ports import larger volumes of capital goods (metals, tools, nails, building materials) (Appendix Figure A.6). This suggests that capital stocks are being accumulated. By 1750, population had recovered in plagued parishes (Appendix Table A.6). Thus, both capital levels and the capital-to-labour ratio were significantly higher in 1750 and in 1810 in plagued parishes.

Fact #2: Plagued regions grow their destination market shares.

I estimate a second event study incorporating the gravity structure of trade:

$$T_{ijt} = \sum_{l=-20\setminus -1}^{30} \beta_l \cdot \mathbb{1}(K_{it} = l) + \alpha_{ij} + \alpha_{jt} + \alpha_{at} + \gamma x_{it} + \epsilon_{ijt}, \quad (5)$$

where T_{ijt} is either the volume of exports from origin i to destination j in year t, or the export share that origin i captures in destination j: $T_{ijt} = \frac{x_{ijt}}{\Sigma_{i \in I} x_{ijt}}$.[15] . K_{it} is the time difference between the current year and the plague, α_{ij} is a bilateral fixed effect, α_{jt} are destination-time fixed effects absorbing the demand side, and α_{at} are area-time fixed effects.[16]. ϵ_{ijt} is the error term. x_{it} are origin controls (annual growing season temperatures, longitude x year, and latitude x year). β_l are the outcomes of interest.

Figure 3 shows that plagued regions increase their export shares by about 1pp, with their pre-plague mean export share at 1.7%. Appendix Figure A.7a shows the volume expansion.[17]

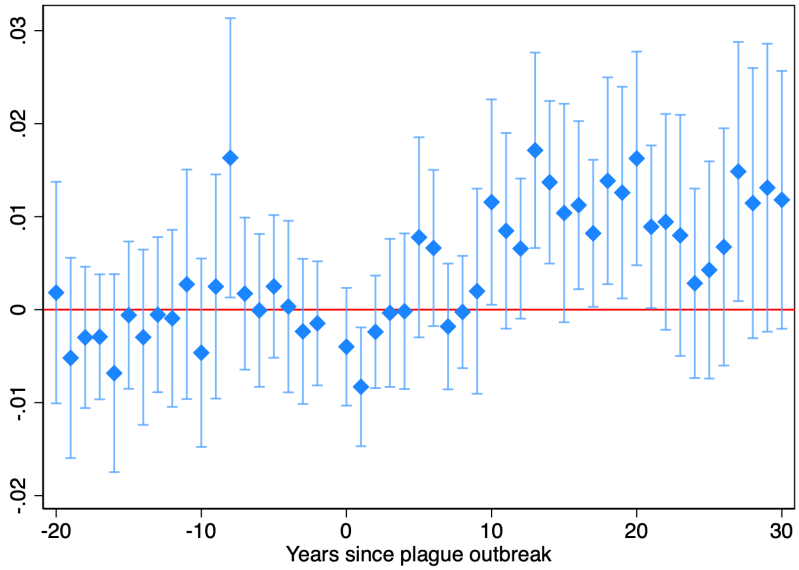
[14]This shift into capital-intensive agriculture runs counter to climate warming, under which regions should reverse their shift into pastoral farming during the Little Ice Age (Degroot et al., 2022, Degroot, 2018).

[15]I set market shares of i in j to 0 when j shows no imports in my data. As the data capture only trade between the North and Baltic Seas, I assume that every region has some (non-Sound crossing) imports.

[16]Areas are displayed in Appendix Figure A.2a. For most of Europe, these are equivalent to NUTS2; for the UK, Germany, France, and Spain, they are NUTS1. This is to ensure similarly sized areas and to avoid areas with a single port (Appendix B). Appendix Figure A.8 finds very similar results with NUTS 1-time and NUTS 2-time fixed effects instead of the proposed area-time fixed effects.

[17]The volume expansion is informative since it is not as susceptible to competitors' exports dropping as plagued regions grow their exports. Note also that any level effects in importers' demand are absorbed by α_{jt}.

Figure 3: Market shares expand



Notes: Estimation of equation (5) on i's share of j's imports, $T_{ijt} = \frac{x_{ijt}}{\Sigma_{i \in I} x_{ijt}}$. Standard errors clustered at the origin level.

The time lag reflects the above-discussed war-related disruptions to shipping that were unrelated to the plague. These results reject that trade simply bounced back. Appendix Figure A.7b extends the pre-period to 30 years and again finds no pre-trends. Appendix Figure A.8c additionally controls for lagged temperatures. Appendix Figure A.8d implements the estimator by Borusyak, Jaravel, and Spiess, 2024. Appendix Table A.20 shows that exports expand across sectors and factor intensities, both when estimating with PPML and with OLS, and it also shows that the export expansion is larger in capital-intensive sectors. The increase in destination market shares indicates rising competitiveness and outlives population recovery (Appendix Figure A.7c).

For both facts, I present heterogeneity results in Appendix Table A.21 by pre-plague export volumes. While the market share effects are larger for regions that exported more before the plague, both effects are significant (marginally so for export shares) even for regions that had no exports before the plague. Finally, I construct a continuous indirect treatment variable and show that a region's own plague treatment affects trade, but their neighbours' treatment does not (Appendix Table A.9).

Rising Competitiveness and Productivity Growth: Evidence from Production Data

To complement the finding of rising competitiveness, I estimate TFP in agricultural data by Andersson Palm, 2025 for Swedish parishes in 1690 and 1750. I provide results using two approaches. For the first, the joint estimation, I estimate for each parish i, year $t \in \{1690, 1750\}$[18] and factor intensity $k \in \{L, K\}$, where labour-intensive output is the harvest and capital-intensive output is the output of animals excluding draught animals,[19]

[18]This exercise cannot be extended until 1810: while Appendix Table A.18 shows more horses per capita in 1810, information on land and harvests is missing for that year, so I cannot estimate productivity.

[19]I assume this to be a proxy for output since output of meat, dairy and others should be proportional to the stock of non-draught animals. The herd is thus the outcome and not an omitted input, while draught animals are excluded from the output measure. The farm-level data from Scania measure production flows directly (calves, rye, and barley) and replicate the shift (Appendix Tables A.16 and A.19).

$$\ln y_{ikt} = c_{kt} + \alpha'_{kt} \ln x_{it} + \lambda_k P_i + \beta_k P_i \times \mathbf{1}[t = 1750] + \varepsilon_{ikt}, \tag{6}$$

where c_{kt} is a factor-intensity-by-year constant, x_{it} collects population, sowed acreage and draught animals, measured as horses, the main draught animal by the 1700s, P_i indicates plagued parishes, λ_k is the pre-plague productivity gap between plagued and unplagued parishes in factor intensity k, and β_k is its change by 1750. Post Plague in Table 1 is β_L, its interaction with capital intensity is $\beta_K - \beta_L$. These are difference-in-differences in TFP, where the input elasticities are estimated jointly with the treatment effect. Column 1 of Table 1 reports the results: productivity in plagued parishes rises relative to unplagued parishes and the difference is larger in capital-intensive agriculture.

The joint estimation is a special case of Loecker, 2013, where the plague takes the position that export entry plays in his framework. It incorporates the notion that the plague may endogenously drive productivity growth, implying that plague status ought to enter the productivity process. Specifically, I implement the linear OLS special case with a deterministic productivity shifter, which his framework nests, since I have no proxy variables such as investment or intermediate inputs, and lags would be implausible given 60-year intervals in the data. The same data constraints thwart me from the estimation approaches in Ackerberg, Caves, and Frazer, 2015, Levinsohn and Petrin, 2003, and Olley and Pakes, 1996.

Importantly, I regress on physical output rather than revenue and thus my estimates do not mix productivity with demand-side price variation and markups (Klette and Griliches, 1996): the recovered object is physical productivity. I argue that input mismeasurement (the quality of labour, capital utilisation, and land quality may have changed) is not a major issue. The post-plague observation is in 1750, four decades after the plague, by which time selection effects among surviving workers had arguably disappeared. Below, I also rule out differential skill acquisition that may have improved human capital. However, the larger TFP growth in capital-intensive agriculture could be squared with sectorally uniform factor quality upgrading when input elasticities differ, and some inputs, such as land services, buildings, and tools remain unmeasured in these historical data. However, while e.g. grazing land is not observed, total farmland is fixed up to costly land reclamations, and sowed acreage in plagued parishes does not contract or expand (Appendix Table A.17). Thus, meadow can only expand through conversion from arable land, which limits the bias from omitted input growth. Finally, both sectors draw on parish-wide inputs, so a reallocation of labour towards livestock would inflate the pastoral residual and deflate the arable residual. The level result is accordingly a lower bound, and since population has recovered by 1750 and land does not shift up, reallocation can only distribute the measured productivity effect across sectors, but it cannot create it. Nonetheless, the estimates might not cleanly separate productivity growth from (unmeasured) input deepening.

A simple second approach recovers productivity as a Solow residual from regressing logarithmic harvests and livestock on logarithmic population, sowed acreage, and draught animals.[20] For comparison, column 2 of Table 1 reports the two-step residual estimate, where the outcome in the second step is the residual of the first step. Recovering productivity in this way is subject to the very issues the joint estimation addresses, and I provide the estimates only as a transparent comparison. Taken together, Table 1 documents significant post-plague productivity growth which is stronger in capital-intensive agriculture.[21]

Summary. This section suggested three possible responses to labour scarcity. First, capital deepening within sectors, which I observe in production and capital data. Second, a composition

[20] The factor elasticity estimates of Appendix Table A.17 come from the same panel and nest this specification.

[21] I recover relative competitiveness growth from gravity below, where I more precisely identify the stronger effect in capital-intensive sectors. The estimation here provides the level effect that the gravity recovery absorbs.

Table 1: Parish-level TFP growth after the plague

	Joint estimation	Two-step residual
	(1)	(2)
Post Plague	0.663***	0.614***
	(0.0541)	(0.0480)
Post Plague x Capital Intensive	0.243***	0.230***
	(0.0896)	(0.0771)
Fixed Effects:		
– Plague Status x Factor Intensity	✓	
– Parish x Factor Intensity		✓
– Factor Intensity x Year	✓	✓
Parishes	869	869

Notes: The dependent variable is log TFP, where labour-intensive output is the harvest and capital-intensive output is the output of animals excluding draught animals, both in physical units. Inputs are logarithmic population, sowed acreage, and draught animals, measured as horses. Column 1 estimates, for each year and factor intensity, a production function that includes a plague dummy. Post Plague is the 1750 plague coefficient minus the 1690 plague coefficient in the output equations and its interaction with capital intensity is the difference in the livestock equations minus that in the harvest equations. The four regressions are combined by seemingly unrelated estimation, which yields their joint covariance matrix. Standard errors of the differences are obtained by the delta method. Column 2 recovers log TFP as the production function's residual and regresses it on a post-plague dummy and its interaction with a dummy for capital-intensive agriculture. Standard errors are clustered at the parish level in both columns. * p <0.10, ** p <0.05, *** p <0.01. The sample is the parishes with complete data on harvests, acreage, population, livestock and horses in both 1690 and 1750.

of production shifting into capital-intensive goods, which I document in plagued regions' export bundles and farms' output composition. Third, productivity growth. I find that plagued exporters' competitveness rises and that plagued parishes see TFP growth.

5 What Drives the Export Expansion?

This section narrows down the search for possible mechanisms and is composed of two parts. First, I argue for elements that the mechanism must deliver. Null findings in areas with serfdom imply a critical role for factor-price transmission and stronger findings in financially more developed regions suggest that capital deepening is in the driver's seat. Second, I exclude alternative mechanisms that fail to rationalise my headline findings.

5.1 Necessary Elements

I study two factor frictions to understand what drives these adjustments to labour scarcity. For labour, I study serfdom which limits the mobility between cities and their hinterlands. For capital, I investigate pre-war variation in the depth of capital markets which affects the feasibility of capital deepening. In both cases, pre-existing labour and capital market frictions provide useful empirical variation that allows me to narrow down the mechanism to factor price-induced capital deepening that drives productivity growth. Finally, I show that capital deepening and productivity growth outlast population recovery by half a century, which implies that the mechanism needs to generate persistence from a temporary factor shock.

Factor-Price Transmission: Evidence from Serfdom

Serfdom limits the mobility of labour across sectors, effectively shielding agriculture from rising wages as workers cannot move to labour-scarce cities. To test whether wage transmission matters, I analyse how my findings differ for regions with serfdom. Under labour mobility, labour scarcity from higher urban mortality rates (Voigtländer and Voth, 2012) is equalised between a city and its hinterland, to which I refer as *absolute* labour scarcity. With serfdom, labour scarcity is not equalised. Beyond absolute labour scarcity, cities suffer from *relative* labour scarcity as workers from the hinterland cannot move. The model in Section 6 sharpens this finding into a quantitative test. In sum, labour is artificially scarce in cities and artificially abundant in rural areas. This *relative* labour scarcity manifests itself in a wage wedge: labour scarcity is transmitted less to agriculture.

Appendix Table A.22 shows how this logic affects production and exports. Urban manufacturing wages increase more under serfdom. Thus, the rising capital intensity of exports is larger in plagued regions with serfdom. For agriculture, I find no significant serfdom interactions, reflecting that agricultural wages under serfdom are not set freely and that agriculture is relatively shielded from labour scarcity. Thus, my findings require that labour scarcity is transmitted into factor prices, suggesting that exposure to changing factor prices sit at the heart of the mechanism.

Capital Deepening: Evidence from Imperfect Capital Markets

In imperfect capital markets, capital deepening is constrained by financial development: ports with deeper pre-war capital markets should undergo stronger capital deepening. Data on loans and interest rates are very limited for this period.[22] Empirical evidence for financial variables is lacking for virtually all ports. Given these constraints, I borrow an insight from Böhm-Bawerk, 1889 and Antràs and Kulesza, 2026: financially developed regions have a comparative advantage in sectors with a high average period of production (APP). In sectors where more time passes between spending on inputs and raising revenue from final goods, financing costs are higher. Thus, ports' pre-plague APP in exports reflects their financial development.

I construct each port's export-weighted pre-war APP.[23] I ask whether the shift into a higher capital intensity and the export expansion are stronger for ports with a higher pre-war APP. Throughout, I control for all interactions with pre-war export levels by sector, as APP is higher in manufacturing and export-weighted APP is mechanically higher in larger ports. In Appendix Table A.23, I find that, even when controlling for export volumes and sectoral structure, plagued ports with a higher pre-war APP see a significantly larger shift into capital-intensive exports and a significantly larger export expansion. Thus, my findings require that regions are able to respond to labour scarcity with capital deepening, since my estimates are significantly larger for ports that were less financially constrained and could more readily engage in capital deepening. This suggests that the mechanism runs through capital deepening.

Temporary Factor Shocks and Persistent Productivity Changes: Evidence from Population Recovery

By 1750, plagued ports and parishes had recovered their populations (Appendix Table A.6). However, exports remained shifted into capital-intensive goods and at higher levels for decades later (Appendix Figures A.4d and A.7c). Agricultural data show that the capital-to-labour

[22]The literature considers Hamburg, Amsterdam, and Bremen as the most developed (Huang, Chilosi, and Sapoznik, 2019, Carlos and Neal, 2011, Denzel, 2010); Danzig less so (Wee, 1977). Stockholm, Königsberg, Riga, Reval, and Lübeck were the least financially developed major ports (Edvinsson, 2011).

[23]All details in Appendix E.1. Ranking ports by their average APP, I find that the major ports are ranked: Hamburg, Bremen, Amsterdam, Reval, Riga, Danzig, Stockholm, Königsberg, Lübeck. This matches the above-described historical evidence very well.

ratio, too, was still significantly higher half a century after population recovery (Appendix Table A.18), and ship ownership data suggest that capital accumulation, too, outlasted population recovery by decades (Appendix Figure A.5). This suggests that the mechanism needs to produce persistent productivity and comparative advantage changes, as well as a persistently higher capital-to-labour ratio and higher capital stocks, from a temporary factor shock.

5.2 What Does not Drive the Export Expansion

My results are not driven by mechanical forces. First, the export expansion is not a reallocation from the Baltic to the North Sea. Sectoral heterogeneity speaks against this, and entering new and far away markets under rising wages requires higher competitiveness, too. Appendix Figure A.11 shows results for a simulated productivity increase in the full and observed samples and finds no bias. Second, territorial changes affecting market access are accounted for by area-time fixed effects, which also absorb famines. I account for harsh winters by controlling for temperatures and for temperatures with a 10 and 20 year lag (Appendix Figure A.8c).

Export hubs may form a third mechanical explanation. As local consumption drops, exports could rise under constant hinterland production.[24] Appendix Figure A.9 shows that plagued regions' imports increase and Appendix Table A.21 shows that also the smallest harbours that are unlikely to have served as export hubs expand their exports. Fourth, there are no exporter-specific declines in trade costs related to the plague or capital intensity. From return trips, I calculate an exporter-specific measure of travel and handling time at harbours, where fixed effects absorb captain-specific and origin-time determinants. Travel and loading times are unaffected by the plague and by capital intensity (Appendix Table A.24). Thus, trade costs are unrelated to the shock.

Fifth, labour scarcity does not affect hinterlands' trade routing. If anything, the bias would be in my favour as treated ports would shed trade to controls. Ports tied to hinterlands through low-cost river transport should be harder to switch away from. Appendix Table A.11 finds no robust evidence of a differential effect. This confirms both that port-level outcomes track the hinterland's own production and the absence of labour scarcity-related port substitution. Finally, I provide evidence in Appendix Table A.10 against both selection and mechanical reallocations by drawing on Hamburg's trade (Pfister, 2012, Baasch, 1929). While the Soundtoll data record trade conditional on passing through the Sound, these data cover trade with all of Hamburg's trading partners. Exports expand after the plague and shift into capital-intensive commodities. This dispels concerns that Baltic exporters simply entered into markets that plagued Hamburg had left. It also confirms my findings for a port for which I have export data for all destinations. As Hamburg was unaffected by naval warfare and sees effects on impact, this suggests that the time lag discussed above is indeed driven by the naval blockade.

Non-Homotheticity

Under non-homothetic demand, demand shifts relatively into more income-elastic goods after labour scarcity leads to real wage gains.[25] I present evidence that non-homothetic demand played no role in this context.[26]

[24] Exports rise if domestic absorption falls by more than income, so production falls less than consumption.

[25] Voigtländer and Voth, 2012 place non-homothetic demand at the centre of their argument. The authors study the much earlier Black Death of 1346, which encompassed all of Europe. Therefore, my paper is not a test of Voigtländer and Voth, 2012.

[26] The only mention reminiscent of non-homothetic demand in the historical literature for this plague outbreak is discussed in Frandsen, 2009: in some places, it was believed that textiles and bedding spread the disease, which led to survivors burning the textiles of those who died. After the plague, this might imply a relative demand shift into textiles, which are summarised as labour-intensive manufacturing here. I show below that there is no indication for a relative demand shift into textiles.

In Appendix E.2, I motivate a test for non-homothetic demand (Fieler, 2011) and present evidence against its presence. Price predictions of non-homothetic demand are empirically rejected as (export) prices for capital-intensive goods decline relatively while relative import prices are unaffected.[27] In a test on relative imports for a demand shift after the plague (Fieler, 2011), I find no evidence of relative demand changes, which also speaks against sector-specific embodied technology driving my TFP findings.[28] Finally, gravity remains separable under non-homotheticity (Fieler, 2011), leaving exporter-sector-time variation, from which I recover competitiveness, robust to non-homothetic demand. Given this evidence, I model homothetic demand below.

Venting Out, Market Power, & Directed Technical Change

Almunia et al., 2021 propose non-constant marginal costs to explain "venting out", for which this historical setting provides an ideal testing ground: the 1708-1714 plague outbreak left most of Europe as unperturbed export markets. Under venting out, regions would increase their exports in line with my findings. However, while Almunia et al., 2021 track exports for four years, this paper studies trade for decades. Short-run marginal costs are unlikely to explain this persistent export boom. Market power may reconcile short-run marginal cost changes with my long-run findings. However, Appendix Table A.26 shows that relative export prices of plagued origins drop and keep falling for 30 years. This pattern is incompatible with market power and instead points to supply-side improvements in competitiveness.

Labour scarcity could lead to directed technical change (Acemoglu, 2002). However, history does not support this: Northern Europe before the Industrial Revolution saw few inventions, let alone R&D responding to market size and price effects. Labour-saving inventions in crop agriculture began succeeding only in the 19th century (Gallardo and Sauer, 2018).[29] Focusing on the 17th century in England, Coleman, 1956 argues that there were no novel labour-saving technologies in agriculture. Those that existed, such as mills, spinning wheels, and water-powered blast furnaces, were centuries old. In manufacturing, Atack, Margo, and Rhode, 2019 argue that labour-saving machinery only came to bear during the 19th century. This points to a fixed menu of technology choices rather than an innovation possibility frontier.

Human and Social Capital Deepening

Plague survivors may invest more in human capital and survivors may be systematically better connected individuals. If these types of capital are sectoral[30] and respond to labour scarcity, I would expect to see a relative export shift into goods that use these factor intensely. In Appendix E.2, I construct measures of goods' skill and contract intensity (Nunn, 2007). In Appendix Table A.27, I show that across specifications the only consistently positive and significant estimates are for a higher *physical* capital intensity. There is no shift into a higher contract or skill intensity of exports.[31] The evidence presented here suggests that it is physical capital deepening, and not human or social capital deepening, that explains the post-plague

[27] These price effects reconcile rising meat and dairy consumption after the plague (Broadberry et al., 2014) with homothetic demand.

[28] Constant relative imports provide further support against a pure Heckscher-Ohlin framework. In the latter, labour scarcity should shift exports into more capital-intensive sectors, mirrored by imports shifting into more labour-intensive sectors.

[29] An early example of a crop harvesting machine that explicitly substituted labour for capital was the cotton gin by American inventor Eli Whitney, patented in 1794.

[30] When quantifying my model below, any sector-neutral increase in human or social capital is absorbed. In the early modern period, sector-specific human capital was very limited.

[31] This analysis has to be caveated by the fact that both measures of human and social capital intensity are based on modern data for the U.S. I assume that there is a sufficiently high correlation over time in goods' relative skill and relative contract intensities.

export expansion.

Pure Ricardian or Heckscher-Ohlin Forces

Neither plain vanilla Ricardian nor factor proportions theories can explain my findings. In a Ricardian world of fixed productivity, higher wages mechanically lower exports, which is at odds with my results. Pure Heckscher-Ohlin forces also struggle to rationalise this result. Reduced labour supply increases wages, and under standard assumptions, GDP falls. Plausibly, imports and domestic absorption then fall, too, and under balanced trade, exports should also fall. Contradicting falling imports, I find in Appendix Figure A.9 that imports rise after the plague, aligned with my finding of rising exports (Figure 3 and Appendix Figure A.7a). Additionally, Appendix Table A.20 shows that the export expansion is larger in capital-intensive sectors. The absorption story cannot account for this sectoral pattern. In the absence of falling domestic absorption, exports would have to fall. Further, a pure Heckscher-Ohlin framework mechanically implies changes in relative imports, which is contradicted by the data. Finally, factor theories predict factor price equalisation, which is at odds with well-documented wage increases after the plague (Allen, 2005).[32] Thus, pure Heckscher-Ohlin forces could explain the shifting export composition, but not the export boom.

Instead, my findings suggest that both Ricardian and Heckscher-Ohlin forces are operating. The rising share of capital-intensive exports is reminiscent of the Rybczynski effect in Heckscher-Ohlin theory: as capital becomes relatively more abundant, exports shift into more capital-intensive goods. Rising exports, on the other hand, are suggestive of a level effect in productivity, large enough to dominate plague-induced wage growth. The productivity growth result in Table 1 is direct evidence of these Ricardian forces. Specifically, productivity growth that is tilted towards capital-intensive sectors can rationalise the shift towards capital-intensive sectors, the export boom, and constant relative imports.

6 A Ricardian Model of Endogenous Technology

I propose a model in which factor price changes drive technology adoption and thus productivity growth. Technological change is factor-biased (Habakkuk, 1962, Hicks, 1932) and labour scarcity triggers adoption from a fixed menu. My model combines Ricardian elements, Melitz-style monopolistic competition, and Krugman-style love of variety. The model is Ricardian in the sense of Costinot, Donaldson, and Komunjer, 2012: the pattern of trade is determined by relative sectoral productivity. The microfoundation beneath it is monopolistic competition among heterogeneous producers in the tradition of Melitz, 2003, carrying the technology margin of Bustos, 2011, extended to an ordered menu of techniques. Monopolistic competition provides the operating profits needed for technology adoption. However, through no entry, the trade block is observationally equivalent to the perfect competition Ricardian benchmark (Arkolakis, Costinot, and Rodríguez-Clare, 2012). Further, there is no selection into markets: every variety sells everywhere, as in Krugman, 1980, predicting strictly positive, though arbitrarily small, trade in all cells. Ricardian forces of relative productivity then set the pattern and volume of trade. In a nutshell, love of variety generates trade, heterogeneity drives technology adoption, and comparative advantage scales and directs trade.

[32] With more than two countries, Helpman and Krugman, 1985 show that FPE continues to hold if all countries fall within the FPE set, which comes down to endowments not being too dissimilar. Schott, 2003 shows that FPE is achieved for many countries if they are in the same diversification cone: as countries become more similar in endowments, FPE holds approximately also for n>2, and precisely when endowments are identical.

Environment and primitives There are regions $i = 1, \ldots, I$, time t, suppressed for static results, and two factors, labour L_{ik} and capital K_{ik}.[33] Capital is rented from a world market at the rental rate r, described below, and labour is immobile across regions[34] but mobile across sectors within a region. Rural-to-urban migration leads to wage equalisation, while under serfdom, agricultural labour is coerced and workers cannot move, producing a wedge ϕ_{it},

$$w_{iMt} = \begin{cases} w_{iAt}, & \text{without serfdom,} \\ (1 + \phi_{it})\, w_{iAt}, & \text{with serfdom,} \end{cases} \tag{7}$$

where A and M stand for agriculture and manufacturing. In support of this modelling, Appendix Figure A.1 shows that a wedge ϕ_i appeared in Denmark after the re-introduction of serfdom in 1733 (Gary et al., 2022). I assume that plague-related changes in labour coercion and thus ϕ_{it} are at the area level. Therefore, area-year fixed effects below account for both the re-introduction of Danish serfdom and increased labour coercion in Estonia (Raster, 2023).

There are five sectors k (labour-/capital-intensive agriculture and manufacturing, plus unclassified): cities produce manufacturing goods and hinterlands agricultural goods.[35] Each region-sector hosts a fixed measure N_{ik} of differentiated varieties[36], each supplied by a single producer, whose Cobb-Douglas technology in labour and capital is specified in (10) once its two components, the exogenous productivity draw and technology choice, have been introduced.

Agents and the capital market. There are workers, a representative household owning the region's producers and an endowment $\bar{K}_i$ of capital, and producers. A world capital market supplies capital: it is a homogeneous good, the numeraire, and depreciates at rate ζ. I set ζ to 5%, which sits inside the range that historical accounts support: Allen, 2009a puts aggregate capital depreciation at 2.4%, and sectoral figures run from 2% in capital-intensive to 7% in labour-intensive production. Producers rent capital on this market at the rental rate r per unit and period, so r is the user cost of a unit of capital. An owner who rents out a unit receives r but must replace the depreciated fraction ζ to keep the unit intact, so the net return to owning capital is the interest rate $\iota = r - \zeta$. Regions are small in this market and take r as given; households rent their endowments there and earn $\iota \bar{K}_i$. Regional capital demand and the regional endowment need not coincide: the difference is a net position with the rest of the world, and regional capital owners hold claims across regions.[37]

Demand and market structure. Preferences are Cobb-Douglas across sectors with expenditure shares α_k and, within a sector, two-tier CES: region-level bundles enter with elasticity of substitution $\xi > 1$, and within region i's bundle its varieties substitute with elasticity $\sigma_k > 1$. Each variety is atomistic within its bundle, so its producer charges the markup $\mu_k = \sigma_k/(\sigma_k - 1)$

[33] I do not model land, which was abundant in the Baltics (Raster, 2023) where labour was the constraining factor. Had land been a constraining factor, a higher relative land supply should benefit land-intensive goods, for which I find no support in production or trade data (Appendix Figures A.4c and A.4f). Land per worker is also unchanged in plagued Swedish parishes after the plague, while capital per worker rises (Appendix Table A.17). Further, my findings hold within manufacturing, where land intensities should play no role (Appendix Table A.15 and Appendix Table A.20).

[34] Labour mobility across regions implies migration, which is at odds with population recovery only after four decades and significant post-plague wage growth (Jedwab, Johnson, and Koyama, 2022).

[35] Hinterland sizes may differ but are assumed fixed. Hinterlands' inability to switch ports, proxied by proximity to rivers providing low-cost transport, does not lead to a differential export response (Appendix Table A.11). The split between urban manufacturing and hinterland agriculture is historically plausible (Klein and Ogilvie, 2015, Magnusson, 2007).

[36] The varieties measure is a technological primitive. Through love of variety, the model predicts strictly positive (arbitrarily small) flows in every cell. I assume a cell below a recording threshold to appear as zero.

[37] Consistent with elastic capital supply, interest rates inferred from grain prices do not rise after the plague (Appendix Table A.29) and imports of capital goods rise (Appendix Figure A.6).

over marginal cost, $p_{ik}(\omega) = \mu_k\, MC_{ik}(\omega)$, and, with iceberg trade costs $d_{ijk} \geq 1$ on shipments from i to j in sector k, the delivered price in j is $p_{ijk}(\omega) = \mu_k\, MC_{ik}(\omega)\, d_{ijk}$. As μ_k is common to every variety and source region, it cancels from the bilateral expenditure share. Operating profit is the share $1/\sigma_k$ of revenue and profits net of adoption accrue to the regional household. I assume no entry.[38] The household in region j has the utility function

$$U_j = \prod_k C_{jk}^{\alpha_k}, \qquad C_{jk} = \Big[\sum_i Q_{ijk}^{\frac{\xi-1}{\xi}}\Big]^{\frac{\xi}{\xi-1}}, \qquad Q_{ijk} = \Big[\int_{\Omega_{ik}} q_{ijk}(\omega)^{\frac{\sigma_k-1}{\sigma_k}}\, d\omega\Big]^{\frac{\sigma_k}{\sigma_k-1}}, \tag{8}$$

with $\sum_k \alpha_k = 1$, and maximises it subject to its budget, where income Y_j is the sum of wages, capital income, and net profits (Appendix F.1).[39] Spending on i's bundle and varieties are

$$X_{ijk} = \Big(\frac{P_{ijk}}{P_{jk}}\Big)^{1-\xi} \alpha_k Y_j, \qquad p_{ijk}(\omega)\, q_{ijk}(\omega) = \Big(\frac{p_{ijk}(\omega)}{P_{ijk}}\Big)^{1-\sigma_k} X_{ijk}, \tag{9}$$

where P_{ijk} and P_{jk} are the CES price indices of the bundle and of the sector.

Technology choice. Technology choice produces the endogenous component to productivity, beyond the exogenous variety-level productivity draws $z(\omega)$, which are i.i.d. from a distribution F_{ik}, continuous with positive density on a support bounded below by $z_{ik} > 0$ and with a finite $(\sigma_k - 1)$-moment.[40]

The technology set is an ordered menu: in sector k, techniques $\tau = 0, 1, \ldots, n_k$ carry labour shares $\gamma_0^k > \gamma_1^k > \cdots > \gamma_{n_k}^k$, productivity gains $\upsilon_\tau^k > 1$ over the next more labour-intensive technique, and sunk costs F_τ^k. Increments are sorted so that each rung's incremental gain per unit of incremental sunk cost declines along the menu, so no rung is skipped. Since gross operating profits scale with z^{σ_k-1} across techniques (derived below), profit differences across rungs are proportional to z^{σ_k-1}, so the optimal technique is monotone in the draw with interval structure: producers occupy rungs in order of their draws and switch only to adjacent rungs.

The technology menu requires some notation. The efficiency-weighted mass on rung τ is m_{ikt}^τ. Occupancy shares are $s_{ikt}^\tau = m_{ikt}^\tau / B_{ik}$ and the cumulative adopter share of rung τ is $\pi_{ik\tau t} = \sum_{\tau' \geq \tau} s_{ikt}^{\tau'}$. Cumulated productivity gains are $U_\tau^k = \prod_{\tau' \leq \tau} \upsilon_{\tau'}^k$ with $U_0^k = 1$. Revenue shares are $\tilde{s}_{ikt}^\tau = S_{ikt}^\tau / \sum_{\tau'} S_{ikt}^{\tau'}$, with S_{ikt}^τ defined below, and the sector's realised labour share is $\bar{\Gamma}_{ikt} = \sum_\tau \gamma_\tau^k\, \tilde{s}_{ikt}^\tau \in [\gamma_{n_k}^k, \gamma_0^k]$. This is the object the parish production functions of Appendix Table A.17 estimate. For the active margin between rungs $\tau - 1$ and τ, the pair mass is $B_{ik}^{(\text{non,ad})} \equiv m_{ikt}^{\tau-1} + m_{ikt}^\tau$, and on a monotone wage path, only the active margin moves. Two reading conventions follow. In the aggregators (17) and (25), adoption enters through the cumulative shares $\pi_{ik\tau t}$.[41]

The analysis therefore reduces to the choice between adjacent rungs. The pairwise results below (Lemma 1, the cutoffs, the inaction band, the law of motion (16), and Proposition 1) apply to each adjacent pair and the menu enters the quantification through Proposition 3, which caps adoption's cumulated contribution to competitiveness by the menu's overall labour share span. I model rungs rather than a binary adoption response to reconcile pointwise adoption, constrained by the uniqueness condition, with the cumulated response across rungs, recovered

[38] A sector-neutral response of the variety measure cancels from the cross-sector difference when $\rho_k = \rho_{k'}$.

[39] Households consume their income each period. The only intertemporal decision is the producers' installation of adoption capital, which is financed on the world market, so no household saving decision is needed.

[40] Nothing here requires a specific distribution. Truncation (technology adoption) is destination-invariant, as opposed to destination-specific and interacting with bilateral costs as in selection models which typically use the truncation-stable Pareto distribution throughout. Instead, Eaton and Kortum, 2002 use max-stable Fréchet due to sourcing from least-cost origins.

[41] In the pairwise derivations (and in Appendix G.3), π denotes the pair-local share $m_{ikt}^{\text{ad}} / B_{ik}^{(\text{non,ad})}$ and Γ_{ik} (without a bar) denotes the active pair's realised share.

by the quantification below. The exposition accordingly proceeds with the two-technology building block, relabelling the active adjacent pair $(\tau-1,\tau)$ as (non,ad): producers choose a labour-intensive technology (subscript non) or a capital-intensive technology (subscript ad), with sector-specific labour cost shares $\gamma^k_{\text{non}} > \gamma^k_{\text{ad}}$. A producer on technology $\tau \in \{\text{non},\text{ad}\}$ with draw $z(\omega)$ hiring labour ℓ and capital κ produces

$$y(\omega) = \upsilon_\tau \, z(\omega) \, \ell(\omega)^{\gamma^k_\tau} \, \kappa(\omega)^{1-\gamma^k_\tau}, \qquad \upsilon_{\text{non}} = 1, \quad \upsilon_{\text{ad}} = \upsilon_k > 1. \tag{10}$$

Adopting capital-intensive technology requires a sunk investment of F units of capital[42] and scales the exogenous draw $z(\omega)$ multiplicatively by $\upsilon_k > 1$.[43] This capital depreciates at rate ζ, which is each period's required reinvestment for maintenance. The stock is bought at the price of capital (which is one), is productive from the period after installation, and is specific to the producer, so its cost is sunk. Producers own the stock, financed on the world capital market at ι, so holding the stock costs ιF in interest and ζF in maintenance. The total cost is rF per period. Once the stock is installed, only the maintenance ζF can still be saved.

The producer's static problem. Given prices, the producer of variety ω on technology τ chooses its inputs and its price in every market to maximise operating profit,

$$\Pi^{op}_\tau(z) = \max_{\ell,\,\kappa,\,\{p_{ijk}(\omega)\}_j} \; \sum_j p_{ijk}(\omega)\, q_{ijk}(\omega) - w_i \ell - r\kappa, \tag{11}$$

subject to technology (10), where output must cover every shipment inclusive of its iceberg loss ($y = \sum_j d_{ijk} q_{ijk}(\omega)$), and to demand (9).
Marginal costs, where $\tilde{w}_i \equiv w_i/r$ are normalised wages and I drop the wage's sector index, are

$$MC_{\text{non}} = \frac{w_i^{\gamma_{\text{non}}} r^{1-\gamma_{\text{non}}}}{z(\omega)}, \qquad MC_{\text{ad}} = \frac{w_i^{\gamma_{\text{ad}}} r^{1-\gamma_{\text{ad}}}}{\upsilon_k z(\omega)}. \tag{12}$$

Under the constant markup, a producer's operating profit is proportional to $MC^{1-\sigma_k}$ times a demand shifter Ψ_{ik} that is common to all producers from i in sector k,

$$\Pi^{op}_\tau(z) = \Psi_{ik}\,(\upsilon_\tau z)^{\sigma_k-1}\big(w_i^{\gamma_\tau} r^{1-\gamma_\tau}\big)^{1-\sigma_k}, \qquad \Psi_{ik} \equiv \frac{1}{\sigma_k}\sum_j \big(\mu_k d_{ijk}\big)^{1-\sigma_k} P_{ijk}^{\sigma_k-1} X_{ijk}, \tag{13}$$

with $\upsilon_{\text{non}} = 1, \upsilon_{\text{ad}} = \upsilon_k$, where Ψ_{ik} collects, from (9), demand in every destination j at the markup price, which producers take as given.

The producer's dynamic problem. Producers make technology decisions to maximise the present value of operating profits net of installation and maintenance, discounted at ι.
Assumption 1 (Static expectations). Producers expect current prices (w_{it}, r, Ψ_{ikt}) to persist.

Then, the problem has a two-threshold solution (Appendix F.3). A producer moves up a rung if the perpetuity value of the gain net of maintenance covers the sunk outlay, and a producer maintains its stock if the gain covers maintenance,

$$\text{install iff} \quad \Pi^{op}_{\text{ad}} - \Pi^{op}_{\text{non}} \geq (\iota+\zeta)F = rF, \qquad\qquad \text{maintain iff} \quad \Pi^{op}_{\text{ad}} - \Pi^{op}_{\text{non}} \geq \zeta F. \tag{14}$$

Installing capital is profitable when $\Pi^{op}_{\text{ad}} - \Pi^{op}_{\text{non}} \geq rF$, i.e. when the draw clears the entry cutoff $z^*_{ik} \equiv z^*_{ik}(rF)$, and maintaining it is worthwhile above the exit cutoff $z^{*-}_{ik} \equiv z^*_{ik}(\zeta F) < z^*_{ik}$, where, for profit gap $\Pi_k(\tilde{w}_i) > 0$ and a cost C that the gain has to cover,

$$z^*_{ik}(C) = \left[\frac{C\, r^{\sigma_k-1}}{\Psi_{ik}\,\Pi_k(\tilde{w}_i)}\right]^{1/(\sigma_k-1)}, \qquad \Pi_k(\tilde{w}) = \upsilon_k^{\sigma_k-1}\tilde{w}^{-\gamma_{\text{ad}}(\sigma_k-1)} - \tilde{w}^{-\gamma_{\text{non}}(\sigma_k-1)}. \tag{15}$$

[42] Denominating the adoption cost in capital is key. In labour units, a higher wage would raise the cost of adoption.

[43] The Cobb-Douglas unit cost carries $\Lambda_\gamma = \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}$; I fold $\Lambda_{\gamma_{\text{non}}}/\Lambda_{\gamma_{\text{ad}}}$ into υ_k.

The ratio of the two cutoffs, $z^*_{ik}/z^{*-}_{ik} = (r/\zeta)^{1/(\sigma_k-1)}$, depends only on r, ζ and σ_k.

Lemma 1 (Adoption rises with wages, locally)**.** *Capital-intensive technology is the cheaper technology iff* $\tilde{w} > \tilde{w}_0 \equiv \upsilon_k^{-1/(\Delta\gamma^k)}$*, where* $\Delta\gamma^k \equiv \gamma^k_{\text{non}} - \gamma^k_{\text{ad}}$ *and* $\Pi_k(\tilde{w}_0) = 0$*. Let* $\tilde{w}_{\text{crit}} = \left(\gamma_{\text{non}}/\gamma_{\text{ad}}\right)^{1/[(\gamma_{\text{non}}-\gamma_{\text{ad}})(\sigma_k-1)]}\tilde{w}_0 > \tilde{w}_0$*. Holding the demand shifter* Ψ_{ik} *fixed, on the interval* $\tilde{w} \in (\tilde{w}_0, \tilde{w}_{\text{crit}})$ *the profit gap* $\Pi_k(\tilde{w})$ *is increasing, so the cutoff* z^*_{ik} *(and* z^{*-}_{ik}*) is decreasing in* $\tilde{w}$*.*

Lemma 1 holds across the historically observed plague-induced wage growth range[44] and depends crucially on $\gamma_{\text{ad}} \neq \gamma_{\text{non}}$. With equal labour shares, wage growth reduces profits symmetrically and leads to lower adoption. Here, adoption rises in wages since capital-intensive technology adoption allows producers to escape part of the wage increase. I note that adoption responds to factor prices and not to market size.[45]

Technology dynamics. To explain persistent technology choice beyond population recovery, I model adoption capital as a state. As the adoption cost is sunk, the entry and exit margins differ. A producer moves up a rung when its draw clears z^*_{ik}. A producer on a higher rung cannot recover F, so they stay as long as the gain covers maintenance and abandon when the draw falls below a lower cutoff z^{*-}_{ik}, set by $\Pi^{op}_{\text{ad}} - \Pi^{op}_{\text{non}} < \zeta F$, where $z^{*-}_{ik} < z^*_{ik}$. Between the two cutoffs, the inherited choice is retained. Under the menu, adoption capital is incremental: each rung's stock carries its own sunk cost and maintenance, and on a falling wage path exit cascades rung by rung as each increment's maintenance condition fails after each other.

The adopter mass is a state that evolves across periods. Let $\Omega^{\text{ad}}_{ikt} \subset \Omega_{ik}$ be the set of adopters at date t and $\Omega^{\text{non}}_{ikt}$ its complement. Because CES demand weights a variety by z^{σ_k-1}, the relevant measures are the efficiency-weighted masses $m^{\tau}_{ikt} = N_{ik}\int_{\Omega^{\tau}_{ikt}} z^{\sigma_k-1}\, dF_{ik}(z)$, which the two technology sets partition.

Decisions taken at t take effect at $t+1$. Adoption is immediate (any variety crossing z^*_{ik} installs at once) and so is abandonment (any adopter falling below z^{*-}_{ik} stops maintaining and loses the technology). Write m^*_{ikt} and m^-_{ikt} for the adopter masses implied by the entry and exit cutoffs,

$$m^*_{ikt} = N_{ik}\int_{z^*_{ikt}}^{\infty} z^{\sigma_k-1}\, dF_{ik}(z), \qquad m^-_{ikt} = N_{ik}\int_{z^{*-}_{ikt}}^{\infty} z^{\sigma_k-1}\, dF_{ik}(z),$$

with $m^*_{ikt} \leq m^-_{ikt}$ because $z^{*-}_{ikt} < z^*_{ikt}$. The adopter mass then evolves as

$$m^{\text{ad}}_{ik,t+1} = \min\Big\{\max\left\{m^{\text{ad}}_{ikt},\, m^*_{ikt}\right\},\, m^-_{ikt}\Big\}, \tag{16}$$

The adopter mass is unchanged whenever $m^*_{ikt} \leq m^{\text{ad}}_{ikt} \leq m^-_{ikt}$, which is the inaction band.[46]

Technology aggregation. I now introduce sectoral productivity A_{ikt}, which combines the exogenous draws with endogenous adoption and corresponds to the absolute advantage term in

[44] Operating profits under the capital-intensive technology fall more slowly, so $\Pi_k(\tilde{w})$ widens. With both terms approaching zero, the gap narrows after peaking at $\tilde{w}_{\text{crit}}$, beyond which a higher wage raises the cutoff.

[45] Higher firm market access lowers the cutoff, as in Brey, Hutkova, and Smitkova, 2026 and Bustos, 2011. Under Assumption 2 below the plague leaves FMA_{ik} unchanged, so that channel is silent. Sector-biased destination price-index responses could in principle move FMA_{ik} and adoption, which I do not quantify. I note that destination-sector-time fixed effects absorb the direct demand changes below, and each origin's negligible share of its destinations makes the induced feedback on its own adoption second-order.

[46] Under the menu, (16) holds per margin. Writing $M^{\tau}_{ikt} = \sum_{\tau' \geq \tau} m^{\tau'}_{ikt}$ for the mass at or above rung τ and $m^{*\tau}_{ikt}$, $m^{-\tau}_{ikt}$ for the entry- and exit-implied masses of margin τ, the state evolves as $M^{\tau}_{ik,t+1} = \min\{\max\{M^{\tau}_{ikt}, m^{*\tau}_{ikt}\}, m^{-\tau}_{ikt}\}$ for each $\tau =$. The ordering of the cutoffs implies that on monotone wage paths only one margin moves at a time and that exit cascades per rung.

Eaton and Kortum, 2002. The rung masses m^{τ}_{ikt} implied by the law of motion at each margin split the varieties across the menu:

$$A_{ikt} = \sum_{\tau=0}^{n_k} \left(U^k_\tau\right)^{\sigma_k-1} m^{\tau}_{ikt} = B_{ik}\Big[1 + \sum_{\tau=1}^{n_k} \Big(\left(U^k_\tau\right)^{\sigma_k-1} - \left(U^k_{\tau-1}\right)^{\sigma_k-1}\Big)\,\pi_{ik\tau t}\Big], \tag{17}$$

where the second equality follows from summation by parts and $B_{ik} = N_{ik}\,E\big[z^{\sigma_k-1}\big]$ is the total efficiency aggregate of the exogenous draws, which I assume fixed over time.[47] The cutoff-implied masses follow from truncating the draws at z^*_{ikt},

$$m^*_{ikt} = B_{ik} - N_{ik}\int_{z_{ik}}^{z^*_{ikt}} z^{\sigma_k-1}\,dF_{ik}(z), \tag{18}$$

i.e. the piece below the cutoff stays on the labour-intensive technology and adoption happens in the residual of the finite B_{ik}. In sum, A_{ikt} is a bounded moment aggregate, $A_{ikt} \in [B_{ik}, \upsilon_k^{\sigma_k-1}B_{ik}]$, increasing in $\tilde{w}$, and enters gravity below as part of an exporter term bundling productivities and factor prices.

Persistence. The band between the cutoffs turns a temporary factor price shock into a persistent change in productivity and comparative advantage. Let $m_{ik}(z) \equiv N_{ik}\int_z^\infty x^{\sigma_k-1}\,dF_{ik}(x)$ for the adopter mass implied by a cutoff z, so that $m^*_{ikt} = m_{ik}(z^*_{ikt})$ and $m^-_{ikt} = m_{ik}(z^{*-}_{ikt})$.

Proposition 1 (Hysteresis from a temporary factor-price shock)**.** *Fix Ψ_{ik} and r, and take a sector with $\upsilon_k > 1$ and $\gamma^k_{\text{non}} > \gamma^k_{\text{ad}}$. Let the normalised wage follow $\tilde{w}_t = \tilde{w}_{\text{pre}} \in [\tilde{w}_0, \tilde{w}_{\text{crit}})$ for $t < 0$, $\tilde{w}_t = \tilde{w}_1 \in (\tilde{w}_{\text{pre}}, \tilde{w}_{\text{crit}})$ for $0 \le t < T$, and $\tilde{w}_t = \tilde{w}_\infty \in [\tilde{w}_0, \tilde{w}_1)$ for $t \ge T$, with the adopter mass at its entry target $m^*_{ik}(\tilde{w}_{\text{pre}})$ before the shock.*

(i) On impact the entry cutoff falls to $z^*_{ik}(\tilde{w}_1) < z^*_{ik}(\tilde{w}_{\text{pre}})$ and the adopter mass jumps to $m^*_{ik}(\tilde{w}_1)$, so A_{ik} rises by $(\upsilon_k^{\sigma_k-1}-1)\big[m^*_{ik}(\tilde{w}_1) - m^*_{ik}(\tilde{w}_{\text{pre}})\big]$.

(ii) If $\tilde{w}_\infty > \tilde{w}_0$, then for all $t > T$ the adopter mass equals $\min\{m^*_{ik}(\tilde{w}_1),\, m^-_{ik}(\tilde{w}_\infty)\} \ge m^*_{ik}(\tilde{w}_\infty)$, with strict inequality whenever the cutoffs are interior to the support of the draws. Adopters with draws above $z^{*-}_{ik}(\tilde{w}_\infty)$ keep maintaining because their operating gain exceeds ζF, no producer between $z^{*-}_{ik}(\tilde{w}_\infty)$ and $z^*_{ik}(\tilde{w}_\infty)$ exits, and none enters. Provided the entry cutoff at $\tilde{w}_1$ is interior, all producers that adopted at $\tilde{w}_1$ keep the technology if

$$\frac{\Pi_k(\tilde{w}_1)}{\Pi_k(\tilde{w}_\infty)} \le \frac{r}{\zeta}. \tag{19}$$

(iii) If $\tilde{w}_\infty = \tilde{w}_0$, the gain vanishes, maintenance stops, and $A_{ik} = B_{ik}$ at $T+1$.

Proof. (i) is Lemma 1 applied to (15) and with immediate entry in (16). For (ii), $\tilde{w}_\infty < \tilde{w}_1$ gives $m^*_{ik}(\tilde{w}_1) > m^*_{ik}(\tilde{w}_\infty)$ by Lemma 1, and $\tilde{w}_\infty > \tilde{w}_0$ gives $\Pi_k(\tilde{w}_\infty) > 0$, so the exit cutoff ((15) evaluated at ζF in place of rF) is finite and $m^-_{ik}(\tilde{w}_\infty) > m^*_{ik}(\tilde{w}_\infty)$. At $t = T$ the state is $m^*_{ik}(\tilde{w}_1) \ge m^*_{ik}(\tilde{w}_\infty)$, so the entry term of (16) is inactive and (16) returns $\min\{m^*_{ik}(\tilde{w}_1), m^-_{ik}(\tilde{w}_\infty)\}$, a fixed point of (16) at $\tilde{w}_\infty$. The minimum equals $m^*_{ik}(\tilde{w}_1)$ iff $z^{*-}_{ik}(\tilde{w}_\infty) \le z^*_{ik}(\tilde{w}_1)$, which by (15) is (19). For (iii), $\Pi_k(\tilde{w}_0) = 0$ makes both cutoffs infinite, so $m^-_{ik} = 0$ and (16) returns zero. The band width follows from the ratio of (15) at rF and at ζF. □

[47] With a single rung ($n_k = 1$), (17) reduces to the two-technology aggregator $A_{ikt} = m^{\text{non}}_{ikt} + \upsilon_k^{\sigma_k-1}m^{\text{ad}}_{ikt} = B_{ik}\big[1 + (\upsilon_k^{\sigma_k-1}-1)\pi_{ikt}\big]$ with $\pi_{ikt} = m^{\text{ad}}_{ikt}/B_{ik} \in [0,1]$, which is the form the pairwise exposition uses. Two adopter shares appear below, the cumulative draw share $\pi_{ik\tau t}$ and the active pair's revenue share, $s^{\text{ad}}_{ikt} = S^{\text{ad}}_{ikt}/(S^{\text{non}}_{ikt} + S^{\text{ad}}_{ikt})$, where S^{τ}_{ikt} is defined in (20).

I can distinguish (ii) and (iii) empirically. Under (iii), technology does not outlive the wage that induced it. Even if abandoned capital kept operating while wearing out, rather than failing at once, the reversion would be geometric at rate ζ. At $\zeta = 5\%$ the half-life is 14 years, at Allen's aggregate 2.4% it is 29 years, and it is way shorter for e.g. draught animals, whose working lives Langdon, 1982 puts at 5 to 7 years. The capital intensity of exports and the number of horses per capita, which proxies for the capital-to-labour ratio in agriculture and indicates technology adoption (Andersson, Karadja, and Prawitz, 2022), were still higher a century after the plague (Appendix Figure A.4d and Appendix Table A.18), which rejects (iii) and requires (ii) in this model. Persistence thus requires reinvestment, not durability. Within (ii), the proposed values put the bar for the stronger case, (19), low (Appendix F.5): reversion by up to four fifths leaves every plague-era adopter in place.

Gravity's exporter term. For gravity, it is useful to define the exporter term $T_{ik}(\tilde{w}_i)$, which combines sectoral productivity A_{ik} with unit costs, and thus captures competitiveness. Partitioning the varieties at the cutoff, the two groups add up:

$$T_{ikt}(\tilde{w}_i) = \left[S^{\text{non}}_{ikt} + S^{\text{ad}}_{ikt}\right]^{\rho_k}, \;\; S^{\text{non}}_{ikt} = m^{\text{non}}_{ikt}\left(w_i{}^{\gamma_{\text{non}}} r^{1-\gamma_{\text{non}}}\right)^{1-\sigma_k}, \;\; S^{\text{ad}}_{ikt} = \upsilon_k^{\sigma_k-1}\, m^{\text{ad}}_{ikt}\left(w_i{}^{\gamma_{\text{ad}}} r^{1-\gamma_{\text{ad}}}\right)^{1-\sigma_k}, \tag{20}$$

where w_i is the wage and $\rho_k \equiv \frac{\xi-1}{\sigma_k-1}$. The bracketed sum is the CES aggregate of region i's varieties and the bundle's price index up to the markup and trade costs, $P_{ijk}^{1-\sigma_k} = (\mu_k d_{ijk})^{1-\sigma_k}\left[S^{\text{non}}_{ikt} + S^{\text{ad}}_{ikt}\right]$, from substituting the markup price into (9) (Appendix F.4).

Endogenous labour share. The realised sectoral labour share is endogenous,

$$\Gamma_{ikt} = \gamma^k_{\text{non}} - (\gamma^k_{\text{non}} - \gamma^k_{\text{ad}})\, s^{\text{ad}}_{ikt} \in [\gamma^k_{\text{ad}}, \gamma^k_{\text{non}}], \tag{21}$$

where s^{ad}_{ikt} is the adopters' revenue (export-weighted) share.

Gravity. Trade costs d_{ijk} are iceberg and follow $d_{ijk} = d_{ij}d_{jk}$.[48] Aggregating varieties into bundles and bundles into the sector composite, the market access terms are

$$CMA_{jk} = (P_{jk})^{1-\xi} = \chi_k \textstyle\sum_i T_{ik}(\tilde{w}_i)\, d_{ijk}^{1-\xi}, \qquad FMA_{ik} = \chi_k\alpha_k \textstyle\sum_j d_{ijk}^{1-\xi}(CMA_{jk})^{-1}Y_j, \tag{22}$$

with the finite $(\sigma_k - 1)$-moment ensuring the within-bundle price indices exist and $\chi_k = \mu_k^{1-\xi}$. Combining the bundle demand (9) with the bundle price index and (22), trade is gravity:

$$X_{ijk} = \frac{\chi_k\, T_{ik}(\tilde{w}_i)\, d_{ijk}^{-(\xi-1)}}{CMA_{jk}}\, \alpha_k Y_j. \tag{23}$$

Competitiveness $T_{ik}(\tilde{w}_i)$ factors into productivity and factor prices (Appendix G.4),

$$\ln T_{ikt} = \rho_k \ln A_{ikt} \;-\; (\xi-1)\,\bar{\Gamma}_{ikt}\, \ln \tilde{w}_{it} \;-\; (\xi-1)\ln r,$$

where $\bar{\Gamma}_{ikt} \in [\gamma^k_{\text{ad}}, \gamma^k_{\text{non}}]$ is the average of the realised share (21) over the factor-cost interval between $\tilde{w} = 1$ and $\tilde{w}_{it}$ at the date-t masses (Appendix G.4).

Summing (23) over j, sales of (i,k) are $T_{ik}\, FMA_{ik}$. Substituting this and the bundle price index into the demand shifter of (13), and summing the labour demand implied by (12) over producers, gives the objects that the technology block and the labour market use (Appendix F.4),

$$\Psi_{ik} = \frac{1}{\sigma_k}\, T_{ik}^{1-1/\rho_k}\, FMA_{ik}, \qquad w_i L_{ik} = \frac{FMA_{ik}}{\mu_k}\, T_{ik}\, \Gamma_{ik}. \tag{24}$$

[48] Exporter-sector-specific trade costs are unaffected by treatment: in Appendix Table A.24, I show that travel times to and handling times in ports are unchanged by the plague.

Equilibrium. Given endowments L_{it} and $\bar{K}_i$, the measures N_{ik}, the adoption stock F, the trade costs d_{ijk}, and the distribution F_{ik}, the parameters α_k, ξ, σ_k, γ_τ^k, υ_k, and ζ, the world rental r, and initial adopter sets, an equilibrium is a sequence of wages, prices, price indices, technology choices, factor demands, incomes, and trade flows such that $\forall t$

(i) households maximise utility, so that demand is (9) with income (35),

(ii) producers minimise costs and set prices by (12) and the markup rule, and choose technologies by (14), so that adopter masses follow (16),

(iii) the regional labour market clears, $\sum_k L_{ikt} = L_{it}$, with L_{ikt} from (24). Under serfdom the urban market clears alone, $\sum_{k\in M} L_{ikt} = L_{it}^M$, and the rural wage is administered,

(iv) capital is rented on the world market at r in the quantity (12) implies. Adoption stocks are financed there, so a region's net capital position is absorbed by the rest of the world,

(v) income equals factor income plus operating profits net of the adopters' flow cost rF, (35),

(vi) goods markets clear. Sales of origin i in sector k equal $\sum_j X_{ijk}$, and expenditure of every destination equals its income, both of which (9) imposes.

Collecting the conditions, the equilibrium at each date is the system formed by (9), (12), (14) with the state (16), (20), (22), (23), (24), (35), and labour market clearing, in the unknowns w_{it}, P_{ijk}, P_{jk}, X_{ijk}, Y_{it}, L_{ikt}, K_{ikt}, and the adopter sets. Given the state and the prices of every other region, all conditions but labour market clearing determine every object from wages, so the equilibrium reduces to one equation per region.

Below, the wage response, persistence, and uniqueness use one further assumption (Demidova and Rodríguez-Clare, 2009).

Assumption 2 (Small open economy). Region i takes r, the price indices $\{P_{jk}\}_{j\neq i}$, and the incomes $\{Y_j\}_{j\neq i}$ of its destinations as given, and its own absorption accounts for a negligible share of its markets.

This reduces existence and uniqueness to the slope of regional labour demand. Labour demand slopes down, and the equilibrium wage is unique, if the adoption feedback stays below the dispersion forces: sector by sector $R_{ik} < 1 + \Gamma_{ik}(\xi - 1)$, where R_{ik} is the response of T_{ik} to the wage through adoption. This sector-by-sector condition is sufficient rather than necessary (Appendix G). Within a period, the adopter masses are predetermined by (16), so the temporary equilibrium wage is unique under the static technology slope condition alone. Instead, the condition on R_{ik} disciplines the stationary adoption locus. Since only one adjacent margin is active at any wage, this condition applies rung by rung. Since decisions at t take effect at $t + 1$, the dynamic equilibrium is a sequence of temporary equilibria joined by the deterministic update (16): given initial adopter sets, the path of wages, prices, and masses is unique.

Persistence in general equilibrium. The capital-intensive technology saves labour, so at any wage inside the interval of Lemma 1 the adopters inherited from the plague years demand less labour than they would on the old technology (Appendix F.5). Once population has recovered, the same labour supply meets a lower labour demand and, as labour demand slopes down (Appendix G), the wage settles weakly below its pre-plague level, $\tilde{w}_\infty \leq \tilde{w}_{\text{pre}}$. Population recovery therefore does not imply wage recovery. Persistence accordingly comes from the band under two conditions: some producers used capital-intensive technology before the plague and the band absorbs the dip, (19). At $r/\zeta = 2$ and the calibration of Section 7, (19) holds whenever the post-recovery wage retains about a fifth of the plague gain over $\tilde{w}_0$. Below that, the adopters between $z_{ik}^*(\tilde{w}_1)$ and $z_{ik}^{*-}(\tilde{w}_\infty)$ abandon and the surviving wage gap is G_∞ of Corollary 1.

In a nutshell. Sectoral productivity A_{ik} depends on factor prices: the adoption cutoff falls as $\tilde{w}_i = \frac{\Gamma_{ik}}{1-\Gamma_{ik}} \frac{K_{ikt}}{L_{ikr}}$ rises (Lemma 1), so the cumulative adopter shares and sectoral productivity (17) are increasing functions of the capital-labour ratio,

$$A_{ikt} = B_{ik}\Big[1 + \sum_{\tau=1}^{n_k} \Big(\big(U_{\tau}^{k}\big)^{\sigma_k - 1} - \big(U_{\tau-1}^{k}\big)^{\sigma_k - 1} \Big) \pi_{ik\tau t}\big(K_{ikt}/L_{ikt}\big) \Big]. \tag{25}$$

Sectoral variation in changes in A_{ik} require sectoral variation in adoption since under Cobb-Douglas, the elasticity of substitution between capital and labour is one in every sector, and the technology parameters v_{τ}^{k} and B_{ik} are fixed over time. In this model, any differential change in sectoral productivity therefore identifies the adoption margin.

7 Mechanism

I propose that a lower population leads to higher wages which induces technology adoption. The share of adopters is higher in capital-intensive sectors, where labour-saving technologies with a productivity gain sufficient to recoup the sunk investment were more available. Appendix Table A.30 lists examples of capital deepening in this period. In the production of wheat, e.g., modern farm machinery was unavailable beyond limited tools. Generally, the options for a sunk investment to increase agricultural productivity were severely limited, with most productivity-increasing devices centuries old (Coleman, 1956). In labour-intensive manufacturing, Atack, Margo, and Rhode, 2019 concludes that few technologies or techniques were available before the 19th century to increase productivities. Thus, labour-intensive sectors already operated close to the most capital-intensive rungs available in the pre-industrial period. In capital-intensive sectors, on the other hand, there were more labour-saving options worth the sunk cost. In metallurgy and ship building, e.g., these could take the form of furnaces or dry docks, such as the Polhemsdockan in Karlskrona, whose construction was begun in 1717. In capital-intensive agriculture, buildings, fencing, and processing and storage facilities presented available options. Given these technological constraints, I expect the technology adoption response to be stronger in capital-intensive sectors.

7.1 Scarce Labour and Higher Wages

I establish lower populations and higher wages. Appendix Table A.4 shows that the plague reduced populations by 36% at the median and Appendix Table A.6 shows that plagued cities' and regions' populations had recovered by 1750. For more granular data, I propose the number of captains reporting to live in a region as a proxy.[49] Appendix Figure A.12 shows a 50% drop in captains and a 15 year recovery period. This likely overstates both the population shock and recovery, as captains are highly mobile. Thus, labour scarcity is present in plagued regions for 15-50 years.[50]

As there is no wage series available for the Great Northern War plague outbreak, I turn to Allen, 2005 who documents wage increases of 38% for the Black Death, which had a similar mortality rate. These wage increases were real (Galofré-Vilà, Hinde, and Meera Guntupalli, 2018, Broadberry et al., 2014). For a period just decades before this paper's plague outbreak, Pfister, 2017 shows a 30% real wage increase for unskilled urban labourers for a 40% population loss. Data on city-level interest rates in the Baltic and North Sea region are scarce in the early

[49] This variable is the home port as in Marczinek, Maurer, and Rauch, 2025. To count captains once, I omit observations in which first name, last name, and region are duplicates in a given year.

[50] An unsuitable alternative, the Hyde data, are either imputed or at the national level, using linear interpolations between centuries, without providing genuine localised annual population data.

18th century. Even the most extensive collections of sources (Huang, Chilosi, and Sapoznik, 2019, Schmelzing, 2020) cover only one plagued city before and after the plague. Thus, I follow Keller, Shiue, and Wang, 2021 who infer interest rates from monthly variation in grain prices. Appendix Table A.29 finds that interest rates do not increase after the plague. While there is therefore no data available to directly test whether (relative) wages rose, the literature on post-plague wages and infered evidence on interest rates lead me to conclude that wages rise absolutely and relatively.

7.2 Technology Adoption and Capital Deepening

In my model, rising relative wages $\tilde{w}_i$ lead to technology adoption and capital deepening. Specifically, Cobb-Douglas cost minimisation explains the empirical evidence on capital deepening in agriculture.

Across sectors, a shift into capital-intensive sectors occurs. Taking ratios of gravity equation (23) for manufacturing, plugging in equation (50), and writing $\kappa = \frac{\alpha_{CM}}{\alpha_{LM}} \frac{\chi_{CM}}{\chi_{LM}} \left(\frac{d_{ijCM}}{d_{ijLM}}\right)^{-(\xi-1)}$:

$$\frac{X_{ijCM}}{X_{ijLM}} = \kappa \frac{(A_{iCMt})^{\rho_{CM}}}{(A_{iLMt})^{\rho_{LM}}} \tilde{w}_{it}^{-(\xi-1)(\bar{\Gamma}_{iCMt}-\bar{\Gamma}_{iLMt})} \frac{CMA_{jLM}}{CMA_{jCM}}, \tag{26}$$

where CM (LM) is the subscripts for capital- (labour-)intensive manufacturing and where $\bar{\Gamma}_{ikt}$ averages the realised share Γ_{ikt} over the factor-cost interval (Appendix G.4). This allows me to decompose the findings of Figure 2 and Section 4 into three channels.

First, capital-intensive sectors expand as they have smaller labour shares ($\bar{\Gamma}_{iCMt} - \bar{\Gamma}_{iLMt} < 0$). The wage term is the labour cost channel, through which rising relative wages shift exports relatively into capital-intensive sectors. Second, capital-intensive sectors expand as technology adoption is stronger in capital-intensive sectors, which I assume given constraints on available technology in labour-intensive sectors. This productivity channel results in the productivity ratio in equation (26) and further increases the expansion of capital-intensive sectors.

In general equilibrium, $\frac{CMA_{jLM}}{CMA_{jCM}}$ decreases: the labour cost and productivity channels shift up the wage term and relative productivity. By equation (22), this results in a relative consumer market access increase (and price index decline) in capital-intensive sectors, which counteracts the first two channels: plagued regions' productivity advances decrease the relative export prices of capital-intensive goods. While not a direct test of relative price indices, Appendix Table A.26 constructs export prices and finds this pattern.

Evidence on Adoption

While I do not directly observe technology adoption in this period's sources, the data contain the footprints that adoption leaves behind. Six pieces of evidence are consistent with the model's prediction that the adopter share π_{ikt} rises with $\tilde{w}_{it}$. First, capital deepening takes the form of installed capacity: plagued ports import larger volumes of tools, nails, metals, and building materials (Appendix Figure A.6), which are the inputs to the investments listed in Appendix Table A.30. Plagued Karlskrona, specifically, began construction of its dry dock in 1717. In agriculture, significantly more horses are used (Appendix Tables A.17 and A.18), which is a proxy for capital deepening and interpreted by Andersson, Karadja, and Prawitz, 2022 as indicative of technology adoption since horses were needed to use capital-intensive agricultural tools. Second, capital deepening is concentrated where installing technology pays off: in Scania, horsepower differentially substitutes for lost labour on rich soil close to plagued cities (Appendix Table A.19). Third, the differential competitiveness and productivity response (Tables 1 and 2) follows the availability of technologies worth a sunk investment and, assuming sectorally symmetric capital deepening under Cobb-Douglas, identifies the adoption margin

(Section 6). Fourth, the response is larger where capital was cheaper to obtain (Appendix Table A.23) and smaller where factor prices were not transmitted under serfdom. This is how a decision weighing a capital cost against a wage-driven profit gap should behave. Fifth, the capital-to-labour ratio (Appendix Table A.18) and the stock of ships (Appendix Figure A.5) remain elevated well beyond population recovery, consistent with sunk adoption costs rather than ordinary substitution. Sixth, the estimated output elasticities shift from labour toward capital in plagued parishes, most strongly in capital-intensive agriculture (Appendix Table A.17), tracking differential availability of technology. Thus, the capital deepening observed after the plague carries the signature of technology adoption.

7.3 Productivity Growth

Finally, technology adoption results in productivity growth. I propose that productivity growth explains the post-plague export expansion (3), for which Section 4 provides direct production-side evidence. To proceed, I map the trade data to the gravity equation, which allows me to recover competitiveness and to separate factor proportions from productivity. I regress observed trade on the time-expanded set of fixed effects in Costinot, Donaldson, and Komunjer, 2012:

$$X_{ijkt} = \exp(\alpha + \delta_{ijt} + \delta_{jkt} + \delta_{ikt}) \times \epsilon_{ijkt}. \tag{27}$$

I estimate equation (27) using PPML.[51] The importer-sector-time effects δ_{jkt} absorb CMA_{jkt} and inward multilateral resistance (Fally, 2015, Anderson and Wincoop, 2003), whereas the general equilibrium forces driving outward multilateral resistance are captured by wages. δ_{ijt} absorb bilateral time-varying trade costs. The object of interest is δ_{ikt}.

Appendix G.4 details that I recover a triple difference (I write the triple difference of x_{ikt} as $\mathcal{D}\big[x_{ikt}\big]$) of competitiveness that contains both productivity and factor proportions:

$$\delta_{ikt} = \mathcal{D}\big[\ln T_{ikt}\big] = \underbrace{\rho_k \, \mathcal{D}\big[\ln A_{ikt}\big]}_{\text{relative productivity}} \underbrace{+ (\xi - 1)\, \Delta\Gamma_{it}\, G}_{\text{factor proportions}}, \tag{28}$$

where $\Delta\Gamma_{it}$ is the labour share gap between sectors (I compare labour- to capital-intensive sectors within agriculture and manufacturing) and G is the treated relative to control change in $\ln(w/r)$. B_{ik}, N_{ik}, z_{ik}, and every exogenous component of comparative advantage cancel, so

$$\mathcal{D}\big[\ln A_{ikt}\big] = \mathcal{D}\Big[\ln\big(1 + (\upsilon_k^{\sigma_k - 1} - 1)\, \pi_{ikt}\big)\Big] \tag{29}$$

and the recovered productivity component is the endogenous adoption response. The adoption gain ν_k scales how much A_{ikt} moves for a given change in π_{ikt}. As it is constant, it can only generate a change in A_{ikt} through adoption. A differential change in A_{ikt} thus requires π_{ikt} to move, and may also reflect sectoral differences in ν_k.[52]

Bounding the Role of Factor Proportions

I compare the competitiveness change of capital- to labour-intensive sectors. Since their labour shares differ, the factor proportions wedge is non-zero and bounded. It depends only on G (I use $g = e^G - 1$ below), the trade elasticity $\xi - 1$, and $\Delta\Gamma = \Gamma_{iL} - \Gamma_{iC}$, which I evaluate as a fixed-masses path average. The share of differential D that factor proportions can explain is

$$\Sigma = \frac{(\xi - 1)\, \Delta\Gamma\, G}{D}. \tag{30}$$

[51] Juhász, Sakabe, and Weinstein, 2024 and Amiti and Weinstein, 2018 discuss issues in estimating equation (27) under initial zeros. Here, I recover a full panel of exporter-sector-year fixed effects.

[52] Note that (28) concerns the sectoral differential, as the triple difference nets out the level. Since under fixed productivity a plagued region loses competitiveness relative to unplagued competitors, the level expansion of Fact #2 is productivity growth alone. The bound below applies to the cross-sector differential.

I draw on two values for the trade elasticity: 5 (Head and Mayer, 2014) and 6.53 (Costinot, Donaldson, and Komunjer, 2012). For g, data for this period are scarce, so rather than plug in a measured relative wage, I bound (28) over assumed relative wage growth $g \in [0, 0.5]$.[53]

The labour-share gap is the other key object to pin down. I bound it through the range of realised pre-industrial labour shares and present two arguments that $\Delta\Gamma = 0.2$ is a generous upper bound. First, the growth accounting literature considers differences between their preferred weighting and an upper bound over the aggregate labour share of $0.11 - 0.14$ (Antràs and Voth, 2003).[54] Second, the sectoral evidence points the same way: adjusting Allen, 2005 to only labour and capital implies that agricultural labour shares never leave $[0.67, 0.86]$. Clark, 2002 gives an even smaller variation in labour shares. A cross-sector gap of 0.2 therefore exceeds anything the pre-industrial accounts display even after the Black Death.[55] This gap is also contained in the confidence interval of the estimated change in the labour elasticity in labour-intensive agriculture (Appendix Table A.17).

Competitiveness and Productivity

I estimate the plague's effect on the capital-intensity differential in sectoral competitiveness:

$$\delta_{ikt} = \left[\beta_{g(k)} + \beta^s_{g(k)}\, s_i\right] plague_{it} \times K_k + \alpha_{ik} + \alpha_{akt} + \alpha^{ag}_{it} + \epsilon_{ikt}, \tag{31}$$

where δ_{ikt} are the fixed effects recovered from equation (27).[56] K_k is an indicator equal to one for the capital-intensive sector of each sector group, $g(k) \in \{A, M\}$ denotes whether sector k belongs to agriculture or manufacturing, and $s_i \in \{0, 1\}$ indicates whether region i is under serfdom. α_{ik} are region-sector fixed effects, absorbing time-invariant region-sector-specific determinants of competitiveness, such as availability of resources or time-invariant institutions. α_{akt} are area-sector-year fixed effects, absorbing area-time variation related to e.g. area- and country-wide economic and trade policies and temperature change by sector. Changes in serfdom are absorbed by α_{akt} as long as they do not vary within areas (Appendix B). α^{ag}_{it} are region-year-agriculture fixed effects ensuring that the capital-intensity differential is identified separately within agriculture and manufacturing. ϵ_{ikt} is the error term. The coefficient $\beta_{g(k)}$ measures the post-plague change in the competitiveness of the capital-intensive relative to the labour-intensive sector. Columns (1) and (4) of Table 2 impose $\beta_g = \beta$ and $\beta^s_g = 0$. Columns (2) and (5) allow the differential to differ between agriculture and manufacturing, $\beta^s_g = 0$. Columns (3) and (6) add the serfdom interactions β^s_g, so that the differential under serfdom is $\beta_g + \beta^s_g$.

Table 2 shows the results. Competitiveness shifts significantly into capital-intensive sectors overall, which appears to be stronger in manufacturing, though the difference to agriculture is not significant. I find the same results when accounting for spatial correlation (Conley, 1999) in Appendix Figure A.13, and the differential shows no pre-trends before the plague (Appendix Table A.14). While the serfdom interactions are not precisely estimated, they mirror the transmission of labour scarcity into factor prices.[57] In agriculture, the serfdom interaction removes more than half of the differential, which can be explained by rural wages that barely move, leaving z^*_{ikt} and the adoption share π_{ikt} flat. In urban manufacturing, the interaction

[53] Allen, 2005 reports English agricultural real wages rising by 38% between 1300 and 1500; his and Clark's accounts are constructed assuming flat interest rates, so this is a 38% rise in w/r across the plague.

[54] Their baseline labour share of 0.5 compares to their upper bound of 0.64; adjusting shares for dropping land gives shares of 0.59 and 0.7.

[55] Note that this bound refers to the pre-plague gap. Under the technology menu of Section 6, the realised gap widens as the capital-intensive sector adopts more and its realised labour share falls along the menu.

[56] See Appendix G.4 for details.

[57] This is because v_k, the γ's, and $\Delta\Gamma$ are serfdom-invariant and every component of the differential is proportional to the factor price movement.

is close to zero, though the confidence intervals includes the positive differential one would expect, since cities surrounded by serfdom were additionally labour-constrained. While the serfdom-interactions in competitiveness are imprecisely estimated, the capital-intensity shift, differing by serfdom-constrained agriculture and free manufacturing, is estimated in trade data (Appendix Table A.22).

Table 2: Effect of plague on capital-intensity differential in sectoral competitiveness

	Plague Dummy			Mortality Rate		
	(1)	(2)	(3)	(4)	(5)	(6)
Differential capital- minus labour-intensive:						
Pooled	1.297**			3.763**		
	(0.555)			(1.530)		
Agriculture		1.131**	1.768*		3.348***	5.431**
		(0.505)	(0.968)		(1.250)	(2.121)
Manufacturing		1.464*	1.480		4.177*	4.551
		(0.797)	(1.401)		(2.306)	(3.661)
Agriculture x Serfdom			-1.095			-3.698
			(1.048)			(2.413)
Manufacturing x Serfdom			-0.029			-0.664
			(1.704)			(4.643)
Fixed Effects:						
– Region x Sector	✓	✓	✓	✓	✓	✓
– Area x Sector x Year	✓	✓	✓	✓	✓	✓
– Region x Year x Agriculture	✓	✓	✓	✓	✓	✓
Observations	198,868	198,868	198,868	198,868	198,868	198,868

Notes: Standard errors clustered at the region-level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable is log sectoral competitiveness recovered from equation (27). Each coefficient is the post-plague change in the competitiveness of the capital-intensive relative to the labour-intensive sector of the respective group, in plagued relative to unplagued regions. Columns (1) and (4) impose a common capital-intensity differential across agriculture and manufacturing. Columns (2) and (5) estimate the two differentials separately. Columns (3) and (6) add interactions of each differential with serfdom: the uninteracted coefficients give the differential in regions without serfdom, the interactions the additional effect under serfdom, and their sum the differential under serfdom. Columns (1) to (3) use a plague dummy that equals 1 for plagued regions after the plague hit. Columns (4) to (6) use the plague mortality rate after the plague hit (Appendix Table A.5).

Returning to the share factor proportions can explain (equation (30)), I take the proposed trade elasticities, wage growth, and the labour-share gap to the estimated pooled differential of column (1) of Table 2, $D \equiv \hat{\beta} = 1.297$ log points. The following table computes the largest fraction that factor proportions can account for.

Share explained by factor proportions	$g = 10\%$	$g = 25\%$	$g = 38\%$	$g = 50\%$
Baseline $\xi - 1 = 5$, pooled $D = 1.297$	7.3%	17.2%	24.8%	31.3%
Costinot et al. preferred estimate $\xi - 1 = 6.53$	9.6%	22.5%	32.4%	40.8%

For wage growth of 38% as in Allen, 2005, factor proportions account for 25-32%. Even at 50% wage growth, this end of the bracket reaches at most 31% at the baseline elasticity and 41% at the Costinot, Donaldson, and Komunjer, 2012 value. On the model-consistent path, where adoption itself widens the realised gap toward the bottom of the menu (Proposition 3), the decomposition at 38% wage growth attributes about 37-41% to factor proportions. Evaluating the wedge at the full-menu gap, $\Delta\Gamma_{\text{menu}} \equiv 0.7 - \gamma_n$, the cross-sector gap once the capital-intensive sector has descended all its rungs (about 0.4 at the baseline elasticity, 0.3 at the higher, where the menu bottoms out at a higher share), caps it near 49%.

The parish production functions provide an independent check on this path: the plague widens the cross sector gap in estimated labour elasticities by 0.37 (Appendix Table A.17), inside the model-consistent path-end range of 0.3 to 0.4.

Overall, technology adoption accounts for 51-75% of the differential, with a model-consistent point of about 59-63%. The 51% floor is deliberately conservative: the widening of the gap beyond its pre-plague value is caused by adoption, which the floor attributes to factor proportions. Thus, I conclude that factor proportions cannot fully account for the observed shift in comparative advantage.[58]

This conclusion also survives sampling uncertainty. The trade elasticity, the labour-share gap, and wage growth are taken from the literature. The only estimated object in equation (30) is D, so the hypothesis that factor proportions fully account for the differential, $\Sigma \geq 1$, is equivalent to the linear restriction $D \leq (\xi - 1)\,\Delta\Gamma\, G$. At the baseline elasticity and headline wage growth, this threshold is 0.32 log points, against the estimated 1.297 with a standard error of 0.555, giving $t = 1.76$.[59] The hypothesis is rejected with $p = 0.039$. Thus, the conclusion that factor proportions alone cannot generate the observed shift in comparative advantage holds at conventional significance levels.

The bound above concerns the decades immediately after the plague. Over a longer horizon, the differential that outlasts population recovery converges to pure technology:

Corollary 1 (Long-run composition of the differential)**.** *Apply Proposition 1 to sectors k and k_0 facing the same wage path and $G_\infty \equiv \ln(\tilde{w}_\infty/\tilde{w}_{\text{pre}})$ for the wage gap after population recovery. The labour demand comparison of Section 6 makes G_∞ weakly negative. For $t > T$, the cross-sector difference of* (28) *is*

$$\left[\rho_k \ln \tfrac{A_{ikt}}{A_{ik0}} - \rho_{k_0} \ln \tfrac{A_{ik_0t}}{A_{ik_00}}\right] \;+\; (\xi - 1)\,\Delta\Gamma_{it}\, G_\infty.$$

The first bracket is set by the adopter mass surviving under Proposition 1(ii), which requires only that the long-run wage stays above $\tilde{w}_0$, the wage at which the adoption gain vanishes. It equals the plague-era mass whenever the long-run wage retains at least a fifth of its plague-induced growth over $\tilde{w}_0$ (Appendix F.5), and otherwise equals $m^-_{ik}(\tilde{w}_\infty)$: once the maintenance margin binds, a lower recovery wage reduces the surviving mass, so the technology component can shrink with G_∞, while still exceeding the mass the long-run wage would induce on its own. The factor proportions wedge, in turn, does scale with G_∞ and thus vanishes, or turns weakly negative, so the long-run differential is a lower bound on the technology channel.

The result follows by substituting Proposition 1 into (28), noting that $\bar{\Gamma}_{ikt} \ln \tilde{w}_{it}$ equals $\bar{\Gamma}_{ikt} G_\infty$ for $t > T$ under the normalisation $\ln \tilde{w}_{it_0} = 0$, and differencing across sectors.

In Appendix Figure A.14, I presents results estimating equation (31) by decade and over the long-run. The capital-intensity differential shows no pre-trends and rises after the plague, though the first decade's estimate, still during the war and naval blockade (see section 3), is not precisely estimated. From the 1720s onwards, a significant capital-intensity differential is estimated, which rises until the mid of the century. At this point, population levels had recovered, bringing the wage gap to Corollary 1's G_∞. Accordingly, the capital-intensity differential

[58] Another approach is to recover the labour share gap needed to fully explain the differential. For 10%, the gap would have to be beyond 2 at either elasticity. For 25%, beyond 1.1 (or about 0.9 for $\xi - 1 = 6.53$). For 38%, about 0.8 (0.6 for $\xi - 1 = 6.53$). Further, using the bounded aggregator (17) and the recovered differentials net of the factor proportions wedge, $(1 - \Sigma)D \leq \rho_k(\sigma_k - 1) \ln \upsilon_k = (\xi - 1) \ln \upsilon_k$, which is a lower bound on the adoption gain. Thus, the rung-by-rung productivity boost satisfies $\upsilon_k \geq \exp\big((1 - \Sigma)D/(\xi - 1)\big)$, between 12-27% across the calibration grid. It holds with equality only under full post-plague adoption from zero initial adoption.

[59] The test is one-sided since the economically relevant alternative is that the differential exceeds what factor proportions can deliver.

in competitiveness falls, as the factor proportions wedge falls following Corollary 1. This leaves the technology channel to drive competitiveness and thus to govern comparative advantage patterns. The differentials remains significant until the first decade of the 1800s: for a century after the plague outbreak, comparative advantages remain shifted towards capital-intensive sectors. Thus, the post-plague economy remained in case (ii) of Proposition 1 for a century after the plague.

Thus, both in the short-run and in the long-run, sectors' differential competitiveness (and thus comparative advantage) response to plague-induced labour scarcity cannot be explained by factor proportions forces alone. Instead, the model's quantification concurs with my empirical findings: labour scarcity led to productivity growth tilted towards capital-intensive sectors.

8 Conclusion

This paper asks whether labour scarcity leads to productivity growth. Exploiting the last Nordic plague as a natural experiment, I show that plagued regions shift their exports into capital-intensive goods and capture larger market shares abroad. Parish-level production data reveal the capital deepening and productivity growth behind these trade patterns, and both remain in place for half a century beyond population recovery.

I rationalise these findings in a general equilibrium trade model in which higher wages induce the adoption of capital-intensive technology from a fixed menu. Taking the model to the data, I decompose gravity-recovered competitiveness into a factor proportions wedge and an adoption component and find that 51-75% of the differential response of capital-intensive sectors reflects technology adoption. The mechanism is switched off where serfdom shielded producers from higher wages and operates more strongly where capital markets were deeper. Labour scarcity thus raises productivity where producers are exposed to changing factor prices and are able to respond with capital deepening.

The result speaks to a present in which labour is becoming scarce through aging, changing preferences on working hours, and declining labour force participation, and in which capital deepening is proceeding at a historic pace through investment in AI. The pessimistic case, where labour scarcity holds back growth, rests on the premise of fixed productivity. My paper presents historical evidence refuting that premise, instead proposing that higher wages incentivise technology adoption and thus induce productivity growth. My results also suggest that the productivity payoff is conditional: it requires that factor prices transmit to producers, which labour and capital market institutions can prevent.

My paper suggests two scope conditions. The large productivity response documented here arises in an economy producing at low capital intensity and far from the technology frontier, where the fixed menu contained labour-saving techniques worth adopting. This makes my findings particularly relevant to developing economies. Closer to the frontier, the importance of adoption may be rivalled by that of innovation. Further, the model's adoption margin is inferred from its footprints in capital deepening and productivity rather than observed, and settings with direct records of technology adoption would allow a sharper test. Validating the mechanism in other periods and regions, and in settings where the menu is expanding and where technology choice is recorded, is thus the natural next step.

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A Data Appendix

Table A.1 holds key descriptives for the trade panels constructed from the Soundtoll records.

Table A.1: Descriptive Statistics

	Obs.	Mean	Std. Dev.	Min	Max
Panel A: Bilateral Trade, Intensive Margin Overall (origin × destination × year)					
Exports (levels)	502,980	13.424	268.081	0.000	35175.146
Market share	502,980	0.013	0.099	0.000	1.000
Panel B: Bilateral Trade, Intensive Margin by Sector (origin × destination × sector × year)					
Exports: capital-int. agriculture	502,980	4.101	78.956	0.000	8592.281
Exports: capital-int. manufacturing	502,980	1.588	45.716	0.000	7047.719
Exports: labour-int. agriculture	502,980	6.722	206.383	0.000	32110.792
Exports: labour-int. manufacturing	502,980	0.596	20.881	0.000	5183.438
Panel C: Region-Level Exports (origin × year)					
Total exports (levels)	49,800	135.585	1296.781	0.000	40149.656
Capital-intensive share	49,800	0.096	0.276	0.000	1.000

A.1 Commodities

143,855 cargoes are reported in old Danish. I use the Soundtoll Project's list of products translated into English and Danish etymological dictionaries. To complement my manual cleaning, I used Claude's Opus 4.6 model to classify cargoes.[60] For 84% of traded value, the algorithmic assignment is the same as the manual one. When accounting for a few inconsequential disagreements (iron and iron works, planks and wood, cloth and sail, brandy and wine, goods and mercery, and meat and beef), I arrive at 96% overlap between the two approaches. Thus, the two approaches produce almost identical results.

As no standardised commodity codes are available in the source, I code goods as granularly as I can. This granularity reflects the source's time period: many types of fish and textiles are recorded, whereas some shipments are mentioned to contain generic goods. Where the data are insufficiently specific, I use compound categories. Differences in granularity matter little as I aggregate up to sectors. I arrive at 227 goods in five sectors. Capital-intensive goods require a high amount of capital (factories, furnaces, machinery, tools, animals) that cannot be swapped for labour. Labour-intensive goods are limited in the degree to which labour can be saved in their production. Labour-intensive agriculture is all of arable farming, whereas pastoral farming, mining, and processed foods are capital-intensive agriculture. Labour-intensive manufacturing are textiles. Capital-intensive manufacturing are processed metals, tools, and ship building materials. 2% of traded value is in unclassified goods, 99% of which are generic goods and mercery. Thus, 98% of traded value is assigned.

Table A.2 ranks the 30 most common goods traded between 1668 and 1750, which together make up 90% of the value of trade.

[60]The prompt was: "In this folder, you find the file "list from raw unique.dta". It contains a list of raw trade records, contained in the variable "soort", the value of the transaction ("totaldaler") and my manually cleaned commodity classificaton ("comm" and "good id"). Disregarding this manual classification, please make a new classification from scratch."

Table A.2: Top 30 goods by value

Good	Sector	Share of traded value	Cumulative share of traded value	Rank
Hemp	Labour-int. Agriculture	11.25	11.25	1
Rye	Labour-int. Agriculture	10.57	21.82	2
Salt	Capital-int. Agriculture	9.75	31.57	3
Wine	Capital-int. Agriculture	8.84	40.42	4
Wheat	Labour-int. Agriculture	8.63	49.05	5
Flax	Labour-int. Agriculture	6.1	55.15	6
Iron Works	Capital-int. Manufacturing	4.31	59.46	7
Iron	Capital-int. Agriculture	3.9	63.36	8
Tobacco	Labour-int. Agriculture	3.68	67.04	9
Planks	Capital-int. Manufacturing	3.22	70.27	10
Leather	Capital-int. Agriculture	2.81	73.07	11
Sugar	Labour-int. Agriculture	2.05	75.13	12
Unclassified	Unclassified	1.32	76.44	13
Ash	Capital-int. Manufacturing	1.22	77.67	14
Goods	Unclassified	1.22	78.88	15
Linnen	Labour-int. Manufacturing	1.14	80.03	16
Cloth	Labour-int. Manufacturing	1.12	81.14	17
Herring	Capital-int. Agriculture	1	82.14	18
Wax	Capital-int. Agriculture	.97	83.12	19
Brandy	Capital-int. Agriculture	.97	84.09	20
Barley	Labour-int. Agriculture	.8	84.89	21
Tow	Capital-int. Agriculture	.69	85.58	22
Wool	Capital-int. Agriculture	.68	86.26	23
Dye	Capital-int. Agriculture	.63	86.88	24
Tools	Capital-int. Manufacturing	.61	87.49	25
Silk	Labour-int. Manufacturing	.58	88.07	26
Wood	Labour-int. Agriculture	.56	88.62	27
Pepper	Labour-int. Agriculture	.55	89.17	28
Cotton	Labour-int. Agriculture	.48	89.65	29
Train Oil	Capital-int. Agriculture	.45	90.1	30

Notes: This table shows the top 30 goods by value of trade between 1668 and 1750.

A.2 Currencies, Units, and Duties

Foreign currency is a non-issue in this data set as virtually all trade is recorded in Danish coins. From 1668 onwards, 48.8% of coin transactions are recorded in Daler and 51.2% are in Skilling (0.0104 Daler). Out of 4.38 million transactions, 347 use the Ort (0.25 Daler) and one uses the Rosenobel (3.94 Daler based on gold content). Thus, only 0.008% of transactions are in non-Danish currency. Table A.3 lists the conversion of historical weight units into kilograms.

Table A.3: Conversion of units of account to kilograms

Unit	Commodity	kg	Unit	Commodity	kg	Unit	Commodity	kg
Boeter	Butter	3.96	Sacker		165	Qtr. Kister	Tea	6.25
Centner		50	Half Sacker		82.5	Pund		0.5
Skippund		159	Tonder	Grain	100	Lispund		8
Qtr. Skippund		39.75	Tonder	Butter	112	Last		1905
Drompt		35.6	Half Tonder	Grain	50	Last	Hemp, Hops	952
Faad		950	Half Tonder	Butter	56	Skaeppe	Grain	12.5
Half Faad		475	Kister	Tea	25	Steen		15
Qtr. Faad		237.5	Half Kister	Tea	12.5	Woger		17.856

For every passage, the type (fixed and proportional) and duty amount is recorded. As an example, take passage 304: this ship paid 4 Daler of fire money ('fyrpenge', a fixed fee for lighthouse maintenance). Its captain received a captain's allowance of 1.5 Daler, which was deducted from the total duty payable. This is a proportional tax free allowance. The relevant value is the proportional duty of 37.5 Daler: when subtracting the allowance and adding the fixed fee, the payable duty was 40 Daler; 37.5 Daler are the relevant proportional duty.

B History and Geography Appendix

Only ships passing the Oresund, between Sweden and Denmark's main island Zealand, appear in the data; trade between ports on either side of the Sound is unobserved.[61] Virtually all trade between the Baltic and North Sea areas are covered. Other straits, such as the Little Belt between Jutland and Funen, were difficult and dangerous to navigate for the large ships used for trading with the West. The main alternative to passing the Sound was sailing through the Great Belt, between Zealand and Funen. Degn, 2017 makes a number of arguments against this being an issue: there was a prohibition on sailing through the Belt; local sailing knowledge was required to pass through; and toll records from the Great Belt show only a small number of passages. The toll revenue from the Great Belt and the Little Belt combined equalled but a few percent of the Sound Toll revenues (Gøbel, 2010). Alternatively, trade over land was slow and expensive and faced a high number of tolls.

My source for identifying cities in the trade data are two files provided by the Soundtoll team, which convert 90,737 raw port names to 3,085 unified port names. My sample are coastal cities (less than 25 kilometres distance to the coast, or those along a major river up to 100 kilometres from the coast) between Ireland to the West, Saint Petersburg to the East, the Arctic Circle to the North, and Bayonne to the South.

Figure A.2b shows digitised army marching routes. Table A.4 shows information on plague outbreaks in my sample. As armies spread the plague across Europe, regions' proximity to army routes dated before the local outbreak is used in Table A.5 to predict mortality rates when this information is not known.

Table A.6 shows that plagued cities did not grow more or less than other cities (Buringh, 2021). As the closest data point after the plague is in 1750, I conclude that cities' populations had recovered to the trend after four decades. For regional rather than urban populations, I draw on Swedish parish-level population data (Andersson Palm, 2025). Both rural and urban populations have recovered by 1750.

B.1 Serfdom

When encoding serfdom, I follow Raster, 2023 rather than Peters, 2022 in coding Estonia, Livonia, and Pomerania as featuring serfdom. Turning to Denmark's reintroduction of serfdom, I use data by Gary et al., 2022 to show that there was no wage gap between unskilled urban and rural workers before serfdom returned in Denmark in 1733, consistent with free movement. After 1733, a wage gap appears, with scarcer urban labour compensated higher. Figure A.1 compares nominal day wages in Danish skilling for farmhands and unskilled workers in Copenhagen, specifically the ratio $\frac{wage_{urban} - wage_{farm}}{wage_{farm}}$. There is a post-serfdom urban wage premium indicating labour immobility.

[61] I do not impose zero trade for these pairs but omit them from the panel.

Table A.4: List of plagued cities and mortality estimates

City	Modern Country	Plague Time	Population	Mortality	Source
Helsingør	Denmark	1710 - 1711	4000	.40775	Frandsen, 2009
København/Copenhagen	Denmark	1711	60000	.28333333	Kroll, 2006, Kroll and Grabinsky, 2007
Kuressaare	Estonia	1710			Frandsen, 2009
Narva	Estonia	1710 - 1711	3000		Kroll, 2006, Kroll and Grabinsky, 2007
Pärnu	Estonia	1710	2350	.52916667	Kroll, 2006, Kroll and Grabinsky, 2007
Saaremaa	Estonia	1710			Frandsen, 2009
Tallinn	Estonia	1710	9900	.66057673	Kroll, 2006, Kroll and Grabinsky, 2007
Borgå	Finland	1710			Vourinen, 2007
Helsinki	Finland	1710		.66	Engström, 1994
Kokkola	Finland	1710 - 1711			Vourinen, 2007
Turku	Finland	1710 - 1711	6000	.33	Vourinen, 2007
Uusikaupunki	Finland	1710 - 1711			Vourinen, 2007
Anklam	Germany	1710			Wieden, 1999
Bremen	Germany	1712 - 1713	28000	.00685714	Frandsen, 2009
Flensburg	Germany	1710			Ulbricht, 2004
Glückstadt	Germany	1712			Frandsen, 2009
Greifswald	Germany	1711			Wieden, 1999
Hamburg	Germany	1712 - 1714	70000	.13035714	Kroll, 2006, Kroll and Grabinsky, 2007
Altona	Germany	1712			Frandsen, 2009, Winkle, 1983
Itzehoe	Germany	1712			Frandsen, 2009
Kiel	Germany	1712			Frandsen, 2009
Rendsburg	Germany	1712			Ulbricht, 2004
Stade	Germany	1712			Frandsen, 2009, Winkle, 1983
Stralsund	Germany	1710 - 1711	7250	.31416667	Kroll, 2006, Kroll and Grabinsky, 2007
Wolgast	Germany	1710 - 1711		.4	Wieden, 1999
Riga	Latvia	1710 - 1711	10477	.65139885	Kroll, 2006, Kroll and Grabinsky, 2007
Klaipeda	Lithuania	1709 - 1710			Kroll, 2006, Kroll and Grabinsky, 2007
Elblag	Poland	1709 - 1710		.3	Frandsen, 2009
Gdansk	Poland	1709	50000	.53283	Kroll, 2006, Kroll and Grabinsky, 2007
Kamien Pomorski	Poland	1710			Wieden, 1999
Szczecin	Poland	1709 - 1711	11250	.17119565	Kroll, 2006, Kroll and Grabinsky, 2007
Kaliningrad	Russia	1709 - 1710	36250	.23916667	Kroll, 2006, Kroll and Grabinsky, 2007
Karlshamn	Sweden	1710 - 1712			Persson, 2011
Karlskrona	Sweden	1710 - 1712			Persson, 2011
Malmö	Sweden	1712	5000	.35	Kroll, 2006, Kroll and Grabinsky, 2007
Stockholm	Sweden	1710 - 1711	53750	.3777381	Kroll, 2006, Kroll and Grabinsky, 2007
Visby	Sweden	1710 - 1711	2375	.23294444	Kroll, 2006, Kroll and Grabinsky, 2007
Ystad	Sweden	1712	1950	.38461538	Kroll, 2006, Kroll and Grabinsky, 2007

Notes: Sources are indicated in the last column. Population and mortality estimates, in the case of several available sources, denote the mean estimate. For Anklam, Flensburg, and Kamien Pomorski I set the onset to 1710. The modern names and countries of cities are used. Seven cities are both besieged and plagued: Altona, Copenhagen, Narva, Riga, Stettin (Szczecin), Stralsund, and Wolgast. For Copenhagen, Narva, and the first siege of Riga, there was no association with the plague, as these sieges occurred at the beginning of the war in 1700 and 1704 before the plague spread.

Table A.5: Determinants of urban mortality

	Mortality rate
	(1)
Year of plague outbreak	-0.001
	(0.001)
Latitude	0.016
	(0.021)
Longitude	0.014*
	(0.008)
Distance to closest army	-0.052**
	(0.018)
East of the Sound	0.181***
	(0.048)
Observations	19

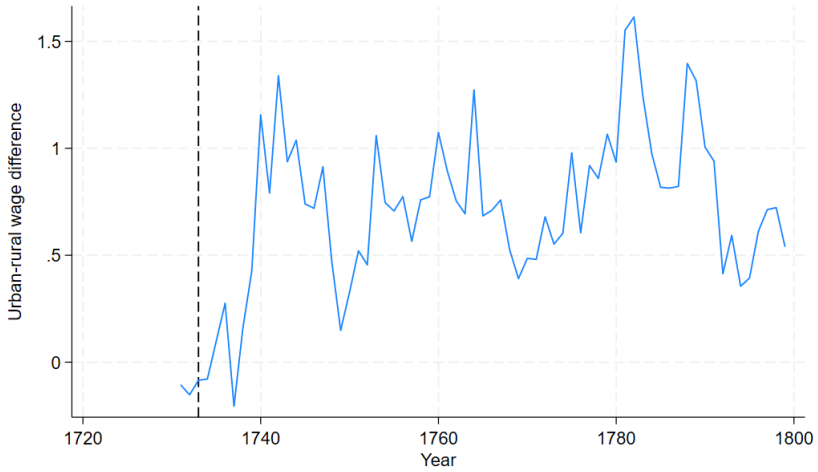
Notes: Standard errors in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable is the median urban mortality estimate. The independent variables are the year of the plague outbreak, latitude, longitude, distance to the closest army route between 1706 and 1714 dated before the local outbreak, and a dummy for the city being to the East of the Sound. The rationale for including army marching routes is that armies spread the plague.

Table A.6: Impact of plague on urban and regional population.

	Urban populations		Regional populations	
	(1)	(2)	(3)	(4)
	Population	Log Population	Population	Log Population
Plague x Year=1500	-4.220*	-0.128		
	(2.179)	(0.112)		
Plague x Year=1550	-3.302*	-0.0670		
	(1.900)	(0.0960)		
Plague x Year=1600	-1.618	-0.00822		
	(1.329)	(0.0907)		
Plague x Year=1650	-1.127	0.0276		
	(0.811)	(0.0426)		
Plague x Year=1750	0.569	-0.0804	25.51	0.0337***
	(1.125)	(0.0585)	(28.84)	(0.0125)
Plague x Year=1800	2.160	-0.0671		
	(2.129)	(0.0846)		
Plague x Year=1810			-28.01	-0.0324**
			(59.52)	(0.0151)
Plague x Year=1850	3.732	-0.0761		
	(4.370)	(0.100)		
Fixed Effects:				
– City	✓	✓		
– Parish			✓	✓
– Year	✓	✓	✓	✓
Observations	8,722	8,722	7,032	7,032

Notes: Standard errors, clustered at the city level (columns 1-2) and the parish level (columns 3-4), are reported in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The table brings together two different sources. Columns 1-2 use city level urban populations from Buringh (2021), restricted to the Baltic and North Sea areas, specifically to cities between the tip of Norway, the northern end of the Alps, the Irish west coast and Arkhangelsk, and to the 1500-1850 period. Columns 3-4 use Swedish parish data from Andersson Palm (2025), observed in 1690, 1750 and 1810. The dependent variable is the (log) population of the respective unit of observation, a city in columns 1-2 and a Swedish parish in columns 3-4. The independent variables are a plague dummy interacted with year dummies, with 1700 the omitted reference in columns 1-2 and 1690 the omitted reference in columns 3-4. The two sources are observed on different year grids, so the row for 1810 is identified by the Swedish data alone and all remaining rows other than 1750 by the urban data alone.

Figure A.1: Urban-rural wage differential in Copenhagen



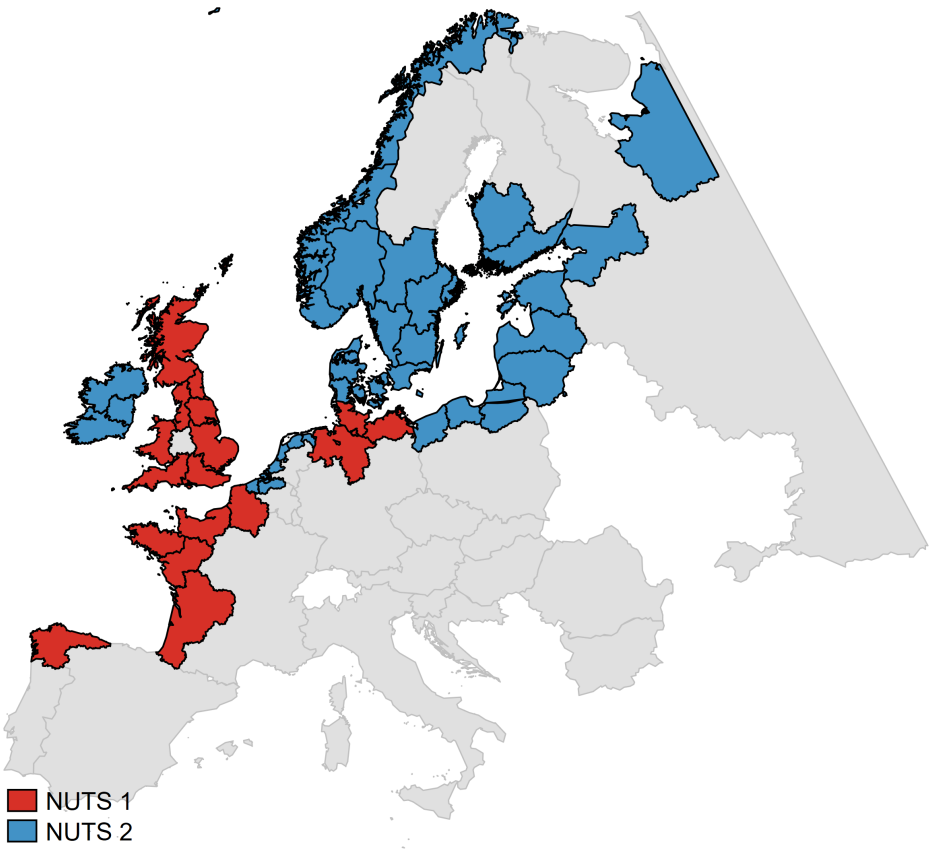
Notes: Difference between urban and farm day wages divided by farm day wages. Data by Gary et al., 2022. The dotted line signifies the 1733 re-introduction of serfdom.

B.2 Classification of Areas

Areas are based on European NUTS regions but do not correspond to one level everywhere as their sizes vary greatly and as Russia and Norway are not EU members. NUTS 1 is too aggregate for Northern Europe, with Finland and Denmark as each only one region. NUTS 2 is too granular in Western Europe, prompting a sixth of areas to only contain one port city. At NUTS 3, a third of all areas contain only one port. Instead, I construct similarly-sized areas: I use NUTS 2 (or a close equivalent) throughout, other than for the United Kingdom, France, Germany, and Spain, where I use NUTS 1. When aggregating areas below, this is to ensure that no area has only one port. Figure A.8 shows that the export expansion result goes through also when instead including NUTS 1-time and NUTS 2-time fixed effects.

Figure A.2: Areas and army marching routes

(a) Areas



(b) Army marching routes, 1706-1714



Notes: Figure A.2a shows the 56 areas into which regions are aggregated. The choice of NUTS 1 or NUTS 2 areas avoids areas with an individual port city and generates areas of comparable sizes. Figure A.2b is a map of army marching routes between 1706 and 1714, when the plague was recorded to have spread across Eastern and later Northern and Central Europe. No distinction is made between armies, as both allied and Russian troops are known to have spread the plague. I base these routes on maps by Spruner and Menke, 1880 and Barraclough, 1997.

C Identification Appendix

C.1 Conditional parallel trends (A1)

I argue that the plague hit regions for reasons uncorrelated with their pre-existing trade or productivity trajectories to support assumption (A1). I show that to-be-plagued cities do not differ in population growth before the plague (Table A.6). In Appendix Figure A.14, I also show that there are no significant pre-trends in the model-recovered competitiveness terms (and thus in comparative advantage). The absence of pre-trends on trade is shown in Figures 2 and 3. While the propensity-score design (Table A.7) shows that to-be-plagued regions have higher trade volumes in levels, these level differences never enter identification, as fixed effects absorb them in every specification. They are also not a threat to the identifying assumptions since identification does not require similar levels of trade but the absence of differential trends, which I confirm. Further, the propensity-score exercise below addresses the observable component of exposure, including level differences, directly: controlling for, reweighting on, and trimming by the score leaves the estimates unchanged (Table A.8). The absence of pre-trends in event studies also directly speaks to assumption (A2). Taken together, these findings support the conditional parallel trends assumption and the no anticipation assumption.

Additionally, I regress the plague dummy on pre-determined covariates in Table A.7: plague outbreaks are significantly likelier close to armies. To ensure that these covariates are strictly pre-determined, I measure a plagued region's army proximity as the distance to the nearest army movement of 1706-1714 dated before its outbreak (for never-plagued regions, the nearest movement of 1706-1714), using digitised annual campaign maps. 1706-1714 is the period when the plague was present in Northern Europe. I code a siege indicator that only includes sieges before the local outbreak. Determinants varying by area and year, such as harsh winters and famines that may increase plague vulnerability, are absorbed by area-year fixed effects in the next step. I use the predicted plague probabilities as a control interacted with a post plague dummy. This allows me to control for the observable component of non-random plague exposure. Further, I apply inverse-propensity weighting to recover ATE and ATT (following Abadie, 2005). This identifies the ATT under conditional parallel trends (assumption (A1)) by reweighting outcome differences on the estimated propensity score. I then re-estimate the baseline specification on the subsample of regions whose propensity score exceeds the 20th, 40th, 60th and 80th percentile (in the spirit of Crump et al., 2009). Throughout, my empirical findings are barely affected (Table A.8) and in all specifications other than for the ATE, the point estimates are statistically indistinguishable from the baseline estimates. Thus, even when controlling for the observable portion of non-random exposure to the plague in numerous ways, the results go through. This suggests that selection into treatment does not pose a threat and provides support for assumption (A1).

Table A.7: Predicting plague exposure from pre-determined covariates

	Plague
Log army distance	-0.228***
	(0.061)
Land suitability	-0.546
	(0.388)
Pre-plague trade decile	0.200***
	(0.034)
Pre-plague siege	-0.047
	(0.451)
Log distance to river	-0.199***
	(0.064)
Pre-plague cap.-int. share	-0.512*
	(0.284)
Model:	Probit
Observations	682
Pseudo R^2	0.330

Notes: Robust standard errors in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable is an indicator equal to one for regions struck by the plague. All covariates are pre-determined: log distance to the nearest army route dated before the local plague outbreak, land suitability, pre-plague trade volume deciles, an indicator for pre-plague sieges, log distance to the nearest river, and the pre-plague share of capital-intensive exports. The predicted probabilities from this Probit regression serve as the propensity score used in the subsequent tables. For precision, the score is estimated on all 682 ports of the sample frame, of which 82 ports are without recorded pre-plague trade and never enter the outcome panels. Estimating on the 600 panel ports leaves the score essentially unchanged (correlation 0.9997).

Table A.8: Plague effect on exports: capital-intensity shift and market-share expansion, conditional on observable exposure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Fact #1 – Shift into Capital-Intensive Exports								
Post Plague	0.355***	0.530***	0.259**	0.378***	0.355***	0.355***	0.361***	0.415***
	(0.075)	(0.132)	(0.114)	(0.076)	(0.075)	(0.075)	(0.075)	(0.074)
Post Plague × Capital-Int.	0.817***	0.778***	1.181***	0.827***	0.824***	0.825***	0.823***	0.599***
	(0.254)	(0.289)	(0.402)	(0.261)	(0.254)	(0.254)	(0.253)	(0.114)
× (PS − 0.287)		-0.624*						
		(0.378)						
× Capital-Int. × (PS − 0.287)		0.160						
		(0.855)						
p-value: Capital-Int. shift equal to column (1)	–	0.739	0.036	0.883	0.238	0.225	0.514	0.317
Specification:	Baseline	+ PS heterogeneity	IPTW (ATE)	IPTW (ATT)	Pr>0.003	Pr>0.008	Pr>0.025	Pr>0.086
– Origin x Sector	✓	✓	✓	✓	✓	✓	✓	✓
– Sector x Year	✓	✓	✓	✓	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓	✓	✓	✓	✓
Observations	52,463	52,463	52,463	52,463	47,084	43,344	36,074	21,433
Panel B: Fact #2 – Market Share Expansion								
Post Plague	0.014***	0.013***	0.009***	0.014***	0.014***	0.015***	0.015***	0.018***
	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
× (PS − 0.287)		0.018						
		(0.011)						
p-value: Post Plague equal to column (1)	–	0.288	0.030	0.702	0.397	0.367	0.071	0.056
Specification:	Baseline	+ PS heterogeneity	IPTW (ATE)	IPTW (ATT)	Pr>0.003	Pr>0.008	Pr>0.025	Pr>0.086
– Origin x Destination	✓	✓	✓	✓	✓	✓	✓	✓
– Destination x Year	✓	✓	✓	✓	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓	✓	✓	✓	✓
Observations	361,059	361,059	361,059	361,059	338,123	313,296	277,611	188,185

Notes: Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. *Panel A:* The dependent variable is exports at the region-sector-year level, estimated by PPML. *Panel B:* The dependent variable is the export share of an origin within its trade partner's total imports for a given year, estimated by OLS. Columns 1-4 use the full sample. Column (1) is the baseline. Column (2) interacts the treatment terms with the propensity score, centred at its mean among plagued regions (0.287), so that the Post Plague rows report the effect at the typical plagued region and the rows marked × (PS − 0.287) report how the effect changes per unit of the score. Columns (3)-(4) apply inverse-propensity weighting (ATE, ATT). Columns 5-8 re-estimate the baseline specification on the subsample of regions whose propensity score exceeds the 20th, 40th, 60th and 80th percentile of the score across origins in the Panel A sample (Pr>0.003, 0.008, 0.025, 0.086). The *p*-value row reports a Wald test that the coefficient of interest (Post Plague × Capital-Int. in Panel A, Post Plague in Panel B) equals the baseline estimate, from a regression stacking the baseline and the respective specification with all regressors and fixed effects interacted with a specification indicator, clustered at the origin level. The propensity score is estimated for these covariates: logarithmic army distance (before the local plague outbreak), land suitability, pre-plague trade volume deciles, pre-trade share of capital-intensive exports, pre-plague sieges, logarithmic distance to the nearest river.

C.2 No interference (A3)

Interference could threaten (A3) through two distinct channels. The first is geographic contagion: the plague could spread from a treated port to its neighbours, leaving controls partially treated. I test for this using a continuous treatment that augments a region's own plague dummy with the population- and distance-weighted exposure to 106 recorded outbreaks in all other European NUTS3 regions. A NUTS 3 region counts as plagued if it has at least one plagued location. The continuous plague treatment is constructed as

$$plague_{it}^{cont} = plague_{it}^{own} + \Sigma_j \frac{Population_j}{Distance_{ij}} plague_{jt},$$

where i are ports, j are regions in Europe at the NUTS 3 level, $plague_{it}^{own}$ denotes a region's own plague status, and $plague_{jt}$ is a NUTS 3 region's plague status. In the absence of population data at this resolution, I assume a constant population density and proxy with the size of a NUTS 3 region. Distances are computed as straight lines.[62] As a robustness check, I also construct a treatment that weighs all regions the same:

$$plague_{it}^{cont} = plague_{it}^{own} + \Sigma_j \frac{1}{Distance_{ij}} plague_{jt}.$$

In Table A.9, the indirect continuous plague variable is of inconsistent sign and typically insignificant, while the own-plague export expansion remains significant and stable in magnitude. A region's trade responds to its own mortality shock but not to its neighbours', which is the restriction (A3) requires.

The second channel is market reallocation: even absent contagion, a treated region's contraction could reshuffle trade and move control outcomes. Four features of the design bound this. First, the effect on export volumes (Appendix Figure A.7a) mirrors the market share result. Volume changes, unlike shares which sum to one within a destination, are not zero-sum across exporters, and mechanical reallocation should not drive the volume response that I observe. Second, the patterns are not an artifact of observing only trade passing through the Sound: in Hamburg, whose exports I observe across all destinations (Table A.10), the same post-plague export expansion and shift into capital-intensive goods appears. Third, simulations in both a full-route sample and the censored Sound-passing sample (Figure A.11) return point estimates centred on the true effect in both. Finally, ports whose hinterlands are tied to them by low-cost upriver transport, where switching export harbours is most costly, show no differential post-plague export response (Appendix Table A.11), directly contradicting a rerouting channel from hinterlands to different export harbours after the plague that might violate assumption (A3). This supports that the sample restriction and potentially unobserved market reallocation introduce no bias.

[62] When a region, not a town, is mentioned in the source, I use the coordinates of the historical capital.

Table A.9: Impact of Plague on Trade, Accounting for Outbreaks in the Hinterland

	Sectoral Exports (Fact #1)			Market Shares (Fact #2)		
	(1)	(2)	(3)	(4)	(5)	(6)
Own Plague	0.27652***	0.27027***	0.23651***	0.01415***	0.01450***	0.01496***
	(0.05493)	(0.08982)	(0.07030)	(0.00350)	(0.00364)	(0.00377)
	[0.06708]	[0.08355]	[0.08453]	[0.00395]	[0.00400]	[0.00407]
	{0.09479}	{0.08586}	{0.08739}	{0.00404}	{0.00401}	{0.00402}
	⟨0.08652⟩	⟨0.09124⟩	⟨0.08184⟩	⟨0.00412⟩	⟨0.00409⟩	⟨0.00411⟩
Plague in Hinterland		-0.00038	1.05197		-0.00000	-0.00453*
		(0.00035)	(0.81479)		(0.00000)	(0.00244)
		[0.00025]	[0.90777]		[0.00000]	[0.00205]
		{0.00051}	{0.93657}		{0.00000}	{0.00250}
		⟨0.00049⟩	⟨0.87753⟩		⟨0.00000⟩	⟨0.00255⟩
Own Plague x Capital-Int.	0.76797***	1.28043***	0.99534***			
	(0.16733)	(0.42261)	(0.20298)			
	[0.16284]	[0.40229]	[0.20296]			
	{0.20666}	{0.20614}	{0.20702}			
	⟨0.20912⟩	⟨0.20864⟩	⟨0.20995⟩			
Plague in Hinterland x Capital-Int.		-0.00085	-1.33932**			
		(0.00066)	(0.51913)			
		[0.00061]	[0.46912]			
		{0.00067}	{0.52482}			
		⟨0.00067⟩	⟨0.52532⟩			
Hinterland measure	–	Size x Proximity	Proximity	–	Size x Proximity	Proximity
Fixed Effects:						
– Origin x Sector	✓	✓	✓			
– Sector x Year	✓	✓	✓			
– Area x Year	✓	✓	✓	✓	✓	✓
– Origin x Destination				✓	✓	✓
– Destination x Year				✓	✓	✓
Observations	12,524	12,524	12,524	359,839	359,839	359,839

Notes: Difference-in-differences set up dropping the war years of 1700 to 1721. Below each estimate are four kinds of inference. Standard errors clustered at the origin level in parentheses, to which the significance stars refer, * p <0.10, ** p <0.05, *** p <0.01. Standard errors clustered at the area level in brackets: as the constructed hinterland exposure varies smoothly in space, origin-level clustering may overstate the independent variation in the regressor, and clustering at the coarser area level accounts for this, as in Table A.13. In braces and angle brackets, standard errors combining origin clustering with spatial correlation across origins within a year up to a cutoff of 200 km and 500 km, respectively: the origin-cluster variance plus the within-year spatial variance, less the origin-year variance both contain, rescaled to the clustered estimator's degrees-of-freedom correction, as in Appendix Figure A.13. The dependent variable is log sectoral exports in columns 1 to 3 (reproducing Fact #1, as in column 1 of Table A.15) and market shares of i in j in columns 4 to 6 (reproducing Fact #2). Columns 1 to 3 are estimated by OLS on log exports weighted by exports in levels. The area-clustered and spatial standard errors reported here are constructed by Frisch-Waugh-Lovell residualisation and therefore require a linear estimator. Weighting OLS on log trade by levels brings the estimates close to PPML (Mayer, Vicard, and Zignago, 2019, Head and Mayer, 2013), as in equation (2). Columns 1 and 4 estimate the specification without any hinterland terms, so that the own-plague coefficients can be compared directly across columns. Columns 2 and 5 add a continuous plague variable accounting for the size and proximity of other Europe-wide plagued regions, columns 3 and 6 a second such variable accounting only for their proximity, in columns 2 and 3 also interacted with a capital intensity dummy. Annual growing season temperatures are an additional control in all specifications.

Table A.10: Export expansion and shift into capital-intensive goods in post-plague Hamburg

	Absolute	Relative
	(1)	(2)
Post Plague	0.439***	
	(0.103)	
Post Plague x Capital-Int.		0.554***
		(0.185)
Fixed Effects:		
– Year		✓
– Sector	✓	✓
Observations	304	304

Notes: Standard errors, clustered at the sector level, in parentheses. * p <0.10, ** p <0.05, *** p <0.01. Regressions using Hamburg export data to all destinations between 1702 and 1713, based on Baasch (1929), with the plague in 1712 (see Table A.4). 38 goods have been classified into sectors precisely as in the rest of the paper. Both columns are estimated by PPML and the dependent variable is annual exports by sector in levels. The independent variable is a post plague dummy, equal to one after 1712 when Hamburg suffered a plague outbreak, and in column 2 a dummy interacting the post plague dummy with a dummy for capital-intensive exports.

Table A.11: Plague effects on trade are not driven by port substitution

	Volume	Export Share	Log Volume
	(1)	(2)	(3)
Post Plague	0.685***	0.021***	0.573***
	(0.192)	(0.004)	(0.159)
Post Plague x Distance to River	-0.291	-0.005*	-0.221
	(0.249)	(0.003)	(0.174)
Fixed Effects:			
– Area x Year	✓	✓	✓
– Origin x Destination	✓	✓	✓
– Destination x Year	✓	✓	✓
Estimator	PPML	OLS	OLS
Observations	103,532	361,059	14,714

Notes: Difference-in-differences setup dropping the war years of 1700 to 1721. Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable is annual exports in levels in column 1 (estimated by PPML), the market share an origin captures in a destination in column 2, and logarithmic exports, weighted by exports in levels, in column 3 (both estimated by OLS). The independent variables are a post plague dummy, equal to one after the origin suffered a plague outbreak, and its interaction with the origin's distance to the nearest major river (Elbe, Oder, Vistula, Neman, and Daugava). A plague effect operating through port substitution would load on the plague and river distance interaction, as hinterlands close to major rivers are likely stuck with the main port up the river. Annual growing season temperatures are an additional control in all specifications.

C.3 Exclusion (E)

Finally, I address exclusion concerns. My claim is that the plague dummy captures plague-induced labour scarcity, not some other wartime effect that correlates with plague status. To provide support to this claim, I conduct a placebo test: I find in Table A.12 that proximity to armies and sieges does not predict a shift into a higher capital intensity of exports. Army distance correlates with the level expansion of exports in some specifications, though the sign works in the opposite direction of what would be a threat here. These level effects are further not significant under spatially robust Conley standard errors. While sieges are associated with a significant export expansion in levels in some specifications, I note that the besieged ports are among the largest in the Baltic (e.g. Copenhagen, Riga, Narva, Szczecin, Stralsund). Indeed, sieges select on size by definition, as only fortified cities that are large to begin with have fortresses one can besege, so the siege indicator partly proxies for port size. Decisively for exclusion, neither army proximity nor siege status shifts the composition of exports toward capital-intensive goods. Siege status is also incorporated in the propensity score design (as a pre-plague siege indicator, restricted to sieges dated before the local outbreak) (Table A.7) and neither significantly predicts plague outbreaks nor changes any of the point estimates (Table A.8).

Relately, note that the propensity score exercise finds that plague outbreaks were likelier near armies: the propensity score thus picks up precisely the main confounder behind the exclusion restriction yet does not change my estimates. Another exclusion concern arises from potential plague-related policies. As argued above, this only matters if such policies have region-idiosyncratic effects within areas.

A third exclusion concern is that the 1710s plague may correlate with the geography of later wars (the War of the Polish Succession (1733-38) and the Russo-Swedish War of 1741-43), subjecting the same regions to renewed wartime disruption. To test this, I geocode data on all battles and sieges of these wars and regress each region's log distance to the nearest engagement and the number of engagements within 100 km on the plague dummy, unadjusted, controlling for the propensity score, applying inverse-propensity ATT weights, and on the subsample of regions with similar propensity scores, mirroring the design of Table A.8. Because battle exposure is spatially smooth by construction, I report standard errors clustered at the area level alongside Conley, 1999 standard errors with a 200 km cutoff. I argue that later wars do not threaten identification as exposure to battles and sieges in these wars is balanced across plague status conditional on the propensity score (Table A.13), so fighting did not differentially return to plagued regions. I also note that the event studies show effects within a few years of the plague, suggesting that my findings are not driven by wars twenty or thirty years later.

Finally, I address the Danish blockade during the Great Northern War. Figure A.3 shows how blockades and naval raids affected the number of ships passing through the Sound. Table A.14 shows that proximity to the naval base from which the Danish blockade was enforced predicts significantly lower exports during the blockade years, beyond blockades and raids, their plague and factor-intensity interactions, and latitude and longitude interacted with the war period. This proximity effect is symmetric across plague status and sectors' factor intensity. The blockade and raid dummies show some asymmetries, but these are identified from very few ports (the raid interactions with plague status are identified from Stockholm and Visby only). Their compositional interaction is negative, so wartime naval disruption pushed plagued ports out of capital-intensive exports, which clearly cannot produce Fact # 1. Thus, exclusion concerns are not a worry in this setting.

Table A.12: Shift into capital-intensive exports by army proximity and siege.

	PPML		OLS		PPML		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Army Distance x Post War	0.825***	0.820***	0.400	0.379				
	(0.235)	(0.232)	(0.261)	(0.258)				
			[0.220]	[0.221]				
Army Distance x Post War x Capital-Int.	-0.031	-0.009	-0.025	-0.007				
	(0.030)	(0.018)	(0.038)	(0.019)				
			[0.032]	[0.017]				
Post Plague	0.462***	0.430***	0.429***	0.395***	0.164	0.157	0.285***	0.275***
	(0.089)	(0.092)	(0.077)	(0.073)	(0.100)	(0.096)	(0.085)	(0.080)
			[0.123]	[0.119]			[0.131]	[0.127]
Post Plague x Capital-Int.	0.597**	0.860***	0.511**	0.770***	0.632***	0.701***	0.621***	0.757***
	(0.277)	(0.203)	(0.237)	(0.130)	(0.210)	(0.104)	(0.180)	(0.103)
			[0.225]	[0.151]			[0.193]	[0.139]
Siege x Post War					0.458**	0.454**	0.221	0.194
					(0.192)	(0.191)	(0.193)	(0.201)
							[0.212]	[0.223]
Siege x Post War x Capital-Int.					0.497	0.581	0.014	0.052
					(0.376)	(0.388)	(0.326)	(0.362)
							[0.320]	[0.347]
Fixed Effects:								
– Origin x Sector	✓	✓	✓	✓	✓	✓	✓	✓
– Sector x Year	✓		✓		✓		✓	
– Area x Year	✓	✓	✓	✓	✓	✓	✓	✓
Estimator	PPML	PPML	OLS	OLS	PPML	PPML	OLS	OLS
Observations	77,407	77,822	16,933	16,959	79,230	79,664	17,124	17,149

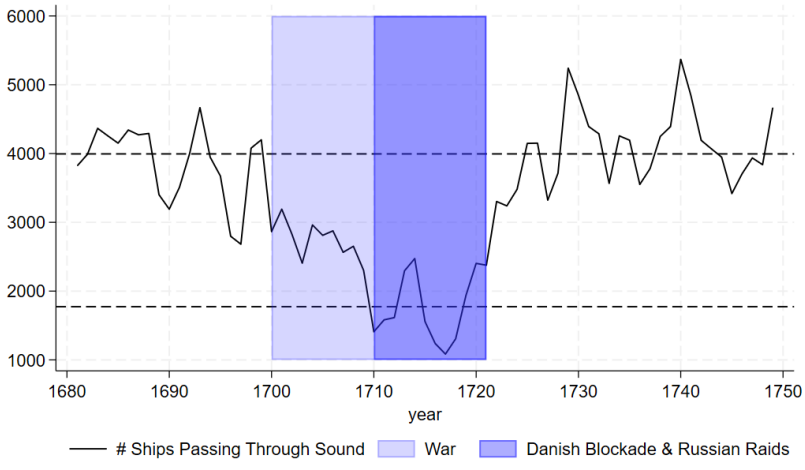
Notes: Standard errors clustered at the origin level in parentheses. Conley standard errors (200 km spatial cutoff, Bartlett serial kernel over the full panel length, computed on the FWL-residualised weighted fit) in brackets, reported for the OLS columns where the linear structure permits them. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable is annual exports by sector in levels (PPML) or logs (OLS). The independent variable is a post plague dummy, equal to one after the origin suffered a plague outbreak, and a dummy interacting the post plague dummy with a dummy for capital-intensive exports. Additionally, in columns 1-4 I include the smallest distance any army had to a region during the Great Northern War, interacted with a post-war dummy (equal to 1 after 1721), and this interaction's interaction with a capital-intensity dummy. The PPML columns show unweighted results, whereas in the OLS columns the weight are exports in levels. Annual growing season temperatures are an additional control in all specifications. There were seven besieged and plagued cities (Altona, Copenhagen, Riga, Narva, Szczecin, Stralsund, Wolgast). Figure A.2b shows army routes.

Table A.13: Balance of later-war battle exposure across plague status

	(1) Raw b/se/conley	(2) P-score control b/se/conley	(3) IPW-ATE b/se/conley	(4) IPW-ATT b/se/conley	(5) Pr>0.002 b/se/conley	(6) Pr>0.007 b/se/conley	(7) Pr>0.020 b/se/conley	(8) Pr>0.074 b/se/conley
Log dist. nearest battle, Polish Succession	-0.113	-0.075	-0.029	-0.115	-0.113	-0.125	-0.146	-0.137
	(0.103)	(0.090)	(0.080)	(0.140)	(0.104)	(0.104)	(0.119)	(0.161)
	[0.093]	[0.070]	[0.062]	[0.105]	[0.093]	[0.094]	[0.110]	[0.142]
Log dist. nearest battle, Russo-Swedish 1741-43	-0.102	-0.081	-0.021	-0.035	-0.102	-0.113	-0.141	-0.136
	(0.098)	(0.102)	(0.021)	(0.030)	(0.098)	(0.106)	(0.138)	(0.112)
	[0.098]	[0.099]	[0.018]	[0.030]	[0.098]	[0.105]	[0.123]	[0.126]
Log dist. nearest battle, either war	-0.197	-0.144	-0.041	-0.139	-0.197	-0.215	-0.254	-0.268
	(0.141)	(0.133)	(0.083)	(0.143)	(0.141)	(0.146)	(0.180)	(0.194)
	[0.135]	[0.118]	[0.066]	[0.111]	[0.135]	[0.139]	[0.165]	[0.194]
N battles within 100km, Polish Succession	0.008	0.006	0.004	0.006	0.008	0.005	0.008	0.013
	(0.029)	(0.034)	(0.016)	(0.049)	(0.029)	(0.030)	(0.034)	(0.050)
	[0.013]	[0.014]	[0.007]	[0.018]	[0.013]	[0.010]	[0.015]	[0.023]
N battles within 100km, Russo-Swedish 1741-43	-0.010	-0.024	-0.004	-0.002	-0.010	-0.010	-0.031	0.000
	(0.009)	(0.020)	(0.004)	(0.002)	(0.009)	(0.010)	(0.032)	(0.000)
	[.]	[0.018]	[.]	[0.002]	[.]	[.]	[0.029]	[.]
N battles within 100km, either war	-0.002	-0.018	-0.000	0.004	-0.002	-0.006	-0.023	0.013
	(0.030)	(0.039)	(0.017)	(0.049)	(0.030)	(0.031)	(0.047)	(0.050)
	[0.012]	[0.022]	[0.006]	[0.018]	[0.012]	[0.008]	[0.032]	[0.023]
Observations	4,074	4,074	4,074	4,074	3,234	2,412	1,572	762

Notes: Each cell reports the coefficient on the plague dummy for the battle-exposure measure named in the row, estimated jointly across measures in a stacked regression with area-by-measure fixed effects. Exposure measures cover all battles and sieges of the War of the Polish Succession (1733–38) and the Russo-Swedish War (1741–43). Because battle exposure varies smoothly in space, standard errors are clustered at the area level rather than the origin level used elsewhere. Conley standard errors with a 200 km cutoff in brackets additionally allow correlation across area boundaries. Only ten regions record a battle of the Russo-Swedish War within 100 km. Where the remaining variation after area fixed effects is insufficient, the coefficient is dropped and no Conley standard error is reported. Significance stars refer to the clustered standard errors. * p <0.10, ** p <0.05, *** p <0.01. Column 1 is the unadjusted comparison and column 2 controls for the propensity score. Columns 3 and 4 apply inverse-propensity weights (ATE and ATT). Columns 5-8 restrict to regions whose propensity score exceeds the 20th, 40th, 60th and 80th percentile of the city-level score (Pr>0.002, 0.007, 0.020, 0.074), mirroring the designs of Table A.8. The propensity score is estimated as in Table A.7.

Figure A.3: Number of ships passing the Sound per year



Notes: This Figures displays the number of ships passing the Sound each year. In light blue, the war years of 1700 to 1721 are shaded; in dark blue, the years of 1710 to 1721 are shaded, during which the Danish navy blocked ships between Sweden and its Northern German posessions and the Russian army raided Baltic coastal areas and frequently intercepted ships. During this second half of the conflict, the number of ships passing the Sound collapsed to about a third, reducing observed trade for both plagued and unplagued ports.

Table A.14: Proximity to Christiansø is associated with significantly lower exports during the blockade, symmetrically across plague status and sectors

	Ports	Ports x Capital Intensity
	(1)	(2)
Log Proximity x War	-0.460**	-0.477*
	(0.217)	(0.283)
Log Proximity x War x Plague	-0.080	0.075
	(0.133)	(0.215)
Log Proximity x War x Capital-Int.		-0.464
		(0.359)
Log Proximity x War x Plague x Capital-Int.		0.130
		(0.254)
War x Plague	-0.675	0.175
	(0.688)	(1.174)
War x Plague x Capital-Int.		0.834
		(1.347)
Danish Blockade	-0.232	0.376
	(0.157)	(0.275)
Danish Blockade x Capital-Int.		-0.568**
		(0.248)
Russian Raid	-0.933***	-2.444***
	(0.207)	(0.203)
Russian Raid x Capital-Int.		2.074***
		(0.257)
Russian Raid x Plague	0.728***	2.675***
	(0.228)	(0.195)
Russian Raid x Plague x Capital-Int.		-2.874***
		(0.390)
Fixed Effects:		
– Origin	✓	
– Area x Year	✓	
– Origin x Capital-Int.		✓
– Area x Year x Capital-Int.		✓
Latitude & Longitude x War	✓	✓
Observations	13,416	19,287

Notes: Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. Both columns are estimated by PPML on the sample through 1721. The dependent variable is the number of ships sailing from an origin port, in column 2 separately by capital intensity. Proximity is the negative log of the great-circle distance in kilometres from each city to Christiansø, the naval base from which the Danish blockade was operated. It enters interacted with a war period indicator equal to one for 1709 to 1721, so that it takes larger values for cities closer to the base during the blockade years. The specification is saturated: proximity, the blockade dummy, and the raid dummy enter together with all of their interactions with plague status, and in column 2 with capital intensity. The interaction of the blockade dummy with plague status coincides with a single port (Karlskrona) and is dropped. The raid and plague interactions are identified from two ports (Stockholm and Visby). Blockades and raids are based on Andersen, 2004, Aldridge, 2009, Ericson Wolke, 2012, Christiansø Administration, 2020, and Frost, 2000.

D Empirical Results Appendix

D.1 Fact #1: Capital-intensive exports increase more than labour-intensive exports.

Figure A.4a shows that reassigning four debatable goods (wine, tow, ash, and planks) as labour-intensive produces virtually identical results. Figure A.4b shows findings by factor intensity using the imputation estimator in Borusyak, Jaravel, and Spiess, 2024. Figure A.4c shows that there is no shift into land-intensive exports after the plague. Figure A.4d shows that the shift into capital-intensive exports lasts for almost 80 years. Figure A.4e repeats the event study with the goods classification produced by Claude's Opus 4.6 model (Appendix A.1) and reproduces the baseline pattern. Table A.15 shows PPML results for Fact #1 in a difference-in-difference-in-difference set up, both overall and within each agriculture and manufacturing.

Table A.15: Plague increases capital-intensive exports more than labour-intensive exports, overall and within sectors.

	All sectors		Agriculture	Manufacturing
	(1)	(2)	(3)	(4)
Post Plague	0.355***	0.332***	0.385***	-0.260
	(0.075)	(0.072)	(0.090)	(0.427)
Post Plague x Capital-Int.	0.817***	1.041***	0.575***	1.088***
	(0.254)	(0.227)	(0.127)	(0.217)
Fixed Effects:				
– Origin x Sector	✓	✓	✓	✓
– Sector x Year	✓			
– Area x Year	✓	✓	✓	✓
Observations	52,463	52,882	25,743	13,010

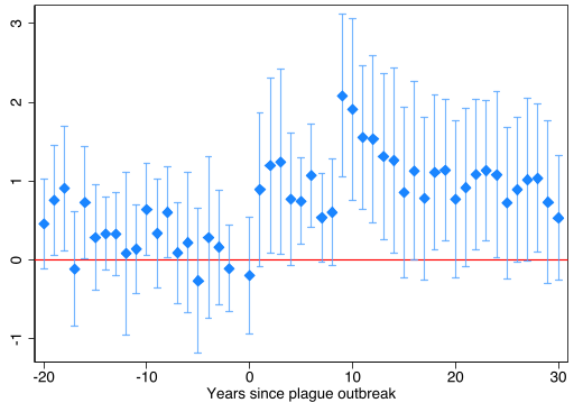
Notes: Difference-in-differences set up dropping the war years of 1700 to 1721. All columns are estimated by PPML with standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable is annual exports in levels. The independent variables are a post plague dummy, equal to one after the origin suffered a plague outbreak, and its interaction with a dummy for capital-intensive exports. Columns 1 and 2 pool all sectors and differ in whether year fixed effects are allowed to vary by sector. Column 3 restricts the sample to capital- and labour-intensive agriculture and column 4 to capital- and labour-intensive manufacturing, so that the interaction compares capital- to labour-intensive exports within each broad sector. Annual growing season temperatures are an additional control in all specifications.

D.1.1 Evidence from Production Data

Figure A.4f shows Scania's overall shift out of arable and into pastoral farming after the plague. Table A.16 shows for individual farms that this shift was stronger the closer a farm was to plagued cities. Table A.18 shows capital deepening in agriculture after the plague as more horses per capita are employed in plagued Swedish parishes. Figure A.5 presents an event study on the number of registered ships, a proxy for capital stocks. Table A.19 shows that labour-scarce farms with rich soils engage in stronger capital deepening. Figure A.6 shows that imports of capital goods increase after the plague.

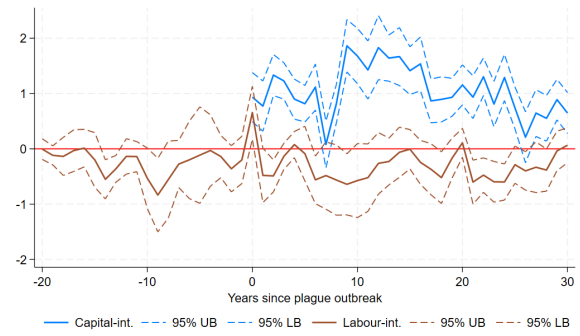
Figure A.4: Robustness of Fact #1 and agricultural production in Scania

(a) Alternative assignment as capital-intensive



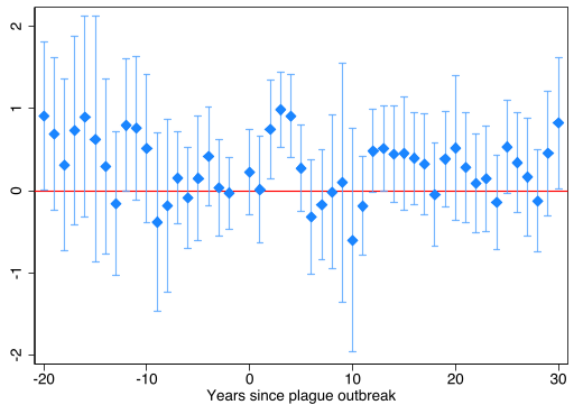
Notes: Among the top 30 goods, I have reassigned four goods (wine and tow from CA to LA, planks and ash from CM to LM) and find almost identical results.

(b) Estimator in Borusyak, Jaravel, and Spiess, 2024



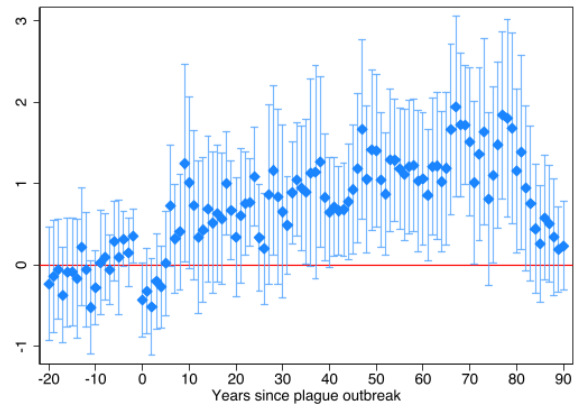
Notes: I estimate equation (2) by factor intensity using the imputation estimator in Borusyak, Jaravel, and Spiess, 2024. These estimates are pooled across both factor intensities.

(c) No shift into land-intensive exports



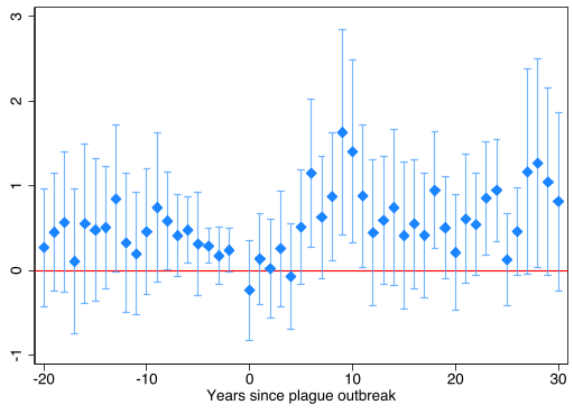
Notes: Distinguishing land-intensive (arable crops, livestock and animal products, plantation goods, raw forestry) from labour-intensive goods (crafted manufactures, iron and metalwork, textiles and clothing). Within only agriculture, the resulting estimates are very similar.

(d) Fact #1 in the long run



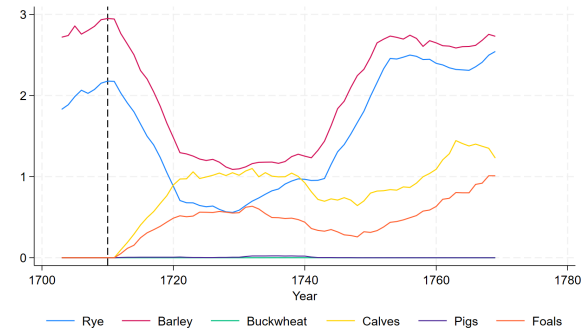
Notes: Estimation of equation (2) in the long run. Standard errors clustered at the origin level.

(e) Classification by Claude



Notes: Estimation of equation (2) with the classification produced by Claude (Appendix A.1). Standard errors clustered at the origin level.

(f) Agricultural production in Scania becomes more capital-intensive



Notes: Figure A.4f covers 119 Scanian farms and the composition of their production.

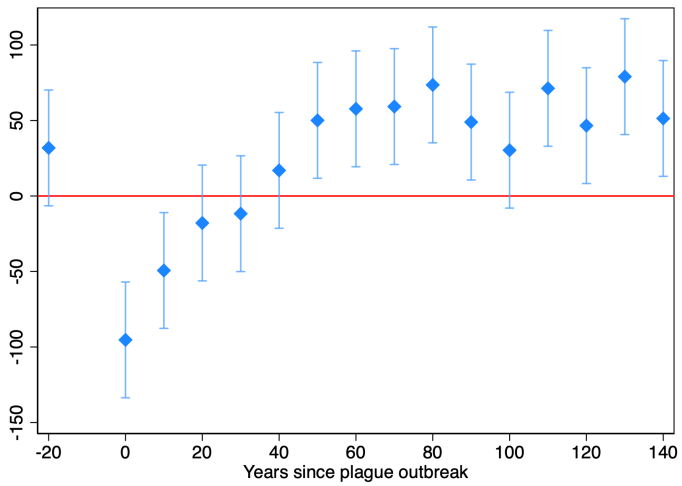
Table A.16: Farm production and distance to plagued cities

	Calves	Pigs	Foals	Rye	Barley	Buckwheat
	(1)	(2)	(3)	(4)	(5)	(6)
Plague Distance x Post Plague	-0.00656**	-0.000395**	-0.00521***	0.0182***	0.0355***	0.000228**
	(0.00281)	(0.000156)	(0.00171)	(0.00138)	(0.00310)	(0.0000965)
	{0.00349}	{0.000187}	{0.00188}	{0.00200}	{0.00304}	{0.000119}
Fixed Effects:						
– Year	✓	✓	✓	✓	✓	✓
– Farm	✓	✓	✓	✓	✓	✓
Observations	5,683	5,683	5,683	5,304	5,304	5,683

Notes: Standard errors clustered at the farm level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variables are farm's production of calves, pigs, foals rye, barley, and buckwheat. The independent variable is the sum of distances multiplied by a plague dummy over the four closest cities in Scania province. Both Ystad and Malmö suffered a plague outbreak in 1712 with a mortality rate of 38% and 35%, respectively, whereas Landskrona and Helsingborg did not suffer plague outbreaks. Conley standard errors in braces (25 km cutoff, spanning every farm pair): spatial HAC across farms within a year plus a Bartlett serial kernel over the full panel length, computed on farm coordinates recovered by least-squares multilateration from the recorded road distances to Ystad, Malmö, Landskrona, and Helsingborg (road-to-straight-line scale 0.70). The farms in the sample all lie within 25 km of each other. Significance stars refer to the farm-clustered standard errors.

Table A.17 estimates equation (3). As agricultural production is land-intensive and acreage correlates strongly with population, I report results with and without land as a factor. In capital-intensive agriculture, the estimates are essentially unchanged, whereas in land-intensive agriculture, precision collapses. The triple interaction nets out level differences in elasticities between plagued and unplagued parishes and every change common to both groups, including aggregate technical change and changes in survey measurements. It is vulnerable to differential input quality changes and to the simultaneity of input choices. I therefore read the estimates as descriptive output elasticities under the Cobb-Douglas technology of Section 6.

Figure A.5: Dynamics of capital: number of ships as proxy for the capital stock



Notes: Event study with area-time, region, and decade fixed effects on the number of ships registered in a harbour in levels. Decade -1 is omitted as the reference decade.

Table A.17: Plague shifts output elasticities from labour to capital, more so in capital-intensive agriculture

Panel A: Output elasticities, triple interactions				
	Labour-int. agriculture (harvest)		Capital-int. agriculture (animal output)	
	(1)	(2)	(3)	(4)
Labour x Post x Plagued	-0.268***	-0.176	-0.633***	-0.640***
	(0.079)	(0.242)	(0.118)	(0.123)
Capital x Post x Plagued	0.146	-0.215	0.448***	0.323**
	(0.133)	(0.359)	(0.168)	(0.148)
Land x Post x Plagued	0.000		-0.168	
	(0.084)		(0.123)	
Land control	✓	–	✓	–
Census years	1690 & 1750	1690 & 1750	1690 & 1750	1690 & 1750
Observations	1,738	1,738	1,738	1,738
Panel B: Factor ratios, long differences 1750−1690				
	Δ log land p.w.	Δ log capital p.w.	Δ log population	Δ log land
Plagued	-0.052	0.127**	-0.013	-0.065
	(0.044)	(0.058)	(0.016)	(0.041)
Parishes	869	869	869	869

Notes: Fully interacted production functions on the parish panel underlying Table 1: log output on log population (labour), log horses (capital), and log sowed acreage (land), each interacted with a post indicator for 1750, a plague indicator, and their product. The two census observations lie sixty years apart, so the triples measure the long-run change in the output elasticity of each factor in plagued relative to unplagued parishes. Standard errors are clustered at the parish level in Panel A, as in Table 1, and robust in Panel B, where the long difference leaves one observation per parish. * p <0.10, ** p <0.05, *** p <0.01. The sample are the 869 parishes with complete data in both years. Panel B regresses parish long differences on the plague indicator.

Table A.18: Parish-level shift into higher capital intensity within agriculture

	Horses	Log Horses
	(1)	(2)
Plague & Year=1750	34.22**	0.232***
	(13.71)	(0.0527)
Plague & Year=1810	51.02***	0.334***
	(9.272)	(0.0524)
Population	0.0858***	
	(0.0135)	
Log Population		0.157***
		(0.0532)
Fixed Effects:		
– Parish	✓	✓
– Year	✓	✓
Observations	7,031	5,695

Notes: Standard errors clustered at the parish level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variables are parish-level logarithmic stocks of horses in 1690, 1750, and 1810. The independent variables are a post-plague dummy interacted with period dummies and parish-level population.

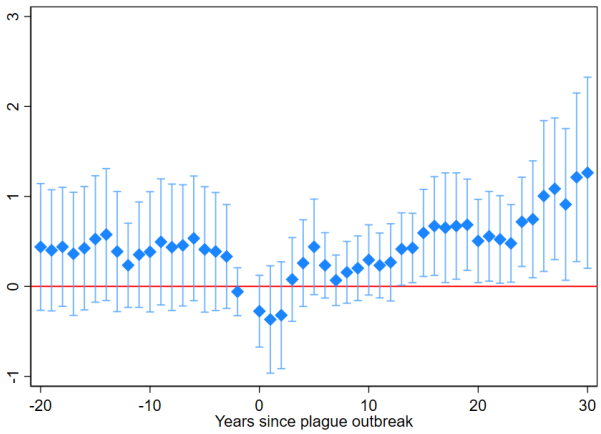
Table A.19: Farm production, distance to plagued cities, and soil quality

	Foals	Rye	Barley	Calves	Pigs	Buckwheat
	(1)	(2)	(3)	(4)	(5)	(6)
Plague Distance x Post Plague	0.00425	-0.0117	0.0405***	-0.00598	0.0000460	-0.000353
	(0.00575)	(0.00894)	(0.0105)	(0.00924)	(0.000181)	(0.000482)
	{0.00611}	{0.00889}	{0.0110}	{0.00928}	{0.000172}	{0.000490}
Soil Quality x Post Plague	0.401**	-1.156***	0.451	0.183	0.0113	-0.00754
	(0.157)	(0.258)	(0.270)	(0.260)	(0.00798)	(0.0130)
	{0.162}	{0.254}	{0.282}	{0.250}	{0.00757}	{0.0133}
Plague Distance x Soil Quality x Post Plague	-0.00284*	0.00913***	-0.00160	-0.000356	-0.000124	0.000155
	(0.00149)	(0.00269)	(0.00293)	(0.00262)	(0.0000845)	(0.000139)
	{0.00156}	{0.00266}	{0.00304}	{0.00250}	{0.0000788}	{0.000148}
Fixed Effects:						
– Year	✓	✓	✓	✓	✓	✓
– Farm	✓	✓	✓	✓	✓	✓
Observations	5,683	5,683	5,683	5,683	5,683	5,683

Notes: Standard errors clustered at the farm level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variables are farm's production of foals, rye, barley, calves, pigs, and buckwheat. The specification interacts a post-plague indicator with (i) plague distance - the sum of distances to the four closest cities in Scania weighted by a plague dummy - (ii) average soil quality - a weighted index over ten soil-quality categories recorded in the land surveys, ranging from 1 (worst) to 10 (best) - and (iii) the product of plague distance and soil quality. Both Ystad and Malmö suffered a plague outbreak in 1712 with a mortality rate of 38% and 35%, respectively, whereas Landskrona and Helsingborg did not suffer plague outbreaks. Conley standard errors in braces (25 km cutoff, spanning every farm pair): spatial HAC across farms within a year plus a Bartlett serial kernel over the full panel length, computed on farm coordinates recovered by least-squares multilateration from the recorded road distances to Ystad, Malmö, Landskrona, and Helsingborg (road-to-straight-line scale 0.70). The farms in the sample all lie within 25 km of each other. Significance stars refer to the farm-clustered standard errors.

Figure A.6: Imports of capital goods increase after the plague, suggesting capital accumulation

(a) Imports of capital goods in levels



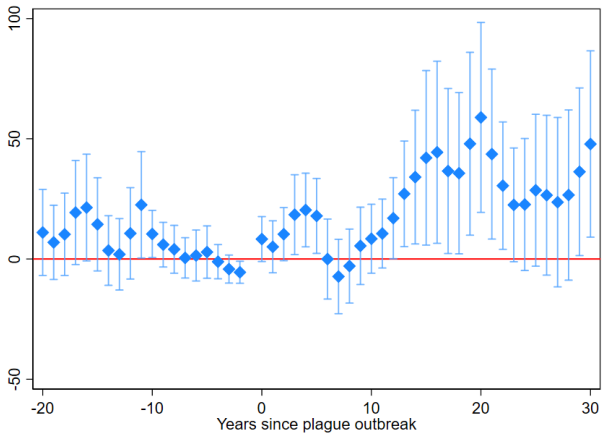
Notes: Capital goods are defined as metals (iron, iron works, steel, lead, copper, tin, sheet metal, brass, brass works), tools (tools, nails), and building materials (planks, tiles, stones, cement, chalk). I regress on imports in levels and include area-time, importer-exporter, and exporter-year fixed effects and control for growing season temperatures. Standard errors are clustered at the importer level.

D.2 Fact #2: Plagued regions capture larger shares of destination markets.

Figure A.7a shows that export volumes expand. Figure A.7b extends the pre-period to 30 years and again finds no pre-trends. Figure A.7c shows the export expansion in the long run. Figure A.8 finds similar results with NUTS 1-time and NUTS 2-time fixed effects instead of area-time fixed effects. Figure A.8c adds lagged temperatures. Figure A.8d repeats the main analysis on trade shares using the imputation estimator by Borusyak, Jaravel, and Spiess, 2024. Table A.20 shows the export expansion by sector, factor intensity, and estimator. Table A.21 presents heterogeneity results by pre-plague export levels.

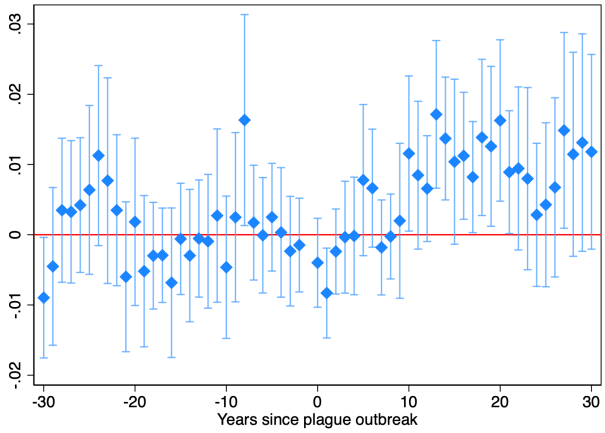
Figure A.7: Robustness of Fact #2

(a) Export volumes



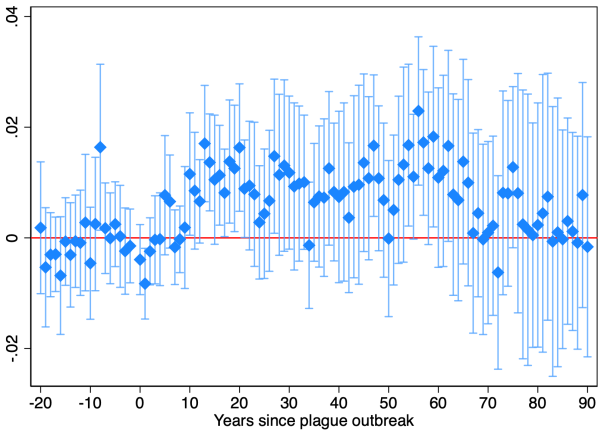
Notes: Estimation of equation (5) on $T_{ijt} = x_{ijt}$. Standard errors clustered at the origin level.

(b) Market shares with 30 pre periods



Notes: Estimation of equation (5) on shares, extending the pre-period to 30 years. Standard errors clustered at the origin level.

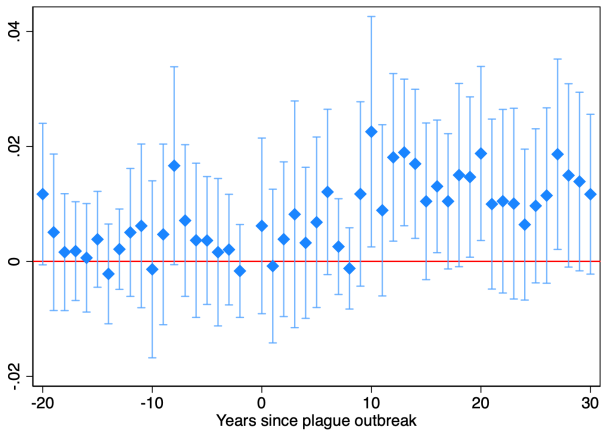
(c) Market shares in the long run



Notes: Estimation of equation (5) on i's share of j's imports, $T_{ijt} = \frac{x_{ijt}}{\Sigma_{i \in I} x_{ijt}}$. Standard errors clustered at the origin level.

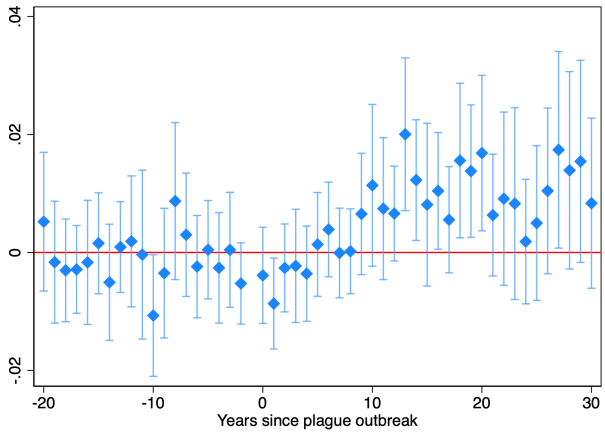
Figure A.8: Robustness results for Fact #2: NUTS 1-time and NUTS 2-time fixed effects

(a) Market shares with NUTS 1-time fixed effects



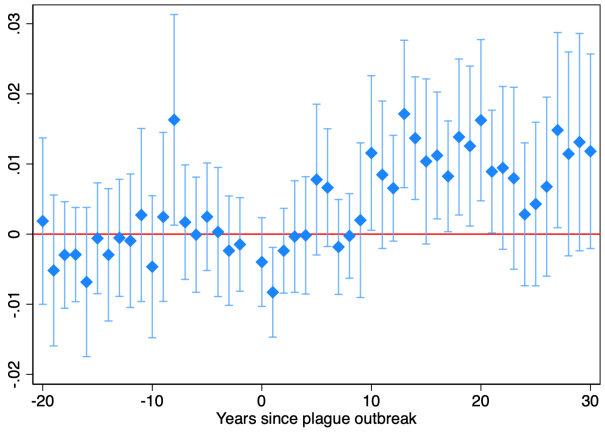
Notes: Estimation of equation (5) on shares. Instead of area-time fixed effects, I use NUTS 1-time fixed effects. Standard errors clustered at the origin level.

(b) Market shares with NUTS 2-time fixed effects



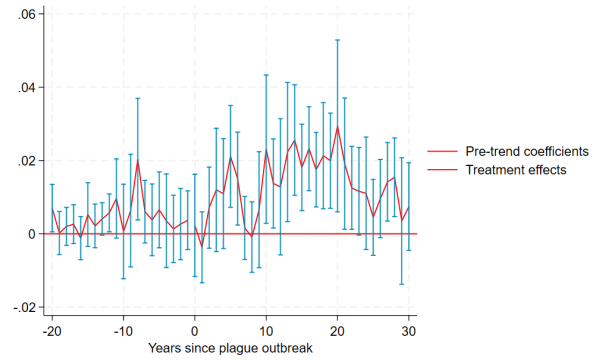
Notes: Estimation of equation (5) on shares. Instead of area-time fixed effects, I use NUTS 2-time fixed effects. Standard errors clustered at the origin level.

(c) Market shares, adding temperature lags of 10 and 20 years



Notes: Estimation of equation (5) on shares, adding growing season temperatures lagged by 10 and 20 years. Standard errors clustered at the origin level.

(d) Market shares, imputation estimator by Borusyak, Jaravel, and Spiess, 2024



Notes: Estimation of equation (5) with 20 pre-trends and 30 post periods using the imputation estimator by Borusyak, Jaravel, and Spiess, 2024. Standard errors clustered at the origin level.

Table A.20: Plague increases export volumes and destination market shares, overall, within sectors, and by factor intensity.

	All sectors		Agriculture		Manufacturing		Capital-intensive		Labour-intensive	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post Plague	0.503***	0.014***	0.391***	0.014***	0.783*	0.010***	1.114***	0.015***	0.355***	0.015***
	(0.086)	(0.003)	(0.082)	(0.003)	(0.436)	(0.003)	(0.391)	(0.004)	(0.081)	(0.004)
Fixed Effects:										
– Area x Year	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
– Origin x Destination	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
– Destination x Year	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Estimator	PPML	OLS	PPML	OLS	PPML	OLS	PPML	OLS	PPML	OLS
Observations	103,532	361,059	93,355	361,059	48,170	361,059	89,966	361,059	56,669	361,059

Notes: Difference-in-differences set up dropping the war years of 1700 to 1721. Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable in odd-numbered columns is annual bilateral exports in levels, estimated by PPML. The dependent variable in even-numbered columns is the share of destination j's imports captured by origin i, estimated by OLS, and set to zero where j records no imports in a given year. Columns 1 and 2 pool all sectors. Columns 3 and 4 restrict to agriculture and columns 5 and 6 to manufacturing. Columns 7 and 8 pool the two capital-intensive sectors and columns 9 and 10 the two labour-intensive sectors. In columns 3 to 10, exports are summed across the two sectors in each group and shares are computed relative to the destination's imports within that group. The independent variable is a post plague dummy, equal to one after the origin suffered a plague outbreak. Annual growing season temperatures are an additional control in all specifications. PPML drops observations that are separated by or are singletons within the fixed effects, so the effective number of observations varies with the sample and fixed effect structure across columns.

Table A.21: Heterogeneity of trade findings by pre-plague export levels.

	Export Volume		Export Share	
	(1)	(2)	(3)	(4)
Post Plague	0.35503670***	0.06628295	0.01419861***	0.00698577*
	(0.07505385)	(0.46240164)	(0.00348704)	(0.00375510)
Post Plague x Cap-Int.	0.81696511***	1.24583691***		
	(0.25374769)	(0.39414205)		
Post Plague x Pre-Plague Exports		0.00000155		0.00000010***
		(0.00000259)		(0.00000002)
Post Plague x Cap-Int. x Pre-Plague Exports		-0.00000141*		
		(0.00000081)		
Fixed Effects:				
– Origin x Sector	✓	✓		
– Sector x Time	✓	✓		
– Area x Year	✓	✓	✓	✓
– Origin x Destination			✓	✓
– Destination x Year			✓	✓
Estimator	PPML	PPML	OLS	OLS
Observations	52,463	52,463	361,059	361,059

Notes: Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable in columns 1-2 is the volume of exports by capital intensity. In columns 3-4, it is the market share origin i captures in j. The independent variables are first a post plague dummy and then additionally its interaction with the level of exports before 1700. In columns 1-2, the independent variables also include a post-plague x capital-intensity dummy and its interaction with pre-plague exports. Columns 1 and 3 reproduce the baseline estimates of Tables A.15 and A.20.

E What Drives the Export Expansion?

E.1 Necessary Elements

Factor-Price Transmission: Evidence from Serfdom

Plagued cities in regions with serfdom are particularly scarce in labour, whereas their rural counterparts are relatively abundant in labour. Thus, the strength of the shift into capital-intensive exports depends on serfdom. The results in Table A.22 show that the shift into capital-intensive goods in manufacturing is significantly stronger under serfdom.

Table A.22: Impact of plague on shift into capital-intensive exports by sector & serfdom.

	Manufacturing	Agriculture
	(1)	(2)
Plague	3.036***	1.598**
	(0.591)	(0.729)
Plague x Serfdom	-1.692**	-1.221*
	(0.849)	(0.727)
Plague x Capital-Int.	-2.652***	0.307
	(0.468)	(1.010)
Plague x Capital-Int. x Serfdom	2.221***	0.340
	(0.510)	(1.014)
Fixed Effects:		
– Origin x Sector	✓	✓
– Sector x Year	✓	✓
– Area x Year	✓	✓
Observations	12,986	25,725

Notes: Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. Both columns are estimated by PPML and the dependent variable is sectoral exports in levels. The independent variables are a plague dummy, a serfdom dummy, a capital-intensive sector dummy, and their interactions. I additionally control for growing season temperatures. War years (1700-1721) are dropped.

Capital Deepening: Evidence from Imperfect Capital Markets

Following Antràs and Tubdenov, 2025, APP can be measured as the ratio of inventories over cost of goods sold. Drawing on data for 6-digit NAICS (2017) industries in the US between 2002 and 2016, I construct this ratio from the Annual Survey of Manufacturers (U.S. Census Bureau, 2002–2016).[63] To assign early modern goods modern industry-level APP values, I check whether a given 6-digit NAICS industry existed in 1700. 165 of 364 are assessed to have existed, clustered in food and drink, textiles, basic materials, and craft-based goods.[64] I then assign goods into these industries, and find that 155 commodities are assigned into 67

[63] Antràs and Tubdenov, 2025 use firm-level data that I do not have access to. However, they validate their findings with industry-level controls from NBER-CES, for which ASM is the upstream source which, as opposed to NBER-CES, includes inventories, and additionally covers private firms. Antràs and Tubdenov, 2025 finds that Census-based data (via NBER-CES) correlate with 0.8+ with their firm-level measure.

[64] All electrical industries, motor vehicles and aircraft, plastics and rubber, most chemicals (except botanical manufacturing and pharmaceutical preparations), petroleum products, pulp-based paper, and most modern precision manufacturing (except clocks) are dropped.

industries, with the rest primary goods.[65] My assumption is not that production technology was the same in the 1700s, but that APP contains a time-invariant component. This is reflected in the 10 goods with the highest APP, which are wines, distilled alcohols, tobacco, tiles, and some manufactured metal items. The 10 goods with the lowest APP are dominated by animal products (beef, meat, ham, bones), bread, beer, cidre, and mead. Thus, APP captures a meaningful dimension of production: for distilled alcohol and wines, more time passes between input costs and sales revenue. Following Antràs and Tubdenov, 2025 and Böhm-Bawerk, 1889, more financially developed regions have a comparative advantage in such goods. Here, I assume that heterogeneity in APP captures differences in financial development rather than within-sector (I control for pre-war sectoral export volumes) differences in product characteristics other than credit access.

Table A.23 shows that ports with higher pre-war financial development, proxied by their export-weighted average period of production (APP), shift more strongly into capital-intensive exports after the plague, and undergo a stronger export expansion.

Table A.23: Pre-war financial development (proxied by export-weighted APP) predicts larger treatment effects

	Sectoral Exports		Exports	
	(1)	(2)	(3)	(4)
Post Plague x APP	-0.00021***	-0.00022***	0.00673**	0.00682**
	(0.00004)	(0.00004)	(0.00284)	(0.00288)
Post Plague x Capital-Int. x APP	0.00025***	0.00026***		
	(0.00006)	(0.00006)		
Fixed Effects:				
– Origin x Sector	✓	✓		
– Sector x Time	✓	✓		
– Area x Year	✓	✓	✓	✓
– Origin x Destination			✓	✓
– Destination x Year			✓	✓
Inventory Measure	Average	End of Year	Average	End of Year
Observations	52,463	52,463	361,059	361,059

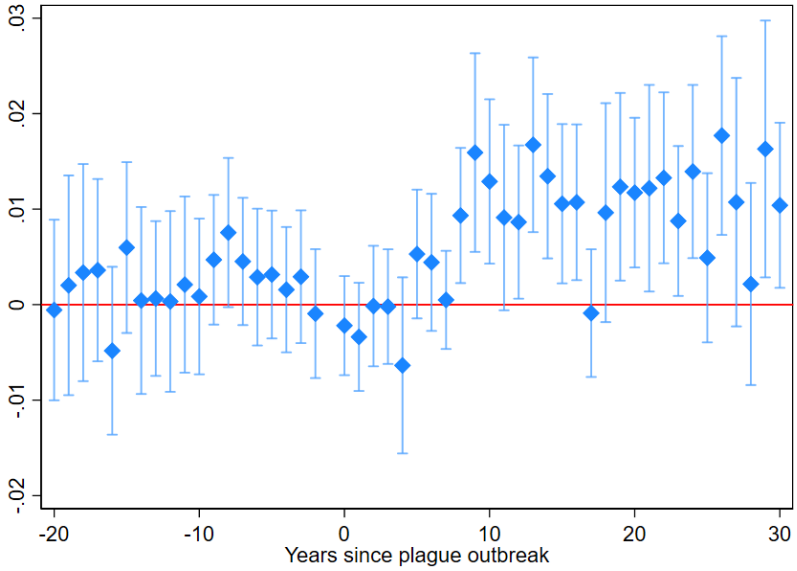
Notes: Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variable in columns 1-2 is the volume of exports by capital intensity. In columns 3-4, it is export volumes. The independent variables are a post plague dummy, the pre-1700 export-weighted APP, their interaction, and annual growing season temperatures. In columns 1-2, I also regress on the interaction of a post plague dummy with a capital-intensity dummy and that interactions' interaction with pre-1700 export-weighted APP. Throughout, I also regress on the interactions of the post plague (and post plague x capital-intensity for columns 1 and 2) dummies with pre-1700 export volumes of all individual sectors. War years (1700 to 1721) are dropped in all columns, as in the main difference-in-differences trade tables.

E.2 What Does not Drive the Export Expansion

Figure A.9 presents evidence that my results are not driven by a demand contraction. In Table A.24, I show that days at sea (a measure of travel time and handling times in harbours) are unrelated to both the plague and to capital intensity.

[65] I exclude primary goods, for which no APP can be calculated, from the construction of port-level APP. The sample in Table A.23 contains the full trade panel, implying an implicit APP of 0 assigned to primary goods.

Figure A.9: Plagued regions also expand their imports



Notes: The outcome are an importer's share of an exporter's exports and I regress on a dummy for the importer suffering a plague outbreak. Standard errors clustered at the level of destinations.

Non-Homotheticity

Fieler, 2011 provides a trade model with non-homotheticity. Under non-homothetic preferences, $\theta_k = \theta \forall k$, and labour share γ_k, the model's gravity equation is:

$$X_{ijk} = \frac{\chi_k A_{ik} (w_{ik})^{-\gamma_k \theta} d_{ijk}^{-\theta}}{CMA_{jk}} X_{jk}. \tag{32}$$

First, relating this equation to the competitiveness measure in my model, note that δ_{jkt} in equation (27) still absorbs the demand side under non-homotheticity. Therefore, the recovery of competitiveness from δ_{ikt} goes through unaffected.

Next, I analyse three types of prices. First, observed regional prices. Under non-homothetic demand capital-intensive goods should become relatively more expensive, as predicted by Voigtländer and Voth, 2012. My mechanism instead produces a relative decline in the price of capital-intensive goods. For plagued Danzig and unplagued Amsterdam, data by Allen and Unger, 2018 hold annual prices on 77 products.[66] Table A.25 shows that prices in labour-intensive sectors rise relative to those in capital-intensive sectors (in agriculture, comparing between columns with the same specified fixed effects) in Danzig compared to Amsterdam. The data show a significant absolute increase in labour-intensive manufacturing good prices. These results point to a supply-driven explanation and reconcile evidence that after the plague consumers received a higher share of their calories from meat and dairy (Broadberry et al., 2014) with my assumption of homothetic demand. As capital-intensive agricultural prices drop, consumers demand higher quantities of meat and dairy under constant consumption shares. Non-homotheticity is not required to make sense of this finding.

[66]No data on capital-intensive manufacturing are containted in the data.

Table A.24: Travel times are unaffected by the plague

	Days at Sea					
	(1)	(2)	(3)	(4)	(5)	(6)
Value of Cap.-Int. Goods	-0.303		-0.006		-0.005	
	(0.409)		(0.004)		(0.004)	
Total Value	1.047		0.015*		0.013	
	(0.777)		(0.008)		(0.008)	
Post Plague x Value of Cap.-Int. Goods	0.333		0.007		0.006	
	(0.653)		(0.009)		(0.008)	
Post Plague x Total Value	-0.346		-0.010		-0.002	
	(1.159)		(0.015)		(0.014)	
Weight of Cap.-Int. Goods		0.278***		0.006***		0.004***
		(0.070)		(0.001)		(0.001)
Total Weight		-0.046		-0.001**		-0.001
		(0.056)		(0.001)		(0.001)
Post Plague x Weight of Cap.-Int. Goods		-0.851		-0.005		-0.009
		(1.410)		(0.010)		(0.016)
Post Plague x Total Weight		0.985		0.011		0.010
		(0.710)		(0.012)		(0.007)
Fixed Effects:						
– Captain	✓	✓	✓	✓	✓	✓
– Origin x Year	✓	✓	✓	✓	✓	✓
Measure	Value	Weight	Value	Weight	Value	Weight
Outcome	Days	Days	Log Days	Log Days	Days	Days
Estimator	OLS	OLS	OLS	OLS	PPML	PPML
Observations	63,359	63,359	63,359	63,359	63,359	63,359

Notes: Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The sample are trips by the same captain (identified by full name and home town) to and from the Sound, where I restrict to trips where the initial destination is the same as the current origin to avoid including trips with unobserved trips within the Baltic or within the North Sea. To the same end, I exclude observations where more than a year has passed since a captain had last been seen at the Sound. War years (1700 to 1721) are dropped, as in the main difference-in-differences tables. The final sample includes 97,078 passages. Days at sea is the difference in days between two observations at the Sound of the same captain. All columns include the post-plague interactions. Cargo is measured by value in columns 1, 3, and 5 and by weight in columns 2, 4, and 6. Columns 1 and 2 estimate days at sea in levels by OLS, columns 3 and 4 log days at sea by OLS, and columns 5 and 6 days at sea by PPML. Cargo value and weight are standardised to mean zero and unit variance, so coefficients give the change in days at sea (columns 1-2), in log days at sea (columns 3-4), and the PPML semi-elasticity of days at sea (columns 5-6) per one standard deviation of cargo.

Table A.25: Product-level prices by sector and year in Danzig and Amsterdam

	Labour-int. Agriculture				Capital-int. Agriculture				Labour-int. Manufacturing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post Plague	10.53***	32.82	114.2*	702.0***	-41.24***	-59.72	-16.44**	4.615***	0.665***
	(3.15e-14)	(21.90)	(16.61)	(8.00e-12)	(6.44e-14)	(10.04)	(0.663)	(1.99e-14)	(3.70e-15)
Fixed Effects:									
– City	✓	✓	✓	✓	✓	✓	✓	✓	✓
– Year		✓	✓			✓	✓		
– Product			✓				✓		
– Product x Year				✓				✓	
Observations	4140	4140	4140	452	5116	5116	5116	248	191

Notes: Standard errors clustered at the city level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variables are prices of products by sector in Amsterdam and Danzig. The independent variable is 1 after the plague in Danzig in 1709. The second, third, and fourth specifications cannot be estimated for labour-intensive manufacturing due to lack of observations. No data on capital-intensive manufacturing products are contained in the data set.

Owing to the exceptional granularity of the Soundtoll data, I can also analyse export and import prices. I construct prices per kilogram whenever weights are recorded (available for 46% of passages).[67] I refer to prices of goods sent to ports as import prices and to prices of goods sent from ports as export prices. Import prices are only indirectly affected: productivities and trade costs in North Sea regions plausibly do not change, and the indirect effects of firm market access changes on wages are small and weighed down by trade costs. In Table A.26, I control for the supply side and show that relative prices of capital-intensive goods are unaffected by the importer's plague status.[68] There is thus no evidence for non-homotheticity. Export prices are directly affected, however. Higher wages lead to a relative decline in capital-intensive export prices. Table A.26 confirms that the (relative) export prices of capital-intensive goods shipped from plagued origins decline significantly.

Finally, I test for demand shifts in plagued regions. I form the ratio of equation 32 between any two sectors, where I asssume $w_{ik} = w_{ik'}$ and $\gamma_k = \gamma_{k'}$. This is conservative, as it interprets any relative import changes as evidence for relative demand changes. I take the ratio of this fraction between a pre-period p, 1700-1710, and any year t after the plague:

$$\frac{X_{ijkt}X_{ijk'p}}{X_{ijk't}X_{ijkp}} = \frac{A_{ikt}A_{ik'p}}{A_{ik't}A_{ikp}}\frac{X_{jkt}X_{jk'p}}{X_{jk't}X_{jkp}}\frac{CMA_{jk't}CMA_{jkp}}{CMA_{jkt}CMA_{jk'p}}. \tag{33}$$

Exporter-time fixed effects absorb relative productivity changes, and relative demand and relative market access are the two demand-side terms, of which importer fixed effects absorb the time-invariant part. Since I found constant relative import prices above, I assume constant relative consumer market access (which is a price index). Thus, equation 33 simplifies as the final fraction drops out.[69]

I present event study results in Figure A.10 where I regress on the log of the left-hand side of equation 33. The independent variable is a dummy for a plague outbreak in the importing

[67]Prices are constructed from the proportional Sound duty, which was levied at 1-2% of declared cargo value (Waldinger, 2022) with treaty-set rates that did not vary across goods or over time in my estimation window (Gøbel, 2010, Marczinek, Maurer, and Rauch, 2025). The duty is thus cargo value up to a constant, and dividing by shipped kilograms yields a price per kilogram up to the same constant. In the logarithmic specifications of Table A.26, this constant is absorbed by the fixed effects, so the estimates are invariant to the duty rate.

[68]I note that these good-level prices do not map neatly to either varieties' prices or price indices in the model and that this is an empirical proxy argument.

[69]This again works against me in that any changes in relative demand will be seen as evidence for non-homotheticity, even if they arise from relative market access changes. The only possibility for this ratio to obscure underlying relative demand shifts is a change in relative demand that is offset by a change in relative consumer market access.

Table A.26: Good-level logarithmic import and export prices

	Import Prices				Export Prices			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Post Plague	0.281***				0.526***			
	(0.0597)				(0.185)			
Post Plague x Capital-Int.	-0.0997		0.0875		-0.511**		-0.664**	
	(0.116)		(0.136)		(0.224)		(0.298)	
0-10 Years Post Plague		0.143				0.253**		
		(0.124)				(0.103)		
11-20 Years Post Plague		0.386***				0.501***		
		(0.0832)				(0.169)		
21-30 Years Post Plague		0.493***				0.546***		
		(0.0982)				(0.183)		
>30 Years Post Plague		0.248***				0.548***		
		(0.0703)				(0.205)		
0-10 Years Post Plague x Cap-Int.		0.0200		0.303		-0.246		-0.375*
		(0.224)		(0.195)		(0.153)		(0.224)
11-20 Years Post Plague x Cap-Int.		-0.0243		0.0534		-0.558***		-0.814***
		(0.129)		(0.176)		(0.203)		(0.296)
21-30 Years Post Plague x Cap-Int.		-0.122		0.0776		-0.639**		-0.801**
		(0.123)		(0.138)		(0.248)		(0.341)
>30 Years Post Plague x Cap-Int.		-0.107		0.0603		-0.505**		-0.657*
		(0.117)		(0.152)		(0.252)		(0.340)
Fixed Effects:								
– Area x Sector x Year	✓	✓	✓	✓	✓	✓	✓	✓
– Origin x Destination x Year			✓	✓			✓	✓
– Origin x Destination x Sector	✓	✓	✓	✓	✓	✓	✓	✓
– Origin x Sector x Year	✓	✓	✓	✓				
– Destination x Sector x Year					✓	✓	✓	✓
Observations	132411	132411	89334	89334	118108	118108	73091	73091

Notes: Standard errors in parentheses, clustered at the destination level for import prices (columns 1-4) and at the origin level for export prices (columns 1-4). * p <0.10, ** p <0.05, *** p <0.01. The dependent variables are prices of goods by sector, origin, destination, and year as recovered from the Soundtoll data. Specifically, the recorded duty amount per good is divided by the weight whenever it is recorded (46% of observations), thus creating comparable prices per kilogram. The duty is the proportional component of the Sound toll, levied at 1-2% of declared cargo value (Waldinger, 2022) with treaty-set rates that did not vary across goods or over time in the estimation window (Gøbel, 2010, Marczinek, Maurer, and Rauch, 2025). Duty per kilogram is therefore proportional to the price per kilogram, and the proportionality constant is absorbed by the fixed effects in these logarithmic specifications. The independent variables are a post plague dummy that is 1 after the plague and the interaction of this post plague dummy with a dummy for a capital-intensive sector. Columns 2 and 4 within each panel split up this dummy and the interaction by time horizon after the plague.

region.[70] I find no evidence that the plague shifted demand relatively into capital-intensive sectors.

Human and Social Capital Deepening

I construct measures of goods' skill intensity and contract intensity. Following Berman, Bound, and Griliches, 1994 and Autor, Katz, and Krueger, 1998, I measure skill intensity as the share of non-production workers in total employment from the Annual Survey of Manufactures (U.S. Census Bureau, 2002–2016). I use 2002-2016 data at the NAICS 6-digit level, which covers 155 of the commodities of this paper following the mapping of goods in 1700 to modern-day U.S. industries laid out around Table A.23. Following Nunn, 2007 and Rauch, 1999, I proxy social capital intensity with contract intensity, which I measure using the share of differentiated inputs (liberally or conservatively defined as in Rauch, 1999). I use data by Nunn, 2007 at the NAICS 6-digit level, which covers 155 manufacturing goods, matched to NAICS 6-digit industries. Table A.27 finds no shift into exports of higher human or social capital content after the plague.

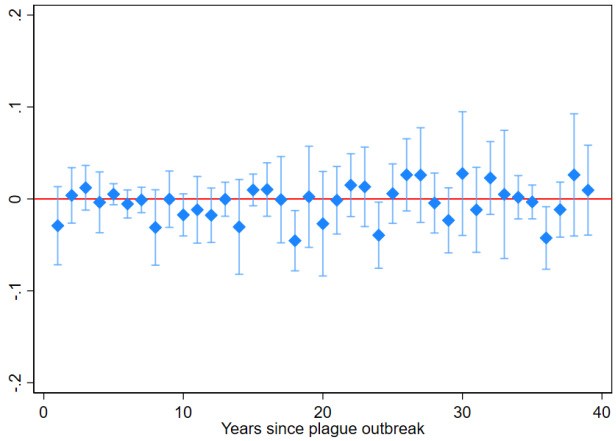
Table A.27: The plague leads to a shift into physical capital intensity, not into human or social capital intensity

	Contract Intensity		Skill Intensity	Both	
	(1)	(2)	(3)	(4)	(5)
Post Plague	-0.09872	-0.06312	1.47851**	1.31778	1.38707*
	(0.33665)	(0.33162)	(0.61863)	(0.82287)	(0.81969)
Post Plague x Capital-Int.	0.58767**	0.56618*	0.99030**	0.98048**	0.98157**
	(0.29780)	(0.29935)	(0.39901)	(0.41430)	(0.42031)
Post Plague x Contract Int.	0.00008	0.00006		0.00003	0.00001
	(0.00010)	(0.00009)		(0.00010)	(0.00009)
Post Plague x Skill Int.			-4.61002**	-4.29949*	-4.42896*
			(1.93770)	(2.26775)	(2.27740)
Fixed Effects:					
– Origin x Good	✓	✓	✓	✓	✓
– Good x Time	✓	✓	✓	✓	✓
– Area x Year	✓	✓	✓	✓	✓
Nunn Measure	Liberal	Conservative		Liberal	Conservative
Observations	25,929	25,929	25,929	25,929	25,929

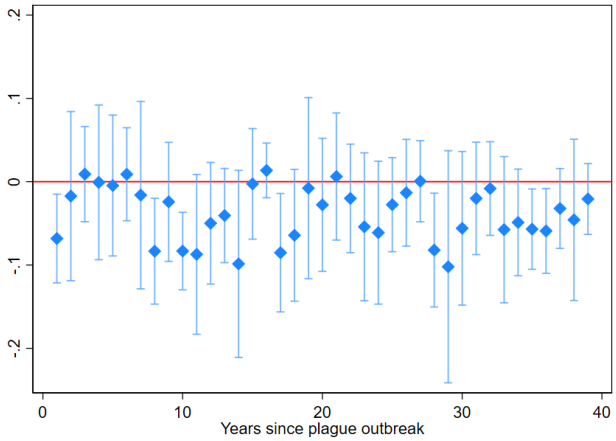
Notes: Standard errors clustered at the origin level in parentheses. * p <0.10, ** p <0.05, *** p <0.01. Results from regressions where the dependent variable is log exports weighted by exports in levels. The independent variables are a post plague dummy, the interaction of a post plague dummy with a capital intensity dummy, and the interaction of the post plague dummy with the liberal measure (columns 1 and 4) or the conservative measure (columns 2 and 5) of the share of a good's input that are differentiated according to Rauch, 1999, which in Nunn, 2007 is a measure of contract intensity, and/or the interaction of the post plague dummy with the share of non-production workers averaged across 2002 to 2016, based on 6-digit NAICS industries in the Annual Survey of Manufacturers (columns 3-5). For columns 1-2 and 4-5, I use Nunn's data at the 6-digit NAICS level for the 1997 classification. Only the 155 goods which are covered by NAICS 6-digit industries are included. As the intensity measures vary at the good level, the fixed effects are the goods-level analogues of the Fact #1 specification: origin x good, good x time, and area x year. As before, war years (1700-1721) are dropped and I control for growing season temperatures.

[70] I also check that there is no evidence for non-homotheticity when using the continuous plague treatment derived in section C.2. Results available on request.

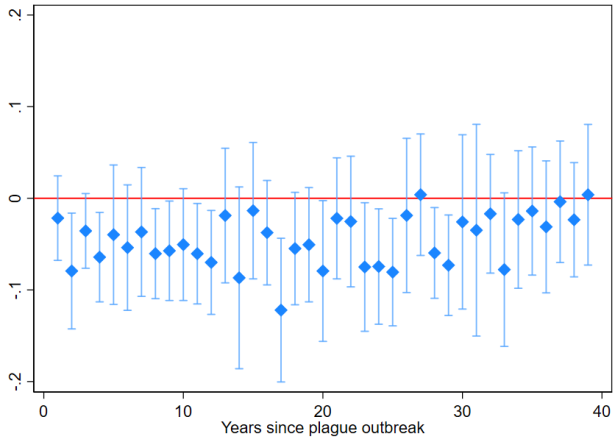
Figure A.10: Event studies testing for non-homotheticity



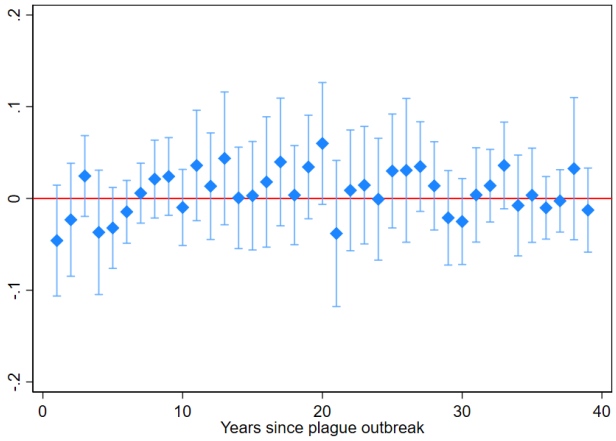
(a) Capital-int. & Labour-int. Manufacturing



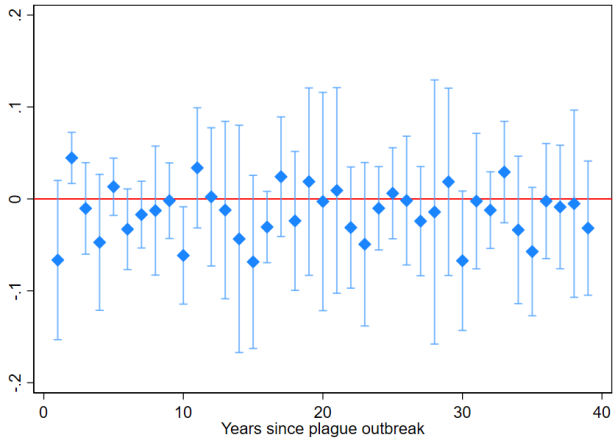
(b) Capital-int. Manufacturing & Labour-int. Agriculture



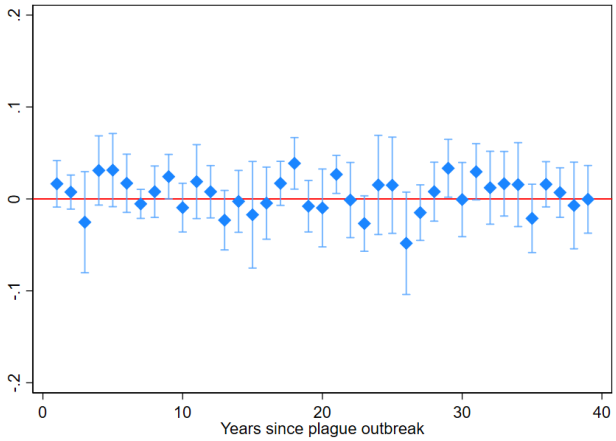
(c) Capital-int. Manufacturing & Agriculture



(d) Capital-int. Agriculture & Labour-int. Manufacturing



(e) Capital-int. Agriculture & Labour-int. Agriculture



(f) Labour-int. Manufacturing & Agriculture

Notes: Event study on the log of the left-hand side of equation 33. The independent variable is a plague dummy on the importer side. Origin-time and destination fixed effects are included and standard errors are clustered at the destination level.

F Model Appendix

This appendix states each agent's problem and derives every result of Section 6. Existence and uniqueness are in Appendix G and Table A.28 collects notation.

Table A.28: Key parameters and variables

Panel A: Preferences, market structure, and parameters			
σ_k	Elasticity of substitution across varieties within a bundle	ρ_k	Productivity pass-through into competitiveness, $\rho_k \equiv \frac{\xi-1}{\sigma_k-1}$
$\xi-1$	Trade elasticity, set to 5 or 6.53	μ_k, χ_k	Markup $\sigma_k/(\sigma_k-1)$ and $\chi_k = \mu_k^{1-\xi}$
r, ι, ζ	User cost of capital $r = \iota + \zeta$, world interest rate, depreciation rate	$\tilde{w}_i$	Normalised wage w_i/r
Panel B: Technology choice, labour shares, and productivity			
γ^k_{non} (γ^k_{ad})	Labour shares without (with) technology adoption, the adjacent-pair building block	Γ_{ikt}	Realised sectoral labour share; weighted average of γ^k_{non} and γ^k_{ad}
$\gamma^k_0 > \cdots > \gamma^k_{n_k}$	Labour shares along the ordered menu, rungs $\tau = 0, \ldots, n_k$; span γ_0/γ_n caps the technology channel (Proposition 3)	$\Delta\Gamma$	Cross-sector labour-share gap $\bar{\Gamma}_{iL} - \bar{\Gamma}_{iC} \leq 0.2$ pre-plague; full-menu gap $\Delta\Gamma_{\text{menu}} = 0.7 - \gamma_n \approx 0.3$-$0.4$ at the path end
$\Delta\gamma$	Within-sector adjacent-pair gap, $\gamma^k_{non} - \gamma^k_{ad}$	B_{ik}	Exogenous efficiency aggregate
A_{ikt}	Sectoral productivity under adoption	$T_{ik}(\widetilde{w}_i)$	Exporter competitiveness (sectoral productivity and unit costs)
z^*_{ik}, z^{*-}_{ik}	Entry and exit cutoffs, (15)	υ_k, F	Productivity gain and capital stock of adoption (per rung: υ^k_τ, F^k_τ)
Ψ_{ik}	Demand shifter, (13) and (24)		
Panel C: Factor prices and recovered objects			
G	Treated vs control change in $\ln \widetilde{w}$	δ_{ikt}	Recovered triple difference of $\ln T_{ikt}$
g	$g = e^G - 1 \in [0, 0.5]$	D	Observed differential $\delta_{iCt} - \delta_{iLt}$

Notes: i, j, k, t denote origin, destination, sector, and time. Subscripts C and L denote the capital- and labour-intensive sector.

F.1 Households: deriving demand (9)

The household in region j maximises (8) subject to its budget constraint,

$$\sum_k \sum_i \int_{\Omega_{ik}} p_{ijk}(\omega)\, q_{ijk}(\omega)\, d\omega = Y_j, \tag{34}$$

where income is the sum of wages, the rental income from the capital endowment net of the replacement of its depreciation, $(r-\zeta)\bar{K}_j = \iota \bar{K}_j$, and the net profits of the producers the household owns,

$$Y_{jt} = \sum_k w_{jkt} L_{jkt} + \iota \bar{K}_j + \Pi_{jt}, \qquad \Pi_{jt} = \sum_k \Big[\int_{\Omega_{jk}} \Pi^{op}_{\tau(\omega)}\big(z(\omega)\big)\, d\omega - rF\, n^{\text{ad}}_{jkt} \Big], \tag{35}$$

with wages differing across sectors only under serfdom, n^{ad}_{jkt} the count of adopters, and rF their flow cost of interest and maintenance (F.3). Sales, factor costs, and operating profits add up (F.5), so Π_{jt} is what the household receives after paying for factors and the stocks that adopters hold. Adoption stocks are financed on the world capital market (equilibrium condition

(iv)), and abandonment scraps the stock: the financing position is extinguished against the scrapped capital, with the write-off absorbed in the region's net capital position with the rest of the world in the period of exit. Two-stage budgeting applies to (8): first, Cobb-Douglas gives sectoral spending $\alpha_k Y_j$. Second, within sector k, minimising $\sum_i P_{ijk}Q_{ijk}$ subject to C_{jk} and rearranging gives $Q_{ijk} = (P_{ijk}/P_{jk})^{-\xi}C_{jk}$ with $P_{jk}^{1-\xi} = \sum_i P_{ijk}^{1-\xi}$ and $X_{ijk} \equiv P_{ijk}Q_{ijk} = (P_{ijk}/P_{jk})^{1-\xi}\alpha_k Y_j$. Third, within i's bundle, the same gives $q_{ijk}(\omega) = (p_{ijk}(\omega)/P_{ijk})^{-\sigma_k}Q_{ijk}$ with $P_{ijk}^{1-\sigma_k} = \int_{\Omega_{ik}} p_{ijk}(\omega)^{1-\sigma_k}d\omega$. Multiplying by $p_{ijk}(\omega)$ gives (9).

F.2 Producers: deriving markups, revenue, and profits (13)

A τ-producer with draw z minimises $w_i\ell + r\kappa$ subject to (10) producing one unit. The FOC is $\frac{\gamma_\tau}{1-\gamma_\tau}\frac{\kappa}{\ell} = \frac{w_i}{r}$, and substituting gives unit cost $\Lambda_{\gamma_\tau} w_i^{\gamma_\tau} r^{1-\gamma_\tau}/(\upsilon_\tau z)$ with $\Lambda_\gamma = \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}$. Choosing units of z so that $\Lambda_{\gamma_{\text{non}}} = 1$ and folding $\Lambda_{\gamma_{\text{non}}}/\Lambda_{\gamma_{\text{ad}}}$ into υ_k gives (12). By the envelope condition, labour costs the share γ_τ and capital the share $1-\gamma_\tau$ of variable cost.

Facing demand (9) in j, the producer chooses p to maximise $(p - MC_\tau d_{ijk})\, p^{-\sigma_k} P_{ijk}^{\sigma_k - 1} X_{ijk}$, with MC_τ defined in (12). The FOC gives $p_{ijk}(\omega) = \mu_k\, MC_\tau(z)\, d_{ijk}$ with markup $\mu_k = \sigma_k/(\sigma_k - 1)$. Revenue in j is then $(\mu_k MC_\tau d_{ijk})^{1-\sigma_k} P_{ijk}^{\sigma_k-1} X_{ijk}$, variable cost is revenue over μ_k, and operating profit is revenue over σ_k. Summing revenue over j and dividing by σ_k gives (13).

F.3 Technology choice: deriving the maintenance rule (14), cutoffs (15), and Lemma 1

Let $\beta = 1/(1+\iota)$ and write Δ for the gain $\Pi_{\text{ad}}^{op} - \Pi_{\text{non}}^{op}$, which by (13) equals $\Psi_{ik} z^{\sigma_k - 1} r^{1-\sigma_k} \Pi_k(\tilde{w}_i)$ with Π_k as in (15). A technology decision at t affects production from $t+1$. Profits and maintenance accrue at the end of each period of use, and the installation outlay F is paid at the start of the first period of use. Under Assumption 1 Δ is expected to be constant. Let W be the value, at the start of a period of use, of holding an operational stock, and normalise the value of the labour-intensive technology to zero. Maintaining yields $\beta(\Delta - \zeta F + W)$ and abandoning yields zero, so

$$W = \max\{\beta(\Delta - \zeta F + W),\, 0\} \quad \Longrightarrow \quad W = \frac{\Delta - \zeta F}{\iota} \;\text{ if }\; \Delta \geq \zeta F, \qquad W = 0 \;\text{ otherwise.}$$

The maintenance rule is the second part of (14). Installing yields $-F + W$, which is non-negative iff $\Delta - \zeta F \geq \iota F$, the first part of (14). Both rules are monotone in Δ, and Δ is increasing in z when $\Pi_k(\tilde{w}) > 0$, which gives the cutoffs (15).

Lemma 1 follows from differentiating Π_k: $\Pi_k > 0$ iff $\upsilon_k^{\sigma_k - 1} > \tilde{w}^{-(\gamma_{\text{non}} - \gamma_{\text{ad}})(\sigma_k - 1)}$, and $\Pi_k' > 0$ iff $\gamma_{\text{non}}\tilde{w}^{-\gamma_{\text{non}}(\sigma_k-1)} > \gamma_{\text{ad}}\upsilon_k^{\sigma_k-1}\tilde{w}^{-\gamma_{\text{ad}}(\sigma_k-1)}$, which rearrange to $\tilde{w} > \tilde{w}_0$ and $\tilde{w} < \tilde{w}_{\text{crit}}$. A producer that installed and faces a gain between ζF and rF neither abandons nor would install again. If the adopter set is an upper set of draws at t, entry adds the draws in $[z_{ikt}^*, \bar{z}_{ikt})$ and exit removes those in $[\bar{z}_{ikt}, z_{ikt}^{*-})$, so it remains an upper set and (16) describes its mass exactly.

Two remarks. First, anticipated wage reversion lowers the entry value and can reduce installation. Second, the adoption outlay is denominated in capital by assumption, so the annualised hurdle rF has zero local-wage elasticity. If the hurdle instead had a positive wage elasticity, the cutoff would fall only when $\epsilon_\Pi > \theta$.

F.4 Prices, gravity, and the demand shifter: deriving gravity (23)

Substituting the markup price into the bundle price index and using (12),

$$P_{ijk}^{1-\sigma_k} = \int_{\Omega_{ik}} p_{ijk}(\omega)^{1-\sigma_k} d\omega = (\mu_k d_{ijk})^{1-\sigma_k} \sum_\tau \left(w_i^{\gamma_\tau} r^{1-\gamma_\tau}\right)^{1-\sigma_k} \upsilon_\tau^{\sigma_k - 1} N_{ik} \int_{\Omega_{ik}^\tau} z^{\sigma_k - 1} dF_{ik}(z),$$

which is the bracketed sum of (20). Raising to ρ_k and summing over i gives $P_{jk}^{1-\xi} = \mu_k^{1-\xi} \sum_i T_{ik} d_{ijk}^{1-\xi}$, the first part of (22). Substituting both indices into the bundle demand of (9) gives (23), and summing (23) over j gives $\sum_j X_{ijk} = T_{ik} FMA_{ik}$ with FMA_{ik} as in (22).

For the demand shifter, revenue of variety ω in j is $(p_{ijk}(\omega)/P_{ijk})^{1-\sigma_k} X_{ijk} = MC_\tau(z)^{1-\sigma_k} [S_{ik}^{\text{non}} + S_{ik}^{\text{ad}}]^{-1} X_{ijk}$ by (20), so summing over j and dividing by σ_k gives $\Pi_\tau^{op}(z) = \sigma_k^{-1} MC_\tau(z)^{1-\sigma_k} [S_{ik}^{\text{non}} + S_{ik}^{\text{ad}}]^{\rho_k - 1} FMA_{ik}$, which is (13) with $\Psi_{ik} = \sigma_k^{-1} T_{ik}^{1-1/\rho_k} FMA_{ik}$, the first part of (24).

F.5 Factor demands and prices

By F.2, a τ-producer spends the share γ_τ of its variable cost on labour, and variable cost is revenue over μ_k. Sales by technology are $X_{ik}^\tau = T_{ik}^{1-1/\rho_k} FMA_{ik} S_{ik}^\tau$, so

$$w_i L_{ik} = \mu_k^{-1} T_{ik}^{1-1/\rho_k} FMA_{ik} \big[\gamma_{\text{non}} S_{ik}^{\text{non}} + \gamma_{\text{ad}} S_{ik}^{\text{ad}}\big] = \mu_k^{-1} FMA_{ik} T_{ik} \Gamma_{ik},$$

with Γ_{ik} the revenue-weighted share (21). For capital, $rK_{ik} = \mu_k^{-1} FMA_{ik} T_{ik} (1 - \Gamma_{ik})$. Sales, factor costs, and operating profits add up, $T_{ik} FMA_{ik} = w_i L_{ik} + rK_{ik} + \int_{\Omega_{ik}} \Pi_{\tau(\omega)}^{op} d\omega$, which gives net profits in (35).

For the post-recovery wage, write $q \equiv (\tilde{w}/\tilde{w}_0)^{\Delta\gamma(\sigma_k - 1)}$ for the cost advantage of the capital-intensive technology, so that by (20) $S^{\text{non}} + S^{\text{ad}} = r^{1-\sigma_k} \tilde{w}^{-\gamma_{\text{non}}(\sigma_k - 1)} B_{ik} [1 + (q-1)\pi]$ and $s^{\text{ad}} = q\pi/[1 + (q-1)\pi]$. Then $\Gamma_{ik}[1 + (q-1)\pi] = \gamma_{\text{non}}(1 - \pi) + \gamma_{\text{ad}} q \pi$, and labour costs with adopter share π relative to $\pi = 0$ at the same $(\tilde{w}, FMA)$ are

$$\frac{L_{ik}^D(\pi)}{L_{ik}^D(0)} = \big[1 + (q-1)\pi\big]^{\rho_k - 1} \Big[1 - \pi + \frac{\gamma_{\text{ad}}}{\gamma_{\text{non}}}\, q\, \pi\Big], \tag{36}$$

which at $\rho_k = 1$ is below one iff $q < \gamma_{\text{non}}/\gamma_{\text{ad}}$ (the interval of Lemma 1). At $\rho_k > 1$, (36) is below one iff $q^{\rho_k} \gamma_{\text{ad}}/\gamma_{\text{non}} < 1$ at full adoption, that is iff $q < (\gamma_{\text{non}}/\gamma_{\text{ad}})^{1/\rho_k}$ (≈ 1.29 at $\rho_k = 6.53/5$), which I impose on the post-recovery wages considered in Section 6.

I can also show that the long-run wage need only retain a fifth of its plague-induced inccrease for all adopters to keep the capital-intensive technology. To see this, note that at $\Delta\gamma(\sigma_k - 1) = 1$ the profit gap is $\Pi_k(\tilde{w}) = \tilde{w}^{-\gamma_{\text{non}}(\sigma_k - 1)}(q - 1)$ with $q = \tilde{w}/\tilde{w}_0$, so (19) reads $(q_\infty/q_1)^{\gamma_{\text{non}}(\sigma_k - 1)}(q_1 - 1)/(q_\infty - 1) \leq r/\zeta$. At $q_1 = 1.38$ (wage growth in Allen, 2005), $\gamma_{\text{non}} = 0.7$, $\sigma_k - 1 = 5$, and $r/\zeta = 2$, this holds iff $q_\infty - 1 \geq 0.081$, about a fifth of the plague wage gain.

G Existence and Uniqueness

Existence requires only boundedness. (17) confines the technology aggregate to $A_{ik} \in [B_{ik}, \upsilon_k^{\sigma_k - 1} B_{ik}]$, which sandwiches the exporter term $T_{ik}(\tilde{w}_i)$ between two power functions of $\tilde{w}_i$. Labour demand $L_{ik} \propto T_{ik}(\tilde{w}_i)\, \Gamma_{ik}\, w_i^{-1}$ is continuous, diverges as $w_i \to 0$ and vanishes as $w_i \to \infty$. Thus, a crossing with fixed regional labour supply exists, provided only that $B_{ik} > 0$ and the adoption gain υ_k is finite.

Turning to uniqueness is not trivial as the model contains a positive feedback: higher wages induce adoption, which raises A_{ik}, which raises labour demand and wages again. A preliminary remark: the dynamic path itself is always unique. Decisions at t take effect at $t + 1$, so within each period the adopter masses are predetermined: the contemporaneous adoption response is zero, the labour share elasticity at fixed masses is weakly negative, so sectoral labour demand slopes down unconditionally. The law of motion (16) then updates the masses deterministically. The question below is therefore uniqueness of the stationary state. I also note that the recovery of δ_{ikt} needs no uniqueness: (23) holds state by state, and the estimator returns the fixed effects of whichever equilibrium generated the data (Costinot, Donaldson, and Komunjer, 2012).

For uniqueness in the adoption-inclusive equilibrium, I make some key assumptions. These assumptions are discussed below where they bind and presented here to provide an overview. First, producers are atomistic. Second, I assume that producers take equilibrium prices (w_i, r, P_{ik}) as given under static expectations and choose the technology that maximises their profit at those prices. Given prices, the adoption choice is a unique best response and there is no within-sector coordination problem. Also, no producer internalises A_{ik} and T_{ik} as each variety has measure zero: atomicity rules out the strategic coordination source of multiplicity, in which producers condition on each other's choices. It does not rule out aggregate multiplicity through pecuniary feedback: when $\rho_k > 1$, adoption by other producers raises Ψ_{ik} and strengthens each producer's own incentive to adopt. That channel is the feedback that operates through prices and demand, which I discuss next. Third, investments take one period to build and I make a small open economy assumption below, so that r and FMA_{ik} are fixed throughout. Fourth, assumption (A1) below restricts spillovers to be local. Fifth, the derivations use the wage window of Lemma 1 for each adjacent pair.

G.1 Feedback and uniqueness

The wage-adoption loop produces the structure of an industry-level external economy of scale, so the question is whether the regional wage it implies is unique. As labour supply is fixed, it takes downward-sloping labour demand for the regional labour market to have exactly one crossing and a unique wage. The adoption feedback is a threat because it pushes labour demand toward sloping up. Below, I derive uniqueness as a condition on labour demand, maintaining one assumption:

(A1) Local spillover. A_{ik} depends on region i's own factor-price ratio $\tilde{w}_i$ and demand shifter Ψ_{ik}, and on nothing outside the region: the plague's technology response is mediated by local labour scarcity. In the economic geography application of Allen, Arkolakis, and Li, 2024, any cross-location productivity or amenity spillover can generate multiplicity, whereas purely local spillovers leave the standard agglomeration-versus-dispersion conditions intact. (A1) places the model in the local case, which is also what the identification strategy assumes as SUTVA. The local loop is then disciplined by the conditions below.

Step 1: labour demand. Sales by technology are $X_{ik}^{\tau} = T_{ik}^{1-1/\rho_k} FMA_{ik}\, S_{ik}^{\tau}$ with S_{ik}^{τ} from (20), so total sectoral sales are $\sum_{\tau} X_{ik}^{\tau} = T_{ik} FMA_{ik}$. Under markup μ_k and Cobb-Douglas shares, a τ-producer spends $w_i L_{ik}^{\tau} = \gamma_{\tau} X_{ik}^{\tau}/\mu_k$ on labour. Summing the two yields

$$w_i L_{ik} = \frac{FMA_{ik}}{\mu_k}\, T_{ik}^{1-1/\rho_k} \left[\gamma_{\text{non}} S_{ik}^{\text{non}} + \gamma_{\text{ad}} S_{ik}^{\text{ad}}\right] = \frac{FMA_{ik}}{\mu_k}\, T_{ik}\, \Gamma_{ik}, \tag{37}$$

$$\Gamma_{ik} \equiv \frac{\gamma_{\text{non}} S_{ik}^{\text{non}} + \gamma_{\text{ad}} S_{ik}^{\text{ad}}}{S_{ik}^{\text{non}} + S_{ik}^{\text{ad}}} = \gamma_{\text{non}} - (\gamma_{\text{non}} - \gamma_{\text{ad}})\, s_{ik}^{\text{ad}}. \tag{38}$$

with $s_{ik}^{\text{ad}} \equiv S_{ik}^{\text{ad}}/(S_{ik}^{\text{non}} + S_{ik}^{\text{ad}})$ the adopters' revenue share and Γ_{ik} the labour share. In logs,

$$\ln L_{ik}(w_i) = \text{const}_k + \ln FMA_{ik} + \underbrace{\ln T_{ik}(w_i)}_{\text{competitiveness}} + \underbrace{\ln \Gamma_{ik}(w_i)}_{\text{labour share}} - \underbrace{\ln w_i}_{\text{cost of a worker}}. \tag{39}$$

Three of the four terms move with the wage.

Step 2: the slope of labour demand. Differentiating (39) in $\ln w_i$ (with r fixed, so $d\ln \tilde{w}_i = d\ln w_i$) turns on the response of T_{ik}. The active pair's combined mass $B_{ik}^{(\text{non,ad})} = m_{ik}^{\text{non}} + m_{ik}^{\text{ad}}$ is locally fixed on a monotone path as only the active margin moves, so a variety clearing the cutoff leaves the non-adopter pool for the adopter pool, $dm_{ik}^{\text{non}} = -dm_{ik}^{\text{ad}}$. Writing $\epsilon_{\tau} \equiv d\ln m_{ik}^{\tau}/d\ln \tilde{w}_i$

for the equilibrium elasticities of the masses,[71] Lemma 1 gives $\epsilon_{\text{ad}} > 0 > \epsilon_{\text{non}}$: a higher $\tilde{w}$ lowers the cutoff and mass flows into adoption. Each S^τ then moves both because its mass moves and because its factor cost moves. Using the active pair's realised share $\Gamma_{ik} = \gamma_{\text{non}}(1 - s^{\text{ad}}) + \gamma_{\text{ad}} s^{\text{ad}}$ (under the menu, the sector object is $\bar{\Gamma}_{ikt}$ and the identities below apply at the active margin with the pair weight of Appendix G.3) and $\rho_k(\sigma_k - 1) = \xi - 1$,

$$\frac{d \ln T_{ik}}{d \ln w_i} = \underbrace{\rho_k \big[(1 - s^{\text{ad}}_{ik})\, \epsilon_{\text{non}} + s^{\text{ad}}_{ik}\, \epsilon_{\text{ad}}\big]}_{R_{ik}\ \text{(reallocation)}} - \Gamma_{ik}(\xi - 1). \tag{40}$$

The reallocation term is the loop gain R_{ik}: the rate at which a higher wage raises T_{ik} through adoption. This loop gain is the agglomeration force, while $\Gamma_{ik}(\xi - 1)$ is the dispersion force.

The last moving piece in (39) is the labour share. From (37), $d\Gamma_{ik}/d \ln w_i = -(\gamma_{\text{non}} - \gamma_{\text{ad}})\, ds^{\text{ad}}_{ik}/d \ln w_i$, so its elasticity is unambiguously negative: a higher wage tilts the sector toward the capital-intensive technology, lowering the labour share and hence labour demand, which is another dispersion force. Combining (39) and (40) with this term,

$$\frac{d \ln L_{ik}}{d \ln w_i} = R_{ik} + \epsilon_\Gamma - \big(1 + \Gamma_{ik}(\xi - 1)\big) \;\equiv\; -\eta_{ik}, \qquad \epsilon_\Gamma \equiv \frac{d \ln \Gamma_{ik}}{d \ln w_i} = -\frac{\gamma_{\text{non}} - \gamma_{\text{ad}}}{\Gamma_{ik}} \frac{ds^{\text{ad}}_{ik}}{d \ln w_i} \;<\; 0. \tag{41}$$

The "1" is the direct cost of a worker, $\Gamma_{ik}(\xi - 1)$ the competitiveness lost to a higher wage, and R_{ik} the adoption feedback. Writing $\epsilon_\delta \equiv d \ln T_{ik}/d \ln \tilde{w}_i$, (40) reads $R_{ik} = \epsilon_\delta + \Gamma_{ik}(\xi - 1)$, the $\Gamma_{ik}(\xi - 1)$ terms cancel, and the slope collapses to

$$\eta_{ik} = 1 - \epsilon_{\delta,ik} - \epsilon_{\Gamma,ik}. \tag{42}$$

Sector k's labour demand slopes down iff $\eta_{ik} > 0$.[72]

Step 3: from sectors to the region. Regional labour demand is $L_i^D = \sum_k L_{ik}$, so its elasticity is the labour share-weighted average of the sectoral elasticities (42),

$$\eta_i \;\equiv\; -\frac{d \ln L_i^D}{d \ln w_i} \;=\; \sum_k \lambda_{ik}\, \eta_{ik}, \qquad \lambda_{ik} \equiv L_{ik}/L_i. \tag{43}$$

This means a sector-by-sector check is conservative: if every $\eta_{ik} > 0$ then $\eta_i > 0$, though the converse need not hold. Collecting the three steps gives the result.

Proposition 2 (Many sectors in a small open economy)**.** *Suppose region i takes $\{CMA_{jk}\}_{j \neq i}$ and $\{Y_j\}_{j \neq i}$ as given and its own absorption accounts for a negligible share of its markets, so that FMA_{ik} is exogenous. Let the within-bundle condition of footnote 71 hold. Then under (A1) the regional equilibrium wage exists and is unique if labour demand slopes down,*

$$\eta_i = \sum_k \lambda_{ik} \big(1 - \epsilon_{\delta,ik} - \epsilon_{\Gamma,ik}\big) \;>\; 0. \tag{44}$$

Proof. By (39), $\{L_{ik}\}$ is a function of w_i alone once FMA_{ik} is fixed (the within-bundle condition makes $T_{ik}(w_i)$ single-valued (Lemma 2(ii))) so market clearing is the scalar equation $L_i =$

[71] The ϵ_τ are total elasticities: alongside the direct effect of $\tilde{w}$ through Π_k, they embed the within-bundle feedback of (13) (adoption raises T_{ik}, hence Ψ_{ik} when $\rho_k > 1$, hence adoption). I maintain throughout that this inner loop is a contraction, which Lemma 2(ii) in Appendix G.3 shows holds at the calibrated elasticities.

[72] In Kucheryavyy, Lyn, and Rodríguez-Clare, 2023 the externality is linked to industry size, so the trade elasticity governs how a scale-induced productivity gain converts into employment. Here, the externality is linked to factor prices and the trade elasticity enters reallocation and dispersion identically and cancels.

$\sum_k L_{ik}(w_i)$. Its right-hand side is continuous and strictly decreasing iff $\eta_i > 0$ (equation (43)). As T_{ik} is sandwiched between two power functions of $\tilde{w}_i$ through the boundedness of A_{ik}, labour demand diverges as $w_i \to 0$ and vanishes as $w_i \to \infty$, so a crossing exists. Under (44), the right-hand side is strictly decreasing and crosses labour supply exactly once. □

Because the adoption capital is sunk and durable, the adopter share is the state (16), so this is a within-period result. Inside the band of (16), the adopter mass is fixed and $R_{ik} = 0$. At the entry or exit margin, the mass responds to the wage through the respective cutoff. The quantification in Appendix G.3 covers both margins and (44) is slack inside the band. Condition (44) is sufficient rather than necessary: regional demand can slope down without every sector doing so (equation (43)) and demand that fails to slope down globally can still cross the fixed labour supply exactly once, so failure of (44) would not imply multiplicity.

Endogenous market access. Proposition 2 holds the rest of the world fixed, which can be relaxed for a single traded sector. Adding assumption (A2) on quasi-symmetric trade costs, $d_{ijk} = \tau_{ijk}\varsigma_{ik}\varrho_{jk}$ with $\tau_{ijk} = \tau_{jik}$, the model becomes a special case of Allen, Arkolakis, and Li, 2024 without intermediates. Applying their theorem would restrict me to a single traded sector, however. To the best of my knowledge, the many-sector, many-region case is not available yet. Allen, Arkolakis, and Li, 2024 assume the measure of each type of agent exogenously fixed, which would imply fixed sectoral employment here. Instead, assuming a small open economy leaves this paper where Kucheryavyy, Lyn, and Rodríguez-Clare, 2023 arrive.

G.2 Checking the condition

Existence is immediate as my estimates bound the adoption gain υ_k below infinity.

Uniqueness requires $\eta_i > 0$ by Proposition 2. The loop gain peaks near the threshold where the reallocation elasticity R_{ik} is largest. The honest check is therefore at the supremum,

$$\sup_{\tilde{w}} \epsilon_\delta < 1, \qquad \text{for which a sufficient sector-level condition is} \qquad \sup_{\tilde{w}} R_{ik} < \gamma_{\text{ad}}(\xi - 1) + 1, \tag{45}$$

where I have used $\Gamma_{ik} \geq \gamma_{\text{ad}}$ to take the smallest value of the dispersion term on the right, and have conservatively dropped the strictly negative ϵ_Γ. As argued before, checking sector by sector is conservative.

Coefficient $\hat{\beta}$ of (31) estimates D_k, the capital-intensity differential. To express D_k in the elasticities of (40), I integrate along each region's path: sectors face the same $\tilde{w}_{it}$, so differencing across sectors leaves the gap in slopes, and differencing across regions and time gives

$$D_k = \Big[\underbrace{\left(R_{ik} - R_{ik_0}\right)}_{\text{reallocation}} + \underbrace{\left(\xi - 1\right)\Delta\Gamma_{it}}_{\text{factor proportions}} \Big] G. \tag{46}$$

(46) requires the cross-sector difference in slopes to be common to treated and control regions. I do not assume same adoption or same realised labour shares, but rather that γ^k_{ad}, γ^k_{non}, υ_k and σ_k are sectoral features, so the mapping from $\tilde{w}$ into adoption is the same function across regions, conditional on Ψ_{ik}. The importer-side components of Ψ_{ik} are held fixed by the small-open-economy assumption and sector-neutral variation in the remainder differences out of the cross-sector comparison.

A threat to this claim would be a change in treated exporters' FMA that is sectorally biased, which would enter D_k through the cutoff's dependence on Ψ_{ik} rather than wages. Area-sector-year FEs absorb such changes at the area level, so the threat is confined to region-specific disruptions, which in this period would, in particular, be the Danish naval blockade. However, Appendix Table A.14 shows that the blockade's incidence across capital intensities did not differ by plague status, so it cannot affect the comparison that identifies D_k.

From (46), the differential identifies the difference in loop gains net of the factor proportions wedge, $R_{ik} - R_{ik_0} = D_k/G - (\xi - 1)\,\Delta\Gamma_{it}$. Setting the reference sector gain $R_{ik_0} = 0$, which is a stronger version of the assumption of Section 7 that labour-intensive sectors offer no technology worth a sunk investment, identifies the loop gain as

$$R_{ik} = \frac{D_k}{G} - (\xi - 1)\,\Delta\Gamma_{it}. \tag{47}$$

This is a normalisation and the evidence points to $R_{ik_0} > 0$: labour-intensive agriculture shows TFP growth (Table 1) and exports expand across factor intensities (Appendix Table A.20). Under $R_{ik_0} > 0$, (47) is a lower bound on R_{ik}, so it cannot by itself establish the uniqueness condition. Uniqueness rests on Appendix G.3, where Proposition 3 bounds the loop gain from primitives, independently of R_{ik_0}. Instead, (47) identifies the differential adoption response $R_{ik} - R_{ik_0}$ and provides a lower bound showing the adoption channel to be quantitatively large. The headline calibration is Allen, 2005: a 38% rise in w/r, $G = \ln(1.38) = 0.322$. Throughout I use the pooled differential $D_k = D = 1.297$ paired with the manufacturing labour shares as the binding case, since the manufacturing capital-intensive labour share is the lowest and hence its threshold $\gamma_{\text{ad}}(\xi - 1) + 1$ the tightest. At the headline G, the recovery is $R_{ik} = 4.03 - (\xi - 1)\Delta\Gamma$. At $\xi - 1 = 5$ and $\gamma_{\text{ad}} = 0.5$, $R_{ik} = 3.03$ at $\Delta\Gamma = 0.2$ is strictly below the threshold $\gamma_{\text{ad}}(\xi - 1) + 1 = 3.5$. At $\xi - 1 = 6.53$, the condition also holds ($R_{ik} = 2.72$ against a threshold of 4.27). The direct sector condition from (41), $\eta_{ik} = 1 + \Gamma_{ik}(\xi - 1) - R_{ik} > 0$, holds throughout the grid $\Gamma_{ik} \in [0.50, 0.70]$ and for both trade elasticities.[73] Positive reference-sector adoption raises the implied R_{ik} one-for-one, so the threshold would bind if R_{ik_0} exceeded 0.47 at the baseline elasticity (1.55 at the Costinot, Donaldson, and Komunjer, 2012 value). The ceiling of Appendix G.3 caps the loop gain in both sectors regardless.

Condition (45) disciplines the model with data: under the conditions leading to (47), it holds iff $G \geq D_k/\big(1 + \gamma_{\text{ad}}(\xi - 1) + (\xi - 1)\Delta\Gamma\big)$, so, under the reference-sector normalisation, this conservative sufficient condition for a unique equilibrium is met if the plague raised relative wages by at least 33% at the baseline elasticity and by 26% at the Costinot, Donaldson, and Komunjer, 2012 value, with positive reference-sector adoption raising the requirement. The pointwise verification of Appendix G.3 settles uniqueness independently of this threshold. Equivalently, since (47) nets out the factor proportions wedge, only the productivity component of the differential, $(1 - \Sigma)D = R_{ik}G$, enters the check: uniqueness requires it to stay below $G\big(1 + \gamma_{\text{ad}}(\xi - 1)\big)$, which is 1.13 log points at the baseline elasticity and 1.37 at the Costinot, Donaldson, and Komunjer, 2012 value, against 0.98 and 0.88 in the data. In other words, D_k is too large to come out of a stable equilibrium unless factor prices moved substantially. The 38% wage growth figure in Allen, 2005 clears this bar, and the required growth is actually smaller since I made conservative assumptions to arrive at this condition. Viewed from both angles, uniqueness holds.[74]

G.3 Uniqueness, a ceiling on the loop gain, and quantification

Notation and state variables

I fix a region-sector pair and suppress the indices ik, hold the rental r and FMA fixed, and write, within this appendix only, $\mathcal{S}$ for the bracketed sum in (20), so $T = \mathcal{S}^{\rho}$. Under the menu, $\mathcal{S} \equiv \sum_{\tau=0}^{n} S^{\tau}$ collects all rungs' bundles, and on a monotone path only the active margin

[73] This holds with more slack at the agricultural labour shares $[0.67, 0.86]$.

[74] The recovery identifies a path-average of $R_{ik}(\tilde{w})$ while (45) bounds its supremum. The quantification in Appendix G.3 settles this pointwise, and Proposition 3 shows the composite of (47) to be an upper bound on the loop gain. The uniqueness margin is accordingly not as tight as found here, where I conservatively read the composite entirely as loop gain.

$(\tau-1,\tau)$, relabelled (non, ad), moves, so that $\mathcal{S}=S^{\text{non}}+S^{\text{ad}}$ in the single-rung case. Two state variables summarise the active margin: the pair-local adopters' share $\pi=m^{\text{ad}}/B^{(\text{non,ad})}\in[0,1]$ and the cost-side adoption gain $q(\tilde{w})\equiv(\tilde{w}/\tilde{w}_0)^{\Delta\gamma(\sigma-1)}=\upsilon^{\sigma-1}\tilde{w}^{\Delta\gamma(\sigma-1)}$, with $\Delta\gamma\equiv\gamma_{\text{non}}-\gamma_{\text{ad}}$ the pair's share gap.

Lemma 1's interval is $1<q<q_{\text{crit}}\equiv\gamma_{\text{non}}/\gamma_{\text{ad}}$. At the calibrated values, $\Delta\gamma(\sigma_k-1)=1$, so q is $\tilde{w}/\tilde{w}_0$, and the headline $G=\ln(1.38)$ corresponds to the window $q\in[1,1.38]$. Direct substitution into (20) gives $\mathcal{S}=r^{1-\sigma}\tilde{w}^{-\gamma_{\text{non}}(\sigma-1)}B[1+(q-1)\pi]$, $s^{\text{ad}}=q\pi/[1+(q-1)\pi]$, and $\Gamma=\gamma_{\text{non}}-\Delta\gamma\,s^{\text{ad}}$. The profit gap of Lemma 1 is $\Pi=\tilde{w}^{-\gamma_{\text{non}}(\sigma-1)}(q-1)$, so that $\epsilon_\Pi\equiv d\ln\Pi/d\ln\tilde{w}=(\sigma-1)\big[q\Delta\gamma/(q-1)-\gamma_{\text{non}}\big]$ is positive on the interval and zero at q_{crit}. $\epsilon_m(z^*)\equiv-\,d\ln m^{\text{ad}}/d\ln z^*>0$ is the local elasticity of the adopter mass to the cutoff.

The within-bundle feedback: contraction, uniqueness, and elasticities

Three derivatives describe the technology block at given $(\tilde{w},FMA)$. First, how the bundle responds to the cutoff: the marginal variety leaves m^{non} for m^{ad}, and with conservation of B,

$$h\;\equiv\;-\frac{d\ln\mathcal{S}}{d\ln z^*}\;=\;\epsilon_m\,\frac{(q-1)\,\pi}{1+(q-1)\pi}\;\in\;[0,\epsilon_m).$$

Second, how the cutoff responds to the bundle: the cutoff condition (15) reads $\ln z^*=\text{const}-\frac{\rho-1}{\sigma-1}\ln\mathcal{S}(z^*)-\frac{1}{\sigma-1}\ln\Pi(\tilde{w})$, a fixed point in z^* whose slope is $g=(\rho-1)h/(\sigma-1)=(\xi-\sigma)h/(\sigma-1)^2$. Whenever $g<1$ the map is a contraction, so the sectoral equilibrium (z^*,T) exists and is unique. Since $h<\epsilon_m$, it suffices that $(\xi-\sigma)\bar{\epsilon}_m/(\sigma-1)^2<1$ with $\bar{\epsilon}_m\equiv\sup_{z^*}\epsilon_m(z^*)$ (i.e. $\bar{\epsilon}_m<(\sigma-1)^2/(\xi-\sigma)$, which is 16.3 at the calibrated elasticities; under the Pareto parameterisation below, $\nu_k<\sigma-1+(\sigma-1)^2/(\xi-\sigma)=21.3$), and at the corner $\pi=1$ the slope is zero. Third, differentiating the same cutoff condition along the wage path at fixed FMA passes the direct effect of the wage through the feedback multiplier $1/(1-g)$: writing $\tilde{R}\equiv h\chi$ for the pair's reallocation term, under the menu $d\ln\mathcal{S}=[\omega\tilde{R}-\bar{\Gamma}(\sigma-1)]\,d\ln\tilde{w}$, where $\omega=(S^{\text{non}}+S^{\text{ad}})/\mathcal{S}\le1$ is the active pair's revenue weight and $\bar{\Gamma}=\sum_\tau\gamma_\tau\tilde{s}^\tau$ the revenue-weighted realised share over all rungs. At $n=1$, $\omega=1$, $\bar{\Gamma}=\Gamma$, and $d\ln\mathcal{S}=[\tilde{R}-\Gamma(\sigma-1)]\,d\ln\tilde{w}$. Since $\omega\le1$, the two-rung expressions below bound the menu counterparts from above, so every uniqueness check that follows is conservative under the menu. The share drift also generalises: at fixed masses $d\bar{\Gamma}/d\ln\tilde{w}=-(\sigma-1)\,\text{Var}_{\tilde{s}}(\gamma_\tau)\le0$, and the active margin's mass shift moves revenue weight to the lower-γ rung, so $\epsilon_{\bar{\Gamma}}\le0$ holds for the menu at $\xi=\sigma$ as well. In the single-rung notation,

$$\chi\;\equiv\;-\frac{d\ln z^*}{d\ln\tilde{w}}\;=\;\frac{1}{1-g}\Big[\frac{\epsilon_\Pi}{\sigma-1}-(\rho-1)\Gamma\Big],\qquad\epsilon_{\text{ad}}=\epsilon_m\,\chi,\qquad R=\rho\,h\,\chi,$$

where $R=\rho\tilde{R}$ follows from $T=\mathcal{S}^\rho$ and (41). At the baseline $\xi=\sigma$, these collapse to closed forms:[75]

$$R(\tilde{w})=\epsilon_m\,\frac{\pi\left(\gamma_{\text{non}}-q\,\gamma_{\text{ad}}\right)}{1+(q-1)\pi},\qquad\epsilon_\Gamma=-\frac{\Delta\gamma}{\Gamma}\,s^{\text{ad}}(1-s^{\text{ad}})\Big[\Delta\gamma(\sigma-1)+\frac{\epsilon_{\text{ad}}}{1-\pi}\Big]\le0,$$

with $R(\tilde{w}_0)$ finite and $R(\tilde{w}_{\text{crit}})=0$. I record the three steps as:

Lemma 2. *At given $(\tilde{w},FMA)$ with $\tilde{w}$ in the interval of Lemma 1: (i) $h=\epsilon_m(q-1)\pi/[1+(q-1)\pi]$; (ii) the sectoral technology equilibrium exists and is unique whenever the feedback gain $g=(\xi-\sigma)h/(\sigma-1)^2$ lies below one (for all $\bar{\epsilon}_m<16.3$, i.e. under Pareto for all $\nu_k<21.3$), with $g=0$ at $\xi=\sigma$; (iii) the equilibrium elasticities are χ, ϵ_{ad}, and R as displayed; (iv) at $\xi=\sigma$ the loop gain and the share drift take the closed forms displayed, with $\epsilon_\Gamma\le0$.*

[75] Substitute ϵ_Π into $h\chi$, use $q\Delta\gamma-(q-1)\gamma_{\text{non}}=\gamma_{\text{non}}-q\gamma_{\text{ad}}$, and differentiate $\Gamma=\gamma_{\text{non}}-\Delta\gamma\,s^{\text{ad}}$ with $ds^{\text{ad}}=s^{\text{ad}}(1-s^{\text{ad}})\,d\ln[q\pi/(1-\pi)]$.

Two scope notes. First, at $\xi \neq \sigma$ the sign of the equilibrium cutoff response requires $\epsilon_\Pi/(\sigma-1) > (\rho-1)\Gamma$ beyond the contraction in (ii): the monotone-adoption statements are claimed at the baseline $\xi = \sigma$. Second, the sign $\epsilon_\Gamma \leq 0$ in (iv) likewise holds at $\xi = \sigma$.

A distribution-free ceiling on the path-average loop gain

How large can the loop gain be on average over the plague window? Integrating $d\ln\mathcal{S} = [\tilde{R} - \Gamma(\sigma-1)]\, d\ln\tilde{w}$ along a monotone path and substituting $\ln\mathcal{S} = \text{const} - \gamma_{\text{non}}(\sigma-1)\ln\tilde{w} + \ln[1+(q-1)\pi]$ from the notation paragraph, the factor-cost terms cancel through $\Gamma = \gamma_{\text{non}} - \Delta\gamma\, s^{\text{ad}}$ and leave

$$\int \tilde{R}\, d\ln\tilde{w} \;=\; \Delta\ln\big[1+(q-1)\pi\big] \;-\; \Delta\gamma(\sigma-1)\int s^{\text{ad}}\, d\ln\tilde{w}. \tag{48}$$

Even complete adoption, $\pi = 1$, caps the first term at $\ln q_{\text{end}}$, which Lemma 1 caps at $\ln(\gamma_{\text{non}}/\gamma_{\text{ad}})$. The second term subtracts and $R = \rho\tilde{R}$ pointwise. Since (48) is an identity (it uses only the two-technology structure and conservation of B, so it holds with the feedback included and for any draw distribution), it bounds each adjacent pair's cumulated contribution. Telescoping across the menu proves:

Proposition 3 (Ceiling across the menu)**.** *Let the sector's technology set be the ordered menu of Section 6, with labour shares $\gamma_0 > \gamma_1 > \cdots > \gamma_n$ and each adjacent pair satisfying Lemma 1 on its own interval. Along any monotone rising wage path, for any draw distribution and any initial adoption state, the path average loop gain satisfies*

$$\bar{R} \;\equiv\; \frac{1}{G}\int R\, d\ln\tilde{w} \;\leq\; \rho\,\frac{\ln\left(\gamma_0/\gamma_n\right)}{G}.$$

Proof. By the interval structure of Section 6, only adjacent margins are active, and the sector's adoption response is the sum of the responses at each active margin. Applying (48) to the pair $(\tau-1,\tau)$ bounds that margin's cumulated contribution along the path by $\rho\ln(\gamma_{\tau-1}/\gamma_\tau)$; summing over $\tau = 1,\ldots,n$ telescopes to $\rho\ln(\gamma_0/\gamma_n)$. □

The cap on the within-sector technology channel is the labour share span of the entire menu: a sequence of switches cannot save more than the factor cost difference between the most and the least labour-intensive technique.

Corollary 2 (Single switch)**.** *With a single adjacent pair ($n = 1$), $\bar{R} \leq \rho\ln\left(\gamma_{\text{non}}/\gamma_{\text{ad}}\right)/G$, which is 1.04 at $\xi - 1 = 5$ and 1.36 at $\xi - 1 = 6.53$ for $G = 0.322$.*

A single technology switch cannot save more than the factor-cost gap that motivates it, capped at $\gamma_{\text{non}}/\gamma_{\text{ad}}$.

Quantification. Two questions remain to be quantified: how much of the ceiling is attainable? Does uniqueness survive pointwise where section G.2 checks only the path-average? Both turn on the mass elasticity ϵ_m of Lemma 2, which I pin down with a specific distribution. I take the draws to be Pareto, $F_{ik}(z) = 1 - (z/z_{ik})^{-\nu_k}$ on $[z_{ik},\infty)$, with shape $\nu_k > \sigma_k - 1$. Under Pareto, $\epsilon_m = \nu_k - \sigma_k + 1$ and every object of Lemma 2 is computable along wage paths starting from the pre-plague steady state on the adoption locus, with the headline window $q \in [1, 1.38]$, drawing on post-plague wage growth in Allen, 2005. Maximising $\bar{R}$ for a single adjacent pair over the free scale of the adoption cost yields 0.08 at $\nu_k = 6$ ($\epsilon_m = 1$), rising monotonically to 0.51 at $\nu_k = 30$ ($\epsilon_m = 25$) at either elasticity. This is below both the single-switch ceiling of Corollary 2 and the composite of (47).

Proposition 3 and Corollary 2 reconcile these magnitudes. A single switch cannot quantitatively generate the recovered composite. Instead, the single-switch numerics above quantify one rung; the recovered differential requires roughly three. This is the reason why I model

technology adoption not as a binary choice but in rungs. The menu raises the cumulated response without raising the pointwise one. To be specific, the calibrated pair $(\gamma_{\text{non}}, \gamma_{\text{ad}}) = (0.7, 0.5)$ is the first rung of the menu. Attaining the recovered composite requires a menu span $\rho \ln(\gamma_0/\gamma_n) \geq (1-\Sigma)D$, a bottom labour share of about 0.26 at the baseline elasticity and 0.36 at $\xi - 1 = 6.53$: three rungs of the calibrated share ratio 1.4 ($0.7/1.4^3 \approx 0.26$), or two at the higher elasticity ($0.7/1.4^2 \approx 0.36$), so the window calibration and the required span are mutually consistent. While a given rung is active, the marginal pair's realised share lies in that rung's interval $[\gamma_{\text{ad}}, \gamma_{\text{non}}]$. The sector-wide realised share $\bar{\Gamma}_{ikt} = \sum_\tau \gamma_\tau \tilde{s}^\tau_{ikt}$ lies in $[\gamma_n, \gamma_0]$.

These bottom shares evaluate the factor-proportions wedge at the pre-plague gap of 0.2. On the model-consistent path of Section G.4, the wedge and the span are determined jointly: with the labour-intensive sector (at most) at 0.7, the realised cross-sector gap widens from at most 0.2 to $0.7 - \gamma_n$ as the capital-intensive sector descends the menu. Taking the gap to rise linearly in $\ln \tilde{w}$, the path-average gap and the bottom share solve a fixed point, $\gamma_n \approx 0.31$ with an average gap of about 0.3 at the baseline elasticity and $\gamma_n \approx 0.40$ with an average gap of about 0.25 at $\xi - 1 = 6.53$. Evaluating the wedge at these path averages attributes 37-41% of D to factor proportions, which is the model-consistent 59-63% adoption attribution of Section G.4. Evaluating it at the full menu gap $\Delta\Gamma_{\text{menu}} = 0.7 - \gamma_n$, the cross-sector gap once the capital-intensive sector has descended all its rungs, about 0.4 and 0.3 respectively, caps the wedge near 49%, which is the 51% floor.

Holding the labour-intensive sector at 0.7 throughout is itself a conservative evaluation choice as there may be some adoption. The normalisation $R_{ik_0} = 0$ is relaxed in section G.2) and any adoption narrows the cross-sector gap and shrinks the wedge. Uniqueness is unaffected as it is confirmed sector by sector. Minimising the pointwise labour demand elasticity over the path, the cost scale, and $\nu_k \leq 30$ returns exactly the static-technology floor at the capital-intensive share, $\eta = 1 + \gamma_{\text{ad}}(\xi - 1)$ (3.50 and 4.27 at the two elasticities), attained only at the full-adoption corner, where $R = \epsilon_\Gamma = 0$. The worst case for uniqueness is thus a point where the adoption margin has shut down, since the loop can push sectoral labour demand no lower than an economy with technology fixed at the capital-intensive share delivers. Uniqueness therefore holds at every wage the transition visits, for any adoption level, and any admissible ν_k, with a margin of $1 + \gamma_{\text{ad}}(\xi - 1)$. This margin is the first rung's; at the menu's full-adoption corner the floor is $1 + \gamma_n(\xi - 1)$, between 2.3 and 3.3 at the implied bottom shares, still bounded away from zero, so the deeper menu leaves uniqueness intact.

G.4 Recovering Productivities

Matching (27) to (23) delivers:

$$\delta_{ikt} = \widehat{\ln T}_{ikt} \;\equiv\; \ln \frac{T_{ikt}/T_{ik0t}}{T_{i0kt}/T_{i0k0t}} \;-\; \ln \frac{T_{ikt_0}/T_{ik0t_0}}{T_{i0kt_0}/T_{i0k0t_0}}. \tag{49}$$

The exporter term (20) factors exactly into productivity and a factor term,

$$\ln T_{ikt} = \rho_k \ln A_{ikt} \;-\; (\xi - 1)\,\bar{\Gamma}_{ikt}\,\ln \tilde{w}_{it} \;-\; (\xi - 1)\ln r, \tag{50}$$

where $\bar{\Gamma}_{ikt} \in [\gamma^k_{\text{ad}}, \gamma^k_{\text{non}}]$ is an efficiency-weighted average labour share, defined in the footnote. It is the average of the realised revenue-weighted share Γ_{ik} over the factor cost interval between $\tilde{w} = 1$ and $\tilde{w}_{it}$, holding the date-t technology masses fixed.[76]

[76]Write $S^{\text{non}}_{ikt} + S^{\text{ad}}_{ikt} = r^{1-\sigma_k} A_{ikt} \sum_\tau \left(\Phi^\tau_{ikt}/A_{ikt}\right) \tilde{w}_{it}^{-\gamma_\tau(\sigma_k - 1)}$ with $\Phi^{\text{non}}_{ikt} = m^{\text{non}}_{ikt}$ and $\Phi^{\text{ad}}_{ikt} = \upsilon_k^{\sigma_k - 1} m^{\text{ad}}_{ikt}$, so that $A_{ikt} = \sum_\tau \Phi^\tau_{ikt}$. The sum lies between $\tilde{w}^{-\gamma_{\text{non}}(\sigma_k - 1)}$ and $\tilde{w}^{-\gamma_{\text{ad}}(\sigma_k - 1)}$ and its logarithm equals $-(\sigma_k - 1)\bar{\Gamma}_{ikt} \ln \tilde{w}_{it}$ for an efficiency-weighted average $\bar{\Gamma}_{ikt} \in [\gamma_{\text{ad}}, \gamma_{\text{non}}]$. Holding the date-$t$ masses fixed, the derivative of the logarithm of the sum in $\ln \tilde{w}$ is $-(\sigma_k - 1)\Gamma_{ik}(\tilde{w})$ with Γ_{ik} the revenue-weighted share of (21), so $\bar{\Gamma}_{ikt} =$

The identified object is a triple difference. I write the triple difference of an object x_{ikt} as

$$\mathcal{D}\big[x_{ikt}\big] \;\equiv\; \Big[\big(x_{ikt} - x_{i_0kt}\big) - \big(x_{ik_0t} - x_{i_0k_0t}\big)\Big] \;-\; \Big[\big(x_{ikt_0} - x_{i_0kt_0}\big) - \big(x_{ik_0t_0} - x_{i_0k_0t_0}\big)\Big], \tag{51}$$

so that $\widehat{\ln T}_{ikt} = \mathcal{D}\big[\ln T_{ikt}\big]$. Substituting (50) into (49), $(\xi-1)\ln r$ drops out, leaving

$$\widehat{\ln T}_{ikt} = \underbrace{\rho_k\,\mathcal{D}\big[\ln A_{ikt}\big]}_{\text{relative productivity}} - (\xi-1)\,\underbrace{\mathcal{D}\big[\bar{\Gamma}_{ikt}\,\ln\tilde{w}_{ikt}\big]}_{\text{factor proportions}}. \tag{52}$$

The general form of the productivity term is $\mathcal{D}[\rho_k \ln A_{ikt}]$, with each sector carrying its own weight. In the quantification I impose $\sigma_k = \sigma$, so $\rho_k = \rho$ and the two expressions coincide. As $\bar{\Gamma}_{ikt} \in [\gamma^k_{\text{ad}}, \gamma^k_{\text{non}}]$ is strictly positive, the factor price term in (50) is weakly negative in every sector, so the level expansion of market shares in Fact #2 at rising $\tilde{w}_{it}$ requires $\Delta \ln A_{ikt} > 0$. The triple difference (52) nets this level out and identifies only the cross-sector differential. The menu makes the level expansion quantitatively feasible since a single switch is capped by Corollary 2 at $\rho\ln(\gamma_{\text{non}}/\gamma_{\text{ad}})$ and cannot offset the wage penalty.

Bounding the role of factor proportions. The factor-proportions term in (52) collapses to a single wedge in two steps. First, I compare capital- to labour-intensive sectors within manufacturing or within agriculture, where wages carry no sector subscript, leaving

$$\mathcal{D}\big[\bar{\Gamma}_{ikt}\ln\tilde{w}_{ikt}\big] = -\Delta_t\big(\Delta\Gamma_{it}\,\ln\tilde{w}_{it}\big) + \Delta_t\big(\Delta\Gamma_{i_0t}\,\ln\tilde{w}_{i_0t}\big), \tag{53}$$

where Δ_t denotes the change between base period t_0 and t, and $\Delta\Gamma_{it} \equiv \bar{\Gamma}_{ik_0t} - \bar{\Gamma}_{ikt} > 0$ is i's labour-share gap between the labour- and capital intensive sector.

Second, the time difference. I set the treated region's pre-period normalised wage to unity, $\ln\tilde{w}_{it_0} = 0$, so the first term of (53) is $-\Delta\Gamma_{it}\,\Delta_t \ln\tilde{w}_{it}$. For the second term, I assume that control region factor prices move little, $\Delta_t \ln\tilde{w}_{i_0t} \approx 0$. With unchanged wages, there is no adoption, so labour shares are unchanged and the term vanishes. I can then write $G \equiv \Delta_t \ln\tilde{w}_{it} - \Delta_t \ln\tilde{w}_{i_0t} = \ln\tilde{w}_{it}$ for the relative normalised wage change. The exporter term recovered from gravity then has a single term capturing factor proportions:

$$\widehat{\ln T}_{ikt} = \underbrace{\rho_k\,\mathcal{D}\big[\ln A_{ikt}\big]}_{\text{relative productivity}} \underbrace{+\,(\xi-1)\,\Delta\Gamma_{it}\,G}_{\text{factor proportions}}. \tag{54}$$

G.5 Simulations

Figure A.11 compares simulated competitiveness growth between the hypothetical full sample and the observed sample, where only trade passing the Sound is registered. I simulate a labour-intensive and a capital-intensive sector and compute trade costs as $d_{ij}^{-\xi-1}$, where d_{ij} is distance over sea.[77] I simulate two time periods, where in the first all regions have the same capital-intensive sector competitiveness. Plagued regions then experience competitiveness growth in the capital-intensive sector, where the simulated effect is $ln2 = 0.693$. I show results for a log-normal specification of noise, where I take the log of simulated trade, add standard normally distributed noise, and take the exponential. Results with multiplicative and uniformly

$(\ln\tilde{w}_{it})^{-1}\int_0^{\ln\tilde{w}_{it}}\Gamma_{ik}(u)\,du$, a fixed-masses average over the factor-cost dimension, whose masses evolve. I fix one reference convention throughout the quantification: the normalisation $\ln\tilde{w}_{it_0} = 0$ of section G.4, under which the control region term vanishes because both of its factors are time-invariant, and all reported productivity and wedge components refer to this convention. Multiplying by ρ_k and using $\rho_k(\sigma_k - 1) = \xi - 1$ gives (50).

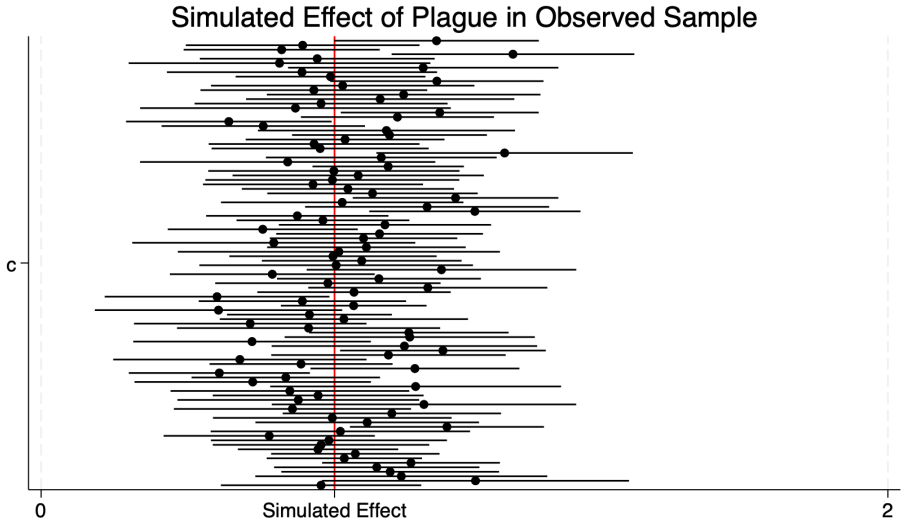
[77] I compute cost distances over sea for each city pair in the dataset using a 30 arc-second raster (about 1km side length) and the CostDistance tool in ArcGIS.

distributed noise display the same pattern. I then recover sectoral competitiveness as before, recover competitiveness growth, and regress it on a plague dummy and its interaction with a capital-intensity dummy, while including region-time, region-sector, and sector-time fixed effects and clustering standard errors on regions.

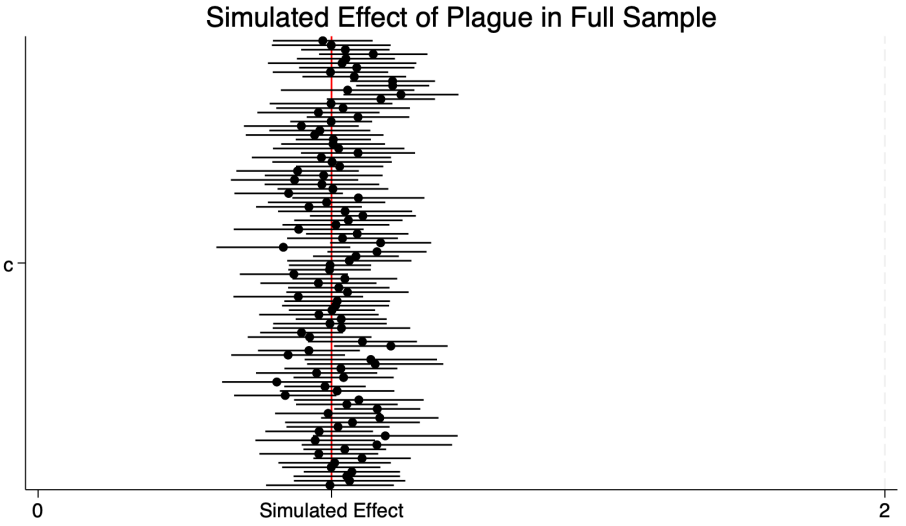
The results are displayed in Figure A.11. Both the observed and the full sample produce point estimates that are centred around the simulated effect.

I also simulate trade following equation (23) and find similar results when regressing simulated trade volumes in both samples on a plague dummy and its interaction with a capital-intensity dummy, including origin-destination and destination-time fixed effects. The results for the capital-intensity interaction, presented in the bottom panels of Figure A.11, show that the point estimates are very similar when comparing the full to the observed sample and again centred around the simulated effect. I conclude that only observing ships passing through the Sound does not introduce a bias in my estimates.

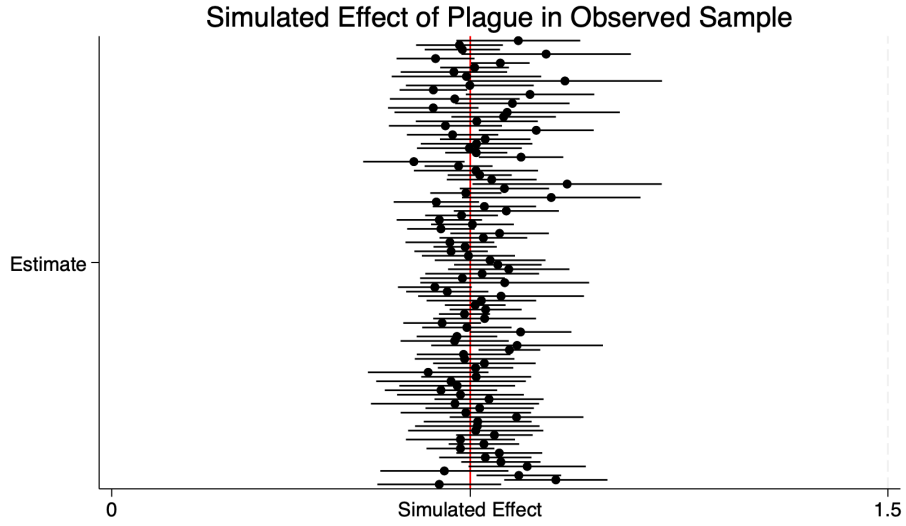
Figure A.11: Simulated competitiveness growth and trade increase in observed vs. full sample



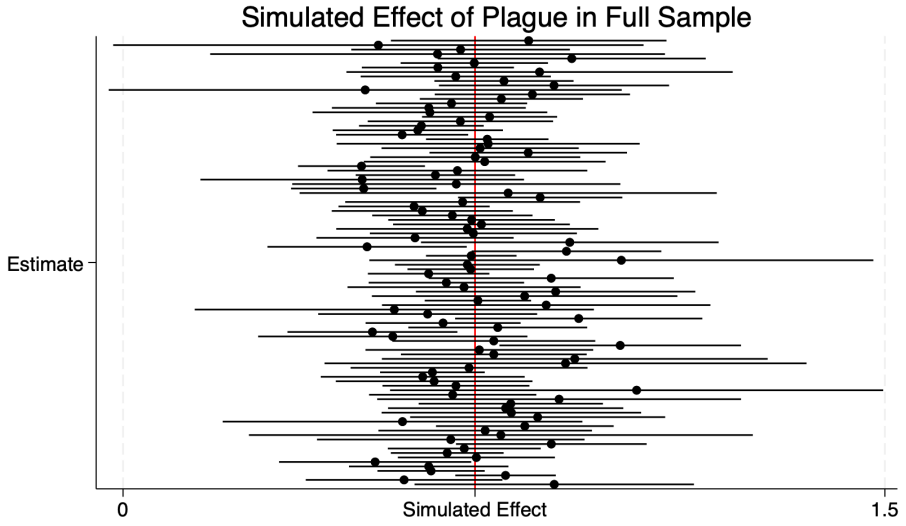
(a) Estimated competitiveness growth in observed sample



(b) Estimated competitiveness growth in full sample



(c) Estimated trade growth in observed sample



(d) Estimated trade growth in full sample

Notes: Simulation results described in Appendix G.5. The simulated effect is $\ln 2 = 0.693$. Standard errors are clustered on regions.

H Mechanism Appendix

H.1 Scarce Labour and Higher Wages

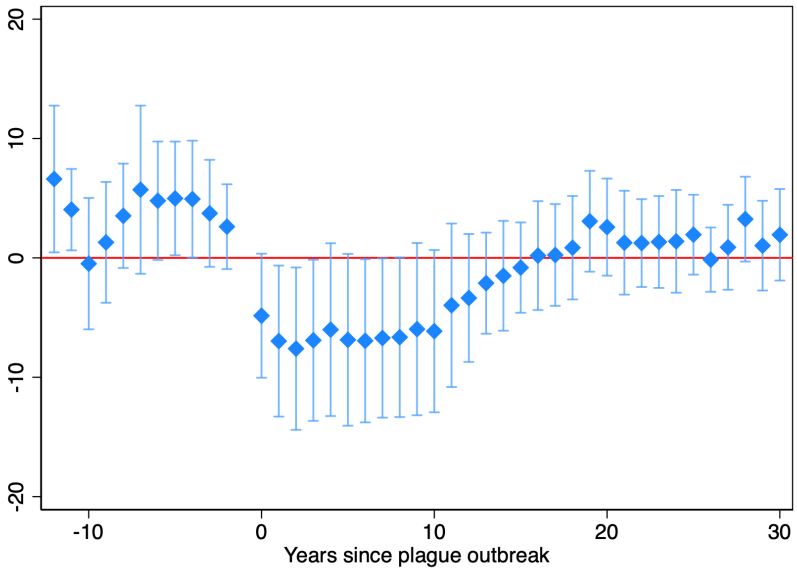
Table A.29 shows that grain price-inferred interest rates in Danzig did not rise after the plague. Figure A.12 shows an event study for the number of captains living in a region.

Table A.29: Grain price-inferred interest rates in Danzig do not increase

	Wheat	Rye	Barley
	(1)	(2)	(3)
Post Plague	-0.606	-0.589	0.245
	(0.397)	(0.390)	(0.333)
Crop years	71	75	65

Notes: Robust standard errors in parentheses. * p <0.10, ** p <0.05, *** p <0.01. The dependent variables are annualised grain price-inferred interest rates by crop year (September to August) in Danzig between 1704 and 1793, using the methodology introduced by Keller, Shiue, and Wang, 2021: twelve times the mean month-to-month log price change over months with positive price changes within the crop year. Prices are the midpoint of monthly minimum and maximum quotations. I assume that storage costs are constant. Grain price data by Allen and Unger, 2018. The independent variable is a post-plague dummy that takes the value 1 from crop year 1709 onwards, when Danzig had been plagued.

Figure A.12: Population dynamics after the plague: number of captains as proxy for population



Notes: Event study with area-year and region fixed effects on the number of unique captains' last names living in a region. Captains with the same first and last name from the same region in the same year are assumed to be duplicates and thus dropped.

H.2 Technology Adoption and Capital Deepening

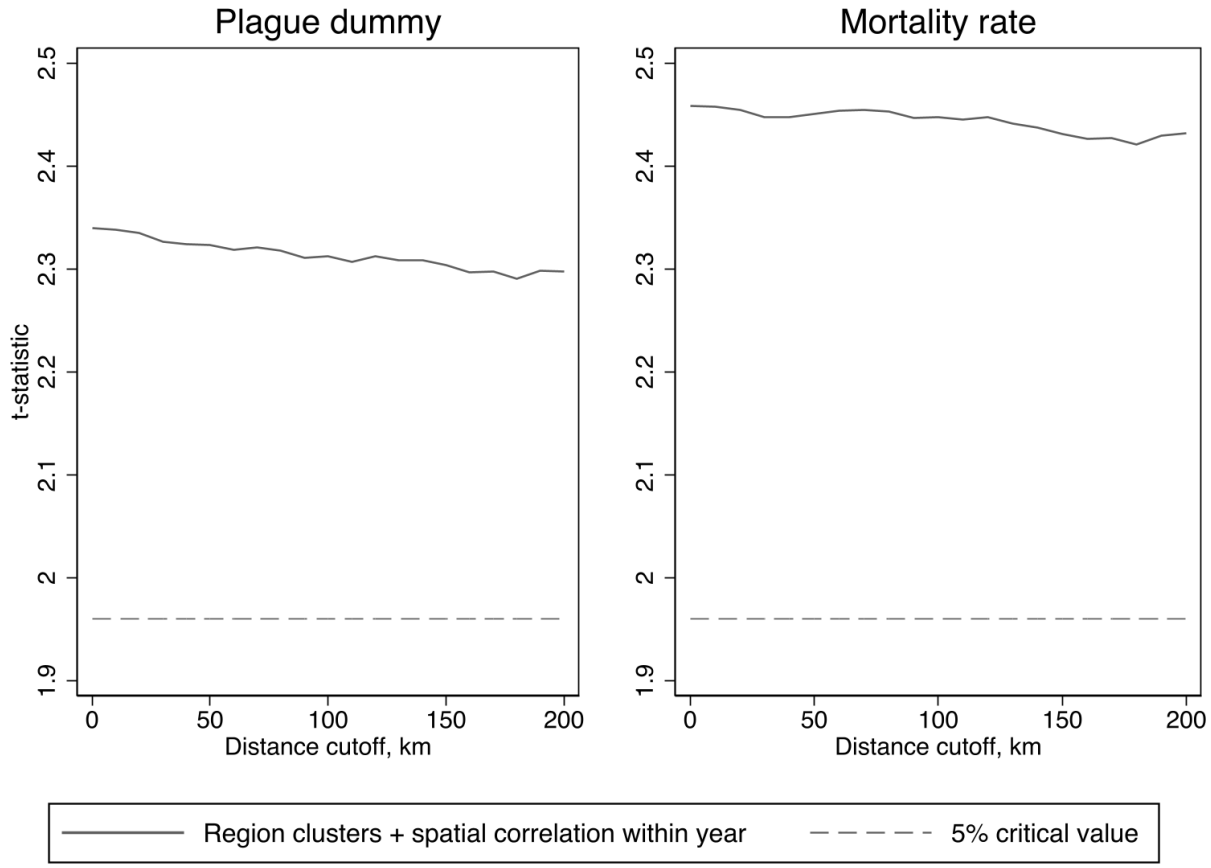
Table A.30 lists examples of capital deepening by sector in this period and area. The character of available investments varies across sectors. The capital-intensive technologies of Panels B and D are dominated by sunk installations such as furnaces, mills, docks, and livestock housing. The labour-intensive entries are largely implements with low sunk costs. Moreover, several Panel C

entries, such as fulling mills, were centuries old and had long diffused to wherever water sites allowed. The menu therefore supports a weaker adoption response in labour-intensive sectors, that is a small R_{ik_0}.

H.3 Productivity Growth

Figure A.13 displays results from estimating the regression underlying column 1 of Table 2 using Conley standard errors (Conley, 1999) and a range of different distance cut-offs. Figure A.14 shows the capital-intensity differential by decade in the long-run.

Figure A.13: Plague and competitiveness under spatial correlation



Notes: The figure plots the t statistics of the pooled capital-intensity differential in columns (1) and (4) of Table 2, for the plague dummy (left panel) and the mortality rate (right panel), against the distance cutoff used in the spatial standard error correction. The variance is region clustering plus cross-region correlation within year, constructed as the sum of the region-clustered variance and the spatial HAC variance following Conley (1999) within year, less the region-year-clustered variance that both contain, and rescaled to the degrees-of-freedom correction of the table. The spatial part uses a Bartlett kernel that declines linearly to zero at the cutoff, computed for distance cutoffs between 10 and 200 kilometres. A cutoff of zero is the region-clustered standard error reported in the table. Point estimates are identical across cutoffs, so all variation reflects the standard errors.

Table A.30: Examples of capital deepening by sector

Panel A: Labour-Intensive Agriculture

Rung	Example	Description
1	Scythe	Replaces sickle in grain harvesting, raises harvest per worker
	Iron plough parts	Iron shares and iron-plated mouldboards on wooden ploughs
2	Horse traction	Horses replacing oxen for ploughing and harrowing
	Grain storage	Fixed granaries reducing post-harvest losses
3	Wind- and watermills	Replacing hand querns for grinding
	Drainage infrastructure	Ditching, dyking, and polder construction

Panel B: Capital-Intensive Agriculture

Rung	Example	Description
1	Dairy processing	Butter churns and cheese presses raising output per worker
	Fencing	Fixed fencing replacing herding labour
2	Livestock housing	Purpose-built winter stalling extending the productive season
	Hay and fodder storage	Barns and winter fodder infrastructure raise carrying capacity
	Fish processing	Smokehouses, salt stores, and drying racks for herring & stockfish
3	Holstein dairy system	Large dairy units with dedicated buildings and equipment, spreading from Holstein

Panel C: Labour-Intensive Manufacturing

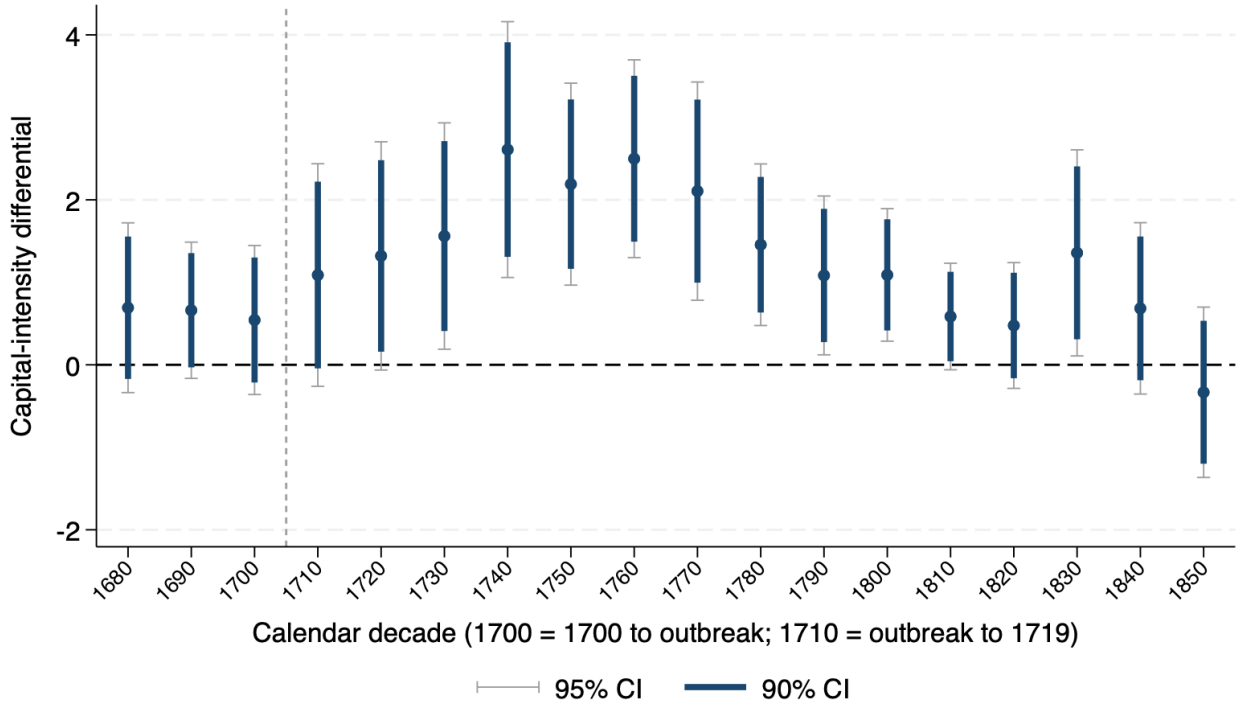
Rung	Example	Description
1	Dutch ribbon loom	Multi-shuttle loom weaving 12 to 24 ribbons at once
	Stocking frame	Frame knitting replacing hand knitting
	Warping mills	Mechanical preparation of warp threads
	Dyeing vats	Dyeing infrastructure with controlled heating
2	Finishing equipment	Mangles, tentering frames, and presses for cloth finishing
	Bleachfields	Dedicated infrastructure for linen and cloth bleaching
3	Fulling mills	Water-powered fulling replaces manual trampling of woollen cloth

Panel D: Capital-Intensive Manufacturing

Rung	Example	Description
1	Shipyard cranes	Treadwheel and horse-powered cranes for heavy lifting
	Ropewalks	Purpose-built long structures for rope production
2	Brick kilns; glassworks	Improved kiln and furnace design
	Water-powered sawmills	Replacing pit sawing with mechanised timber processing
	Slitting and rolling mills	Replacing hand hammering with mechanical processing of bar iron
3	Hammer forges	Water-powered tilt hammers and forge methods to refine bar iron
	Larger blast furnaces	Scaling of charcoal furnaces with water-powered bellows
	Dry docks	Fixed infrastructure for ship construction and repair, such as the Karlskrona dock begun in 1717

Notes: Examples of investments in this period. Within each panel, entries are grouped into rungs of increasing sunk cost and indivisibility, matching the ordered menu of Section 6: rung 1 collects divisible implements and equipment, rung 2 fixed structures and installations, and rung 3 integrated or water-powered works. The capital-intensive sectors (Panels B and D) offer deeper menus, whereas the upper rungs of labour-intensive menus had long diffused.

Figure A.14: Capital-intensity differential in competitiveness (and thus comparative advantage) in the long run



Notes: This figure plots the coefficients on the interaction of a capital-intensity indicator with decade indicators for plagued regions, from a single regression of log sectoral competitiveness recovered from equation (27) on the four classified sectors. It thus mirrors equation (31) in its set up but extends to the end of the data in 1857 and estimates the post-plague term by decade. Each coefficient is thus the change in the competitiveness of the capital-intensive relative to the labour-intensive sector, in plagued relative to unplagued regions, in the respective decade relative to the reference period 1668-1679. Decades before the plague are defined on pre-outbreak years only, so that the 1700s bin runs from 1700 to the year before the region's outbreak, and the 1710s bin runs from the outbreak to 1719. The remaining bins are calendar decades from the 1720s to the 1850s. The vertical dashed line separates pre- and post-plague bins. The regression absorbs region-sector, sector-area-year and region-agriculture-year fixed effects, as in Table 2, and standard errors are clustered at the region level. Gray bars show 95% and navy bars 90% confidence intervals.

Appendix References

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